



Stochastic Analysis

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Preface

These notes grew out of the course on Stochastic Analysis taught by Prof. Jieliang HONG at the Southern University of Science and Technology. After my exchange semester at SUSTech in Fall 2025, I returned to my home institution, yet continued to travel to Shenzhen on Wednesdays to attend the lectures as an auditing student. I write these notes partly to preserve what I learned there, and partly because stochastic analysis has become one of the foundations I must build carefully for my future study in financial mathematics.

The course begins where ordinary stochastic processes cease to be enough. Starting from continuous-time martingales, it develops the Itô integral, and several martingale inequalities. These tools then lead naturally to stochastic differential equations: strong solutions, the Yamada–Watanabe argument, comparison principles, and weak solutions. Finally, the narrative culminates in the profound connection between Brownian motion and partial differential equations. I have tried to keep the exposition close to the lectures, while reorganising some arguments in the way I found most natural when learning them.

This notebook is also a continuation of an earlier one. In Fall 2025, when I first wrote notes for stochastic processes, it began almost out of necessity: I was trying to keep up with a course I did not fully understand, and perhaps also trying to give myself something steady to hold on to. Over time, however, the act of writing became something gentler. I began to enjoy shaping mathematics in my own language, with my own habits, and with music quietly marking the hours. Sometimes a song brings back a whole period of life; perhaps these pages will do the same.

So these notes are not only a record of definitions, theorems, and proofs. They are also a small attempt to leave something honest from a time when I still had enough curiosity, patience, and fire to make things carefully. If, years later, I return to these pages and feel even a little moved by the person who wrote them, then this notebook will have already meant more than a set of lecture notes.

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Chapter 0

Motivation

To initiate, recall the Riemann-Stieltjes Integral:

Definition 0.0.1. If f is continuous and g is of finite variation, then:

$$\int_a^b f(x) dg(x)$$

is well defined to be, with $a \leq x_0 < x_1 < \dots < x_n \leq b$ and $\xi_i \in [x_i, x_{i+1}]$:

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(\xi_i) (g(x_{i+1}) - g(x_i))$$

, and known as Riemann-Stieltjes Integral.

Then the main goal of stochastic analysis is to re-walk through the real analysis by rigorously defining the Itô Integral even for stochastic processes:

$$(H \cdot X)_t = \int_0^t H_s dX_s$$

The original rule for Riemann-Stieltjes fails if we choose integrator to be Brownian motion.

Theorem 0.0.1. For a standard brownian motion, $B = \{B_t, t \geq 0\}$:

$$\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right)^2 \xrightarrow{a.s.} 1$$

Proof. On dyadic partition, for each fixed n , define, $\forall 1 \leq k \leq 2^n$:

$$\Delta_k^{(n)} := B_{k/2^n} - B_{(k-1)/2^n} \implies Q_n := \sum_{k=1}^{2^n} (B_{k/2^n} - B_{(k-1)/2^n})^2 = \sum_{k=1}^{2^n} \left(\Delta_k^{(n)} \right)^2$$

Since B is a SBM, the random variables $\Delta_1^{(n)}, \dots, \Delta_{2^n}^{(n)}$ are independent and

$$\Delta_k^{(n)} \sim \mathcal{N}(0, 2^{-n}) \implies E \left(\Delta_k^{(n)} \right)^2 = 2^{-n}$$

Therefore, the expectation of Q_n :

$$\mathbb{E}Q_n = \sum_{k=1}^{2^n} \mathbb{E} \left(\Delta_k^{(n)} \right)^2 = 2^n \cdot 2^{-n} = 1$$

Besides, by simple manipulation, the variance is given by, using independence:

$$\text{Var}(Q_n) = \mathbb{E} (Q_n - 1)^2 = \sum_{k=1}^{2^n} \text{Var} \left[\left(\Delta_k^{(n)} \right)^2 \right] = 2^n \cdot 2 \cdot 2^{-2n} = 2^{1-n} \rightarrow 0$$

Now fix $\varepsilon > 0$, by Chebyshev's inequality,

$$\mathbb{P}(|Q_n - 1| > \varepsilon) \leq \frac{\mathbb{E} (Q_n - 1)^2}{\varepsilon^2} = \frac{2^{1-n}}{\varepsilon^2} \implies \sum_{n=1}^{\infty} \mathbb{P}(|Q_n - 1| > \varepsilon) \leq \frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} 2^{1-n} < \infty$$

By the Borel–Cantelli lemma,

$$\mathbb{P}(|Q_n - 1| > \varepsilon \text{ i.o.}) = 0.$$

Since this holds for every $\varepsilon = 1/m$, $m \in \mathbb{N}$, we conclude that

$$Q_n \rightarrow 1 \quad \text{a.s.}$$

This completes the proof. □

Using above easy-to-prove theorem, we first to be precise about **finite variation**:

Definition 0.0.2. The **Finite Variation** of a function $g : [a, b] \mapsto \mathbb{R}$ is defined to be:

$$V_g([a, b]) = \int_a^b |dg(x)| = \sup_n \sum_{i=1}^n |g(x_{i+1}) - g(x_i)|$$

Now, we could justify that brownian motion fails in having infinite variation:

Corollary 0.0.2.

$$V_B([0, 1]) = \infty, \text{ a.s.}$$

Proof. Firstly, we notice the following inequality:

$$\begin{aligned} \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right)^2 &\leq \sup_{1 \leq k \leq 2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \cdot \sum_{k=1}^{2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \\ &\leq V_B([0, 1]) \cdot \sup_{1 \leq k \leq 2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \end{aligned}$$

Then if $V_B([0, 1]) < \infty$, then by continuity of B , we have:

$$\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right)^2 \rightarrow 0$$

, which is a contradiction, so $V_B([0, 1]) = \infty$ with probability 1. □

However, Ito found that, if we put some restriction on integrand and integrator, such as:

$$\begin{aligned} H_s &= \text{previsible/ predictable} \\ X_s &= \text{semimartingale} \end{aligned} \implies (H \cdot X)_t = \int_0^t H_s dX_s \text{ is well defined.}$$

Example 0.0.1. Let $H_n \in \mathcal{F}_{n-1}$, and $M = \{M_n, n \in \mathbb{N}\}$ be a martingale, then:

$$(H \cdot M)_n = \sum_{k=1}^n H_k (M_k - M_{k-1})$$

is a martingale, which is known as discrete **martingale transformation**.

Besides rigorously defining the integral for general case, we also want the result integration keeps some desirable property, like **martingale**, this intuition comes from the financial application, where if the integral can't keep martingale, we could replicate an **arbitrage portfolio**. Then next section is to introduce some toolkits and also know the definition of **predictable, semimartingale**.



Chapter 1

Preliminary

Given a probability space $(\Omega, \mathcal{F}, \mathbf{P})$, a stochastic process $X = \{X_t, t \geq 0\}$ is a collection of r.v.s taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ formally, it is a measurable function:

$$\begin{aligned} X : (t, \omega) &\mapsto X_t(\omega) \\ [0, \infty) \times \Omega &\mapsto \mathbb{R}^d \end{aligned}$$

1.1 Sample Paths

Now if we fix ω , then the function:

$$t \mapsto X_t(\omega)$$

is called a sample path. Based on this concept, we could provide some useful definitions:

Definition 1.1.1. For two stochastic processes X and Y , we say Y is a modification of X if $\forall t \geq 0, \mathbf{P}(X_t = Y_t) = 1$

Remark 1.1.1. To be more precise, $\mathbf{P}(\{\omega : X_t(\omega) = Y_t(\omega)\}) = 1$

Most of time, modification is enough for us to inherit some property like continuity, but in order for the smoothness of knowledge and to get better understanding, we present a "slightly" stronger concept:

Definition 1.1.2. We say X and Y are indistinguishable if

$$\mathbf{P}(X_t = Y_t, \forall t \geq 0) = 1$$

Then Y is a version of X .

The reason why we say it is a slightly stronger version, is that due to the countable additivity, we could guarantee all the rational points to be the same, and if we have the "continuity" of the sample path we could approximate other points by rational points, then achieve indistinguishable, as following example gives:

Example 1.1.1. Let Y be a modification of X and suppose X and Y both have a.s. right continuous sample paths, then X and Y are indistinguishable.

Definition 1.1.3. A function f is right-continuous if:

$$f(x) = f(x^+) = \lim_{t \downarrow x} f(t)$$

Proof. Since $\forall t \geq 0, P(X_t \neq Y_t) = 0$, then we get:

$$P\left(\bigcup_{q \in \mathbb{Q}^+} X_q \neq Y_q\right) \leq \sum_{q \in \mathbb{Q}^+} P(X_q \neq Y_q) = 0 \implies P\left(\bigcap_{q \in \mathbb{Q}^+} X_q = Y_q\right) = 1$$

For two right-continuous functions f, g if:

$$f(q) = g(q), \forall q \in \mathbb{Q}^+ \implies f(t) = g(t), \forall t \geq 0$$

Therefore, using this result from measure theory, we land in:

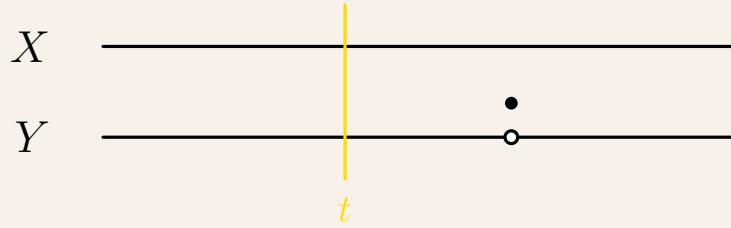
$$P(X_t = Y_t, \forall t \geq 0) = P\left(\bigcap_{t \geq 0} X_t = Y_t\right) = 1$$

Hence, we completed the proof. □

Please notice from above example 1.1.1 that we restrict to a.s. right-continuous sample path, so that we could land in approximation, but the following opposite example shows us the difference of modification and indistinguishable process.

Example 1.1.2. Let $T \sim \text{Exp}(1)$, define $X_t \equiv 0, \forall t \geq 0$, and another one:

$$Y_t = \begin{cases} 0, & t \neq T \\ 1, & t = T \end{cases}$$



Then we know:

$$\forall t \geq 0, P(X_t \neq Y_t) = P(Y_t = 1) = P(t = T) = 0$$

so X is a modification of Y but not a version of, which you should verify.

1.2 Filtration

After talking about process, as processes go the information also increases, then to model this intuitive behavior, we introduce **filtration**: Let $\{\mathcal{F}_t, t \geq 0\}$ be a filtration of $(\Omega, \mathcal{F})$ such that $\forall 0 \leq s < t < \infty$:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$$

Typically, we just adapt to the so called **natural filtration**:

Example 1.2.1. Let $X = \{X_t, t \geq 0\}$ be a stochastic process, define:

$$\mathcal{F}_t^X = \sigma(X_s, 0 \leq s \leq t)$$

, then $\{\mathcal{F}_t^X, t \geq 0\}$ is a filtration.

From definition we could see, natural filtration is the minimum filtration that contains the information of the corresponding process, and also the adaptivity, i.e. $X_t \in \mathcal{F}_t^X$.

Example 1.2.2. Let X be a process, every sample paths of which is **RCLL**, let:

$$A = \{\omega : X(\omega) \text{ is continuous on } [0, t_0)\}$$

for some fixed $t_0 > 0$, then $A \in \mathcal{F}_{t_0}^X$.

Definition 1.2.1. The sample path of a process is **RCLL** if its sample path is **Right-Continuous with Left Limit**, i.e.:

$$f(x) = f(x^+), f(x^-) \exists$$

Proof. Since we already have right continuous, then what remain is show it meets with left limit:

$$A = \{\omega : \forall s \in (0, t_0), X_{s-}(\omega) = X_s(\omega)\}$$

Similar trick, to rewrite event A into "rational number version":

$$A = \{\omega : \forall q \in (0, t_0) \cap \mathbb{Q}, X_{q-}(\omega) = X_q(\omega)\}$$

It boils down into countable cases, to numerate $(0, t_0) \cap \mathbb{Q} = \{q_m, m \geq 1\}$:

$$\begin{aligned} A &= \left\{ \omega : \forall q_m, \forall k \geq 1, \exists N \geq 1, \forall n \geq N, \left| X_{q_m - \frac{1}{n}}(\omega) - X_{q_m}(\omega) \right| < \frac{1}{k} \right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ \left| X_{q_m - \frac{1}{n}}(\omega) - X_{q_m}(\omega) \right| < \frac{1}{k} \right\} \end{aligned}$$

Notice each $\left\{ \left| X_{q_m - \frac{1}{n}}(\omega) - X_{q_m}(\omega) \right| < \frac{1}{k} \right\} \in \mathcal{F}_{t_0}^X$, then $A \in \mathcal{F}_{t_0}^X$. □

Notice that example 1.2.2 requires every sample path to be RCLL, but if we lessen a little bit, even into a.s. RCLL, the conclusion may fail as following example suggests:

Example 1.2.3. Let X be a process whose sample paths are RCLL a.s., with all the same setup as example 1.2.2, then it is possible that $A \notin \mathcal{F}_{t_0}^X$.

Proof. WLOG, let $t_0 = 1$, and U be a non-measurable r.v. such that:

$$\mathbb{P}(U = 1) = 0$$

Define our X by:

$$X_t(\omega) = \begin{cases} 0, & U(\omega) \neq 1 \\ \begin{cases} 0, & t \neq \frac{1}{2} \\ 1, & t = \frac{1}{2} \end{cases}, & U(\omega) = 1 \end{cases}$$

By our construction, we know that:

$$A = \{\omega : X(\omega) \text{ is continuous on } [0, t_0]\} = \Omega / \{U = 1\}$$

, so if $\{U = 1\} \notin \mathcal{F}_{t_0}^X$, then $A \notin \mathcal{F}_{t_0}^X$. □

From example 1.2.3, we see the key of the proof is to construct a zero-measure set such that it is not in our natural filtration, which is too tricky and hopefully be forbidden, then it gives us the motivation to consider broader class of filtration:

Definition 1.2.2. A filtration $\{\mathcal{F}_t, t \geq 0\}$ is said to satisfy the usual condition if:

- (a) **Right-continuous:** $\mathcal{F}_t = \mathcal{F}_{t+} := \bigcap_{s>t} \mathcal{F}_s$.
- (b) $\mathcal{F}_0$ contains all the P-negligible set in $\mathcal{F}$.

To complete the gap, we further provide the definitions of two:

Definition 1.2.3. An event A is P-negligible if $\exists N \in \mathcal{F}$, s.t. $A \subseteq N, \mathbb{P}(N) = 0$.

Another one is "left/right limit" of filtration:

Definition 1.2.4. We say a filtration is right-continuous if $\mathcal{F}_t = \mathcal{F}_{t+}$, where:

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s, \mathcal{F}_{t-} = \sigma \left(\bigcup_{s<t} \mathcal{F}_s \right)$$

After introducing information flow, we need to relate it with process:

Definition 1.2.5. The stochastic process X is adapted to the filtration $\{\mathcal{F}_t, t \geq 0\}$ if $\forall t \geq 0, X_t \in \mathcal{F}_t$. Clearly, X is adapted to the natural filtration.

Adaptivity is too loose, and impedes understanding, thus we consider progressively measurable.

Definition 1.2.6. The stochastic process X is called **progressively measurable** w.r.t $\{\mathcal{F}_t, t \geq 0\}$ if $\forall t \geq 0$, the map following is measurable.

$$(s, \omega) \mapsto X_s(\omega) : ([0, t] \times \Omega, \mathcal{B}([0, t]) \otimes \mathcal{F}_t) \mapsto (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

Remark 1.2.1. To compare, adaptivity is to require measurability to:

$$(s, \omega) \mapsto X_s(\omega) : ([0, t] \times \Omega, \mathcal{B}([0, t]) \otimes \mathcal{F}) \mapsto (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

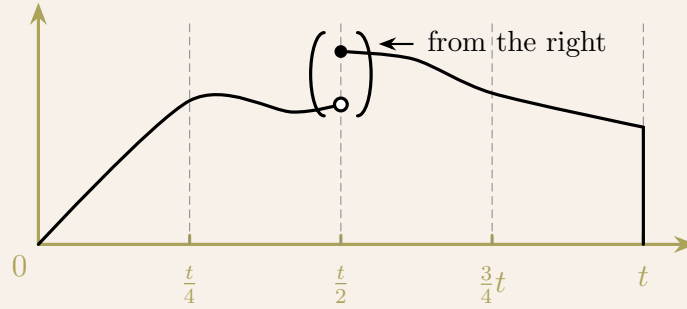
Intuitively speaking, progressively measurable is to require the information is "strictly" enough at this each time, to see the relations:

$$\text{adapted} \supseteq \text{progressively measurable} \supseteq \text{previsible}$$

Most of cases, we don't differentiate them, since it is really close as below example shows:

Example 1.2.4. If X is adapted to filtration $\{\mathcal{F}_t, t \geq 0\}$ and every sample path is right-continuous (same for left-continuous), then X is also progressively measurable.

Proof. Since right continuity, $\forall t > 0, \forall n \geq 1, \forall 0 \leq k \leq 2^n - 1$, we construct:



$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \forall \frac{kt}{2^n} < s \leq \frac{(k+1)t}{2^n}$$

, then $\lim_{n \rightarrow \infty} X_s^{(n)}(\omega) = X_s(\omega)$ for all $(s, \omega) \in [0, t] \times \Omega$. Recall that limit preserve our measurability, then it suffices to prove:

$$(s, \omega) \mapsto X_s^{(n)}(\omega) : ([0, t] \times \Omega, \mathcal{B}([0, t]) \otimes \mathcal{F}_t) \mapsto (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable, however, this is clearly true since, $\forall A \in \mathcal{B}(\mathbb{R}^d)$:

$$\{(s, \omega) : X_s^{(n)}(\omega) \in A\} = \bigcup_{k=0}^{2^n-1} \left(\frac{k}{2^n}t, \frac{k+1}{2^n}t \right] \times \left\{ X_{\frac{k+1}{2^n}t} \in A \right\} \in \mathcal{B}([0, t]) \otimes \mathcal{F}_t$$

Therefore, we completed the proof. □

1.3 Stopping Times

Stopping times are really important subject of interest, since we already known that original brownian motion has infinite variation, but if we use stopping times to replace the original time, then to construct **local martingale**, which can be used as integrator.

Definition 1.3.1. A random time T is a **stopping time** of $\{\mathcal{F}_t, t \geq 0\}$ if $\forall t \geq 0$, $\{T \leq t\} \in \mathcal{F}_t$, another one is **optional time** if $\forall t \geq 0$, $\{T < t\} \in \mathcal{F}_t$.

Intuitively speaking, stopping time is a r.v. such that its property can be checked at each time (i.e. the information $(\{T \leq t\})$ of it contains within time t ($\mathcal{F}_t$)), and optional time is by time t but not included. However, they don't have too much difference as below proposition suggests:

Proposition 1.3.1. *If T is a stopping time, then also a optional time. Conversely, if the filtration is **right-continuous**, then stopping time is same as optional time.*

Proof. First one is easy to see, we focus on latter equivalence statement:

$\Rightarrow$ The forward direction is simple, as we notice below:

$$\{T < t\} = \bigcup_{n=1}^{\infty} \left\{ T \leq t - \frac{1}{n} \right\}$$

, as each $\{T \leq t - 1/n\} \in \mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t$, then $\{T < t\} \in \mathcal{F}_t$.

$\Leftarrow$ For backward direction, it is non-trival, $\forall m \geq 1$:

$$\{T \leq t\} = \bigcap_{n=m}^{\infty} \left\{ T < t + \frac{1}{n} \right\}$$

, as each $\{T < t + 1/n\} \in \mathcal{F}_{t+1/n} \subseteq \mathcal{F}_{t+1/m}$, then $\{T \leq t\} \in \mathcal{F}_{t+1/m}$.

Lastly, we finished the proof by the arbitrariness of m , then $\{T \leq t\} \in \mathcal{F}_{t+} = \mathcal{F}_t$. $\square$

Next, after we defined the stopping time, let us see some properties:

Proposition 1.3.2. *If T and S are stopping times, then:*

$$T \wedge S, T \vee S, T + S$$

are all stopping times.

Proof. The first two are easy once we separate the event properly:

(1) For $T \wedge S$, we could consider as below:

$$\{T \wedge S \leq t\} = \{T \leq t\} \cup \{S \leq t\} \in \mathcal{F}_t$$

, as T, S are stopping times, then $\{T \leq t\}, \{S \leq t\} \in \mathcal{F}_t$.

(2) For $T \vee S$, we do the similar separation, $\{T \vee S \leq t\} = \{T \leq t\} \cap \{S \leq t\}$.

(3) $T + S$ is tricky, where we should transfer using closure under complement:

$$\{T + S \leq t\} \in \mathcal{F}_t \iff \{T + S > t\} \in \mathcal{F}_t$$

After we do this transformation, we could separate this by:

$$\{T + S > t\} = \{T = 0, S > t\} \cup \{S = 0, T > t\} \cup \{T \geq t, S > 0\} \cup \{0 < T < t, T + S > t\}$$

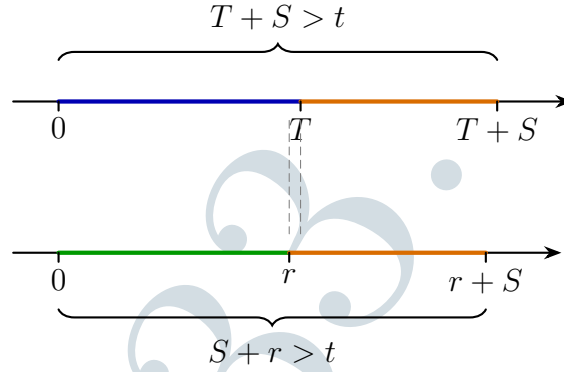
For the first one, it is easy to see:

$$\{T = 0, S > t\} = \{T = 0\} \cap \{S > t\} = \{T = 0\} \cap \{S \leq t\}^c \in \mathcal{F}_t$$

Then using similar method, we could easily check middle two, what remains is:

Lemma.

$$\{0 < T < t, T + S > t\} = \bigcup_{\substack{0 < r < t \\ r \in \mathbb{Q}^+}} \{r < T < t, S > t - r\}$$



Proof. of Lemma:

$\Leftarrow$ Backward direction is easy, as $S > t - r > t - T$, hence $S + T > t$.

$\Rightarrow$ Forward direction relies on the dense of rational number, where since $S + T > t$, then by dense of rational numbers, $\exists r \in (0, T)$, s.t. $S + r > t$.

Last one can be visualised as above by moving r really close to T . □

With the lemma above, we could separate event as before, then complete the proof. □

Lastly, if we have a set of stopping times, then the sup of it is also a stopping time:

Proposition 1.3.3. *If $\{T_n, n \geq 1\}$ are s.t., then $\sup_{n \geq 1} T_n$ is a s.t.*

Proof. This proof is simple and straightforward, $\forall t \geq 0$:

$$\left\{ \sup_{n \geq 1} T_n \leq t \right\} = \bigcap_{n=1}^{\infty} \{T_n \leq t\} \in \mathcal{F}_t$$

Then we complete the proof. □

Since we already have s.t. as a kind of time, it is reasonable to ask its σ algebra:

Definition 1.3.2. The stopping σ -algebra is:

$$\mathcal{F}_T = \left\{ A \in \mathcal{F} : \forall t \geq 0, A \cap \{T \leq t\} \in \mathcal{F}_t \right\}$$

Intuitively speaking, above definition 1.3.2 is to say for every event in stopping σ algebra, given the time t we are at, we could know whether it happens or not.

Proposition 1.3.4. If $T \equiv t_0$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. Firstly, we observe that $\forall t \geq 0$:

$$\{T \leq t\} = \begin{cases} \emptyset, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

$\Leftarrow$ Then we firstly check the backward direction, $\forall A \in \mathcal{F}_{t_0}, \forall t \geq 0$:

(1) If $t \geq t_0$, then:

$$A \cap \{T \leq t\} = A \cap \Omega = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$$

(2) If $t < t_0$, then:

$$A \cap \{T \leq t\} = A \cap \emptyset = \emptyset \in \mathcal{F}_t$$

Therefore, $A \in \mathcal{F}_T$ as definition states.

$\Rightarrow$ Forward direction is similar, as $\forall A \in \mathcal{F}_T$, and $\forall t \geq 0$, then:

$$A \cap \{T \leq t\} \in \mathcal{F}_t \implies A = A \cap \Omega = A \cap \{t_0 \leq t_0\} = A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0}$$

Therefore, we completed the proof. □

Besides, we will also be interested into the relationship of a set of s.t.s,

Proposition 1.3.5. $\forall$ stopping times T and S , $\forall A \in \mathcal{F}_S$, we have:

$$A \cap \{S \leq T\} \in \mathcal{F}_T$$

In particular, if $S \leq T$ for all $\omega \in \Omega$, then:

$$\mathcal{F}_S \subseteq \mathcal{F}_T$$

Proof. The key idea is to use the condition that $A \in \mathcal{F}_S$, so that:

$$A \cap \{S \leq t\} \in \mathcal{F}_t$$

And in order to show, $A \cap \{S \leq T\} \in \mathcal{F}_T$, we need $\forall t > 0$,

$$A \cap \{S \leq T\} \cap \{T \leq t\} \in \mathcal{F}_t$$

Notice here, since on $\{S \leq T\} \cap \{T \leq t\}$:

$$(a) \quad \{S \leq T\} \cap \{T \leq t\} = \{S \leq T \leq t\} \implies S \leq t.$$

$$(b) \{S \leq T\} = \{S \wedge t \leq T \wedge t\}.$$

Then, it suffice to show, $\forall t > 0$:

$$(A \cap \{S \leq t\}) \cap \{T \leq t\} \cap \{S \wedge t \leq T \wedge t\} \in \mathcal{F}_t$$

The first lives in $\mathcal{F}_t$ followed from the fact that $A \in \mathcal{F}_S$, and second comes from the fact that T is a s.t., it remains to show, $\{S \wedge t \leq T \wedge t\} \in \mathcal{F}_t$:

Lemma. *Let T and S be two s.t.s, then $\forall t \geq 0$:*

$$\{S \wedge t \leq T \wedge t\} \in \mathcal{F}_t$$

Proof. of Lemma

This follows the similar structure of proposition 1.3.2's lemma to decompose:

$$\{S \wedge t \leq T \wedge t\} = \bigcup_{r \in \mathbb{Q}^+ \cap [0, t]} \{S \wedge t \leq r\} \cap \{T \wedge t \geq r\}$$

Using the closure of complement of σ -algebra to show its conclusion. $\square$

Therefore, we completed the proof for general situation, and in particular, since $S \leq T$ for all $\omega \in \Omega$, then $\{S \leq T\} = \Omega$, so $\forall A \cap \Omega = A \in \mathcal{F}_S, A \in \mathcal{F}_T \implies \mathcal{F}_S \subseteq \mathcal{F}_T$. $\square$

To reiterate, the reason why we introduce the stopped time $(T \wedge t)$ is to relate the original s.t. with t , so that we could manipulate on it with relationship pf $\mathcal{F}_t$.

Proposition 1.3.6. *Let S and T be two s.t.s, then:*

$$\mathcal{F}_{S \wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$$

, and also:

$$\{T < S\}, \{S < T\}, \{T \leq S\}, \{S \leq T\}, \{S = T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

Proof. The intuition is clear, for σ -algebra for s.t.s which is the smallest of two, is certainly the intersection of two σ -algebra of two s.t.s Let's prove them in order:

(a) Firstly, consider the inclusion:

$\implies$ From proposition 1.3.5, we have:

$$\begin{cases} S \wedge T \leq S \\ S \wedge T \leq T \end{cases} \implies \begin{cases} \mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_S \\ \mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_T \end{cases} \implies \mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_S \cap \mathcal{F}_T$$

$\Leftarrow$ For backward direction, $\forall A \in \mathcal{F}_S \cap \mathcal{F}_T, \forall t > 0$, we need to show:

$$A \cap \{S \wedge T \leq t\} = A \cap (\{S \leq t\} \cup \{T \leq t\}) \in \mathcal{F}_t$$

, above is easy to show by distribution rule and the fact that $A \in \mathcal{F}_S, \mathcal{F}_T$:

$$A \cap (\{S \leq t\} \cup \{T \leq t\}) = (A \cap \{S \leq t\}) \cup (A \cap \{T \leq t\}) \in \mathcal{F}_t$$

(b) Similarly, from proposition 1.3.5, we have, $\forall A \in \mathcal{F}_S$, we have:

$$A \cap \{S \leq T\} \in \mathcal{F}_T \xrightarrow{\text{Choose } A=\Omega} \{S \leq T\} \in \mathcal{F}_T \implies \{S \leq T\}^c = \{S > T\} \in \mathcal{F}_T$$

It remains to show, $\{S < T\} \in \mathcal{F}_T$, since others can be shown by interchanging the place of them, which we will see it later, here notice that, in order to introduce the relations of S and T , we construct a stopped s.t. by defining $R := S \wedge T$, then:

$$\{S < T\} = \{R < T\}, R \in \mathcal{F}_R \subseteq \mathcal{F}_T \implies \{S < T\} = \{R < T\} \in \mathcal{F}_T$$

Lastly, we use the power of interchanging the orders,

$$\{T \leq S\}, \{T > S\}, \{S > T\} \in \mathcal{F}_S$$

, so that other parts can all be expressed by their complements:

(i) For $\{S \leq T\} \in \mathcal{F}_T$, we use $\{S > T\} \in \mathcal{F}_S$:

$$\{S \leq T\} = \{S > T\}^c \in \mathcal{F}_S \implies \{S \leq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

(ii) For $\{S < T\} \in \mathcal{F}_T$, we use $\{T \leq S\} \in \mathcal{F}_S$:

$$\{S < T\} = \{T \geq S\}^c \in \mathcal{F}_S \implies \{S < T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

(iii) Lastly, for $\{T = S\}$, we use (i) and (ii) $\{S \geq T\} = \{S < T\}^c \in \mathcal{F}_S \cap \mathcal{F}_T$:

$$\{S = T\} = \{S \leq T\} \cap \{S \geq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

Then we completed the proof. □

Remark 1.3.1. A short summary of the properties for s.t.s:

- (1) For s.t. T , $T \in \mathcal{F}_T$, i.e. $\forall t > 0$, $\{T < t\} \in \mathcal{F}_T$.
- (2) For two s.t.s, S and T , if $S \leq T$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$.
- (3) For s.t. T , and $\forall t > 0$, then $T \wedge t \in \mathcal{F}_{T \wedge t} \subseteq \mathcal{F}_t$.
- (4) For two s.t.s S and T , then $\mathcal{F}_{S \wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$.

Lastly, we need to justify its validity of changing time with s.t.:

Theorem 1.3.7. *Let $X = \{X_t, t \geq 0\}$ be a progressively measurable process, and T be a s.t., then X_T defined on $\{T < +\infty\}$ is $\mathcal{F}_T$ -measurable. And the stopped process $\{X_{T \wedge t}, t \geq 0\}$ is also progressively measurable.*

Proof. The two statements is equivalent, since if we want to show, X_T defined on $\{T < +\infty\}$ is $\mathcal{F}_T$ -measurable, then it is equal to show, $\forall t \geq 0$, and $\forall B \in \mathcal{B}(\mathbb{R}^d)$:

$$\{X_T \in B\} \cap \{T \leq t\} \in \mathcal{F}_t \iff \{X_{T \wedge t} \in B\} \cap \{T \leq t\} \in \mathcal{F}_t$$

Then it remains to show, $X_{T \wedge t} \in \mathcal{F}_t$, by observing:

$$(s, \omega) \mapsto (T(\omega) \wedge s, \omega)$$

is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable, and since the process is progressively measurable:

$$(s, \omega) \mapsto X_s(\omega)$$

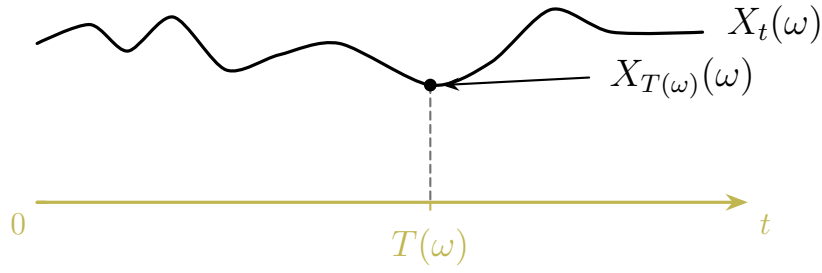
is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable, then by composition rule:

$$(s, \omega) \mapsto (T(\omega) \wedge s, \omega) \mapsto (X_{T(\omega) \wedge s}, \omega)$$

is also $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable, then we completed the proof. $\square$

Remark 1.3.2. To remind, we interpret the stochastic process replaced time with s.t. (or random time), by following composite functions and figure:

$$\omega \mapsto (T(\omega), \omega) \mapsto X_{T(\omega)}(\omega)$$



Chapter 2

Continuous-time Martingale

Martingale sits at an important role of this course, and also stochastic process, for this session, we mainly focus on how to transfer the classical discrete time martingale properties (where most of reader should learn from classical stochastic process like Rick. Durrett's book) to continuous time version, and most powerful tool for proving convergence.

Definition 2.0.1. A martingale is an adapted stochastic process $X = \{X_t, t \geq 0\}$ such that:

- (a) $\forall t \geq 0, E|X_t| < \infty$.
- (b) $\forall t \geq s \geq 0, E(X_t | \mathcal{F}_s) = X_s$

Remark 2.0.1. A simple extension to sub/super-martingale:

- (1) If definition 2.0.1's (b) change to:

$$E(X_t | \mathcal{F}_s) \geq X_s$$

, then it is called submartingale.

- (2) If definition 2.0.1's (b) change to:

$$E(X_t | \mathcal{F}_s) \leq X_s$$

, then it is called supermartingale.

2.1 Martingale Theory in Nutshell

Following we presents some important theorem some of them without proof:

Theorem 2.1.1. Let $\varphi : \mathbb{R} \mapsto \mathbb{R}$ be a convex (concave up) function, if $X = \{X_t, t \geq 0\}$ is a martingale, then $\varphi(X) = \{\varphi(X_t), t \geq 0\}$ is a submartingale.

Proof. By Jensen's inequality:

$$|EX_t| \leq E|X_t| \implies |E(X_t | \mathcal{F}_s)| \leq E(|X_t| | \mathcal{F}_s)$$

Hence, by assuming integrability and adaptivity, it remains:

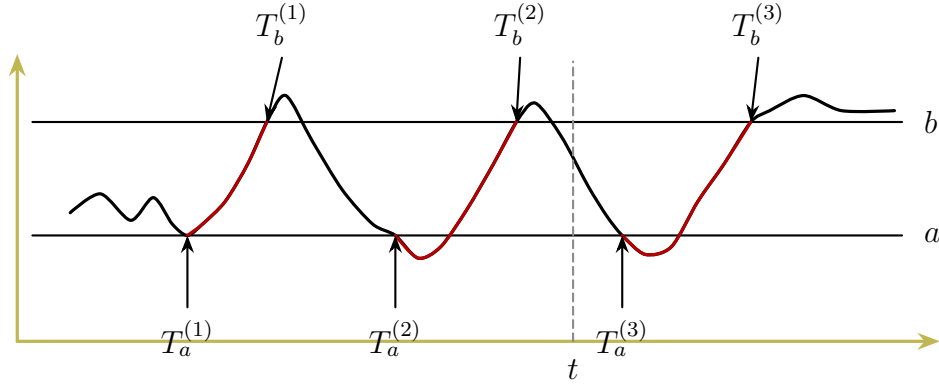
$$\mathbb{E}(\varphi(X_t) \mid \mathcal{F}_s) \geq \varphi(\mathbb{E}(X_t \mid \mathcal{F}_s)) = \varphi(X_s)$$

Then we completed the proof. $\square$

Remark 2.1.1. Some common convex function we used are, where $p \geq 1$.

$$|X_t|, \quad X_t^2, \quad (X_t)^+ = \max(X_t, 0), \quad |X_t|^p$$

Next important theorem is **continuous-time Doob's upcrossing inequality:**



As the figure shows, we could denote the i th upcrossings (or downcrossings) time to be the first time, after the $i-1$ th downcrossings (or upcrossings), more formally:

$$\begin{aligned} T_b^0 &= -\infty \\ T_a^{(1)} &= \inf \left\{ t \geq (T_b^{(0)})^+ : X_t \leq a \right\} \\ T_b^{(1)} &= \inf \left\{ t \geq T_a^{(1)} : X_t \geq b \right\} \end{aligned}$$

After the initial setup, for $\forall \ell \geq 2$:

$$\begin{aligned} T_a^{(\ell)} &= \inf \left\{ t \geq T_b^{(\ell-1)} : X_t \leq a \right\} \\ T_b^{(\ell)} &= \inf \left\{ t \geq T_a^{(\ell-1)} : X_t \geq b \right\} \end{aligned}$$

Then we focus on the # of upcrossing in an interval $[0, t]$ for given two bounds ($a < b$):

$$U_{ab}^X[0, t] = \max \left\{ k \geq 0 : T_b^{(k)} \leq t \right\}$$

Now, we could present our most important theorem about submartingale:

Theorem 2.1.2. (Continuous-time Doob's Upcrossings Inequality)

Let $X = \{X_t, t \geq 0\}$ be a right-continuous submartingale, then for all $T > 0$:

$$\mathbb{E}U_{ab}^X[0, T] \leq \frac{\mathbb{E}(X_T - a)^+}{b - a}$$

Proof. The idea is to how to apply the discrete-time Doob's upcrossing inequality, where we assume to be known, since we presume all readers have studied basic stochastic process. Then we could try to discretise the continuous time, for each $n \geq 1$, we let:

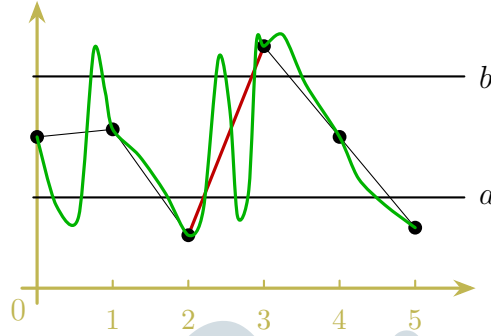
$$D_n = \mathbb{Z}/2^n = \left\{ 0, \frac{\pm 1}{2^n}, \frac{\pm 2}{2^n}, \frac{\pm 3}{2^n}, \dots \right\}$$

Consider the upcrossings of X on $[0, T] \cap D_n$, where:

$$\{X_{j/2^n}, j \geq 0\}$$

, is discrete-time submartingale, then as showed as the figure below, we have:

$$U_{ab}^X([0, T] \cap D_n) \leq U_{ab}^X[0, T]$$



Clearly if we discretise it finer, it will be nondecreasing, then:

$$U_{ab}^X([0, T] \cap D_n) \uparrow U_{ab}^X[0, T]$$

Therefore, by using discrete-time upcrossing inequality, i.e.

$$U_{ab}^X([0, T] \cap D_n) \leq \frac{1}{b-a} \mathbb{E}(X_{T_n} - a)^+ \leq \frac{1}{b-a} \mathbb{E}(X_T - a)^+$$

, where $T_n := \frac{\lfloor T \cdot 2^n \rfloor}{2^n}$, let $n \rightarrow \infty$ by limiting theory, we finished the proof. $\square$

Just as we said, it is powerful tool to show the convergence for submartingale:

Proposition 2.1.3. *If $X = \{X_t, t \geq 0\}$ is a right-continuous submartingale, and:*

$$\sup_{t \geq 0} \mathbb{E}X_t^+ < \infty$$

, then $\exists X \in L^1$ such that, as $t \rightarrow \infty$:

$$X_t \xrightarrow{a.s.} X$$

Proof. Here, we ignore the procedure of showing existence, just focus on convergence:

$$\begin{aligned} \mathbb{E}U_{ab}^X[0, \infty) &= \lim_{n \rightarrow \infty} \mathbb{E}U_{ab}^X[0, n] \\ &\leq \limsup_{n \rightarrow \infty} \frac{1}{b-a} \mathbb{E}(X_n - a)^+ \\ &\leq \limsup_{n \rightarrow \infty} \frac{1}{b-a} (\mathbb{E}X_n^+ + |a|) < \infty \end{aligned}$$

, above is provided by theorem 2.1.2 and condition, hence:

$$\forall a < b, U_{ab}^X[0, \infty) < \infty, \text{ a.s.} \implies \text{w.p.1 } \forall a < b \in \mathbb{Q}, U_{ab}^X[0, \infty) < \infty$$

, this is given by the fact that countable additivity of our measure, therefore:

$$\liminf_{t \rightarrow \infty} X_t < a \leq b < \limsup_{t \rightarrow \infty} X_t$$

does not occur, which is equivalent to say:

$$\liminf_{t \rightarrow \infty} X_t = \limsup_{t \rightarrow \infty} X_t$$

, then we completed the proof. $\square$

To connect relations of different convergence modes, we need one more definition:

Definition 2.1.1. A family of r.v.s $\{X_t, t \geq 0\}$ is uniformly integrable (u.i.) if:

$$\lim_{M \rightarrow \infty} \sup_{t \geq 0} \mathbb{E}|X_t| \mathbf{1}_{\{|X_t| > M\}} = 0$$

Remark 2.1.2. Using Markov Inequality, if $\sup_{t \geq 0} \mathbb{E}|X_t|^p < \infty$, for some $p > 1$, then X is u.i.

With uniformly integrable, we could present the following equivalence:

Theorem 2.1.4. If $X_n \xrightarrow{p.} X$, then the following are equivalent:

- (a) $X_n \xrightarrow{L^1} X$.
- (b) $\mathbb{E}|X_n| \longrightarrow \mathbb{E}|X|$.
- (c) $\{X_n, n \geq 0\}$ is u.i.

Proof. We omit the proof since it is not our focus. $\square$

Last important property is last elements, before that we need:

Proposition 2.1.5. Let $Z \in L^1$, then following is u.i.

$$\{\mathbb{E}(Z | \mathcal{G}) : \mathcal{G} \text{ is a sub } \sigma\text{-field of } \mathcal{F}\}$$

Proof. Proof omitted.

Remark 2.1.3. In particular, if we pick $\mathcal{G}$ to be filtration, i.e.:

$$\{\mathbb{E}(Z | \mathcal{F}_t), t \geq 0\}$$

, is u.i. and also a martingale. $\square$

With the help of u.i., we could show following relations, (last element), providing convenience in proving Optional Sampling Theorem (OST).

Proposition 2.1.6. *A right-continuous martingale $\{X_t, t \geq 0\}$ is u.i. iff:*

$$\exists X_\infty \in L^1, \text{ s.t. } X_t = \mathbb{E}(X_\infty | \mathcal{F}_t)$$

Proof. Firstly, we start with the inclusion:

$\implies$ Firstly, by u.i., we have $\sup_{t \geq 0} \mathbb{E}|X_t| < \infty$, then through proposition 2.1.3:

$$\exists X_\infty, \text{ s.t. } X_t \xrightarrow{a.s.} X_\infty$$

, more important, we know it is L^1 since:

$$\mathbb{E}|X_\infty| = \mathbb{E} \lim_{t \rightarrow \infty} |X_t| \leq \mathbb{E} \liminf_{t \rightarrow \infty} \mathbb{E}|X_t| < \infty$$

, where last inequality comes from **Fatou's Lemma**, next to prove:

$$\forall t \geq 0, X_t = \mathbb{E}(X_\infty | \mathcal{F}_t)$$

is equivalent, by the definition of conditional expectation, to show $\forall A \in \mathcal{F}_t$:

$$\mathbb{E}X_t \mathbf{1}_A = \mathbb{E}X_\infty \mathbf{1}_A$$

Common as usual, we approximate it by sending limit to infinity, for any $s \geq 0$:

$$\mathbb{E}(X_{t+s} | \mathcal{F}_t) = X_t \implies \mathbb{E}X_{t+s} \mathbf{1}_A = \mathbb{E}X_t \mathbf{1}_A$$

For limiting validity, we recall that $u.i. + a.s. \implies L^1$:

$$|\mathbb{E}X_{t+s} \mathbf{1}_A - \mathbb{E}X_\infty \mathbf{1}_A| \leq \mathbb{E}|X_{t+s} - X_\infty| \longrightarrow 0$$

Then by sending $s \rightarrow \infty$ we complete the first part of the proof.

$\Leftarrow$ This direction immediately follows from proposition 2.1.5.

Then we completed the proof by showing both directions. □

After all the preparation, we commence the OST:

Theorem 2.1.7. (Optional Sampling Theorem)

Let $\{X_t, t \geq 0\}$ be a right-continuous martingale, let $S \leq T$ be two s.t.s suppose:

(a) $\{X_t, t \geq 0\}$ is u.i.

(b) $\exists$ constant $K > 0$, s.t. $S, T \leq K$

, either above two satisfied, then we have:

$$\mathbb{E}(X_T | \mathcal{F}_S) = X_S$$

Proof. In order to use proposition 2.1.6, we consider:

(a) Then by proposition 2.1.3, we choose $Z = X_\infty$

(b) We directly pick the "last element", i.e. $Z = X_K$

In either case, we have: $X_t = \mathbb{E}(Z \mid \mathcal{F}_t)$, then it suffices to show:

$$X_T = \mathbb{E}(Z \mid \mathcal{F}_T)$$

Since provided the above satisfied, by symmetry, we could interchanged it to S , so that:

$$X_S = \mathbb{E}(Z \mid \mathcal{F}_S)$$

Hence, by tower rule and proposition 1.3.5, then:

$$\mathbb{E}(X_T \mid \mathcal{F}_S) = \mathbb{E}[\mathbb{E}(Z \mid \mathcal{F}_T) \mid \mathcal{F}_S] = \mathbb{E}(Z \mid \mathcal{F}_S) = X_S$$

Now, we could firstly consider the discrete cases:

(1) If T can be decomposed into discrete cases, i.e. $T \in \{t_1, t_2, \dots\}$, then $\forall A \in \mathcal{F}_T$:

$$\begin{aligned} \mathbb{E}X_T \mathbf{1}_A &= \mathbb{E}Z \mathbf{1}_A = \sum_{n=1}^{\infty} \mathbb{E}X_{t_n} \mathbf{1}_{\{T=t_n\}} \mathbf{1}_A \\ \text{Because } \begin{cases} A \cap \{T=t_n\} \in \mathcal{F}_{t_n} \\ \mathbb{E}(Z \mid \mathcal{F}_{t_n}) = X_{t_n} \end{cases} \\ &= \sum_{n=1}^{\infty} \mathbb{E}Z \mathbf{1}_{A \cap \{T=t_n\}} = \mathbb{E}Z \mathbf{1}_A \end{aligned}$$

(2) For general T , we discretise from right (for the sake of right-continuous), then define:

$$T_k = \frac{\lfloor 2^k \cdot T \rfloor + 1}{2^k}$$

, clearly $T_k \downarrow T$, then $\forall A \in \mathcal{F}_T \subseteq \mathcal{F}_{T_k}$:

$$\mathbb{E}(Z \mid \mathcal{F}_{T_k}) = X_{T_k} \implies \mathbb{E}Z \mathbf{1}_A = \mathbb{E}X_{T_k} \mathbf{1}_A$$

Sending the $k \rightarrow \infty$ to see $\mathbb{E}Z \mathbf{1}_A = \mathbb{E}X_T \mathbf{1}_A$, which is provided by:

$$\begin{cases} X_{T_k} \xrightarrow{a.s.} X_T \text{ by right-continuous} \\ \{X_{T_k}\} \text{ is u.i. by proposition 2.1.5} \end{cases} \implies X_{T_k} \xrightarrow{L^1} X_T$$

Hence, we completed the proof, by the help of dicretisation. □

By closing subsections for martingale, we presents some useful inequality:

Theorem 2.1.8. (Doob's Maximal Inequality)

Let $X = \{X_t, t \geq 0\}$ be a right-continuous submartingale, then $\forall \lambda > 0$:

$$\begin{aligned} \mathbb{P}\left(\sup_{0 \leq s \leq t} X_s > \lambda\right) &\leq \frac{1}{\lambda} \mathbb{E}X_t^+ \\ \mathbb{P}\left(\inf_{0 \leq s \leq t} X_s < -\lambda\right) &\leq \frac{1}{\lambda} (\mathbb{E}X_t^+ - \mathbb{E}X_0) \end{aligned}$$

Proof. Since two of the statements are similar, we only show first one, $\forall \lambda > 0$, define:

$$T = \inf\{t \geq 0 : X_t > \lambda\} \implies \mathbb{P}\left(\sup_{0 \leq s \leq t} X_s > \lambda\right) = \mathbb{P}(T \leq t)$$

, then consider a stopped process, $\{X_{t \wedge T}, t \geq 0\}$, notice that:

$$\begin{aligned} \lambda \mathbb{P}\left(\sup_{0 \leq s \leq t} X_s > \lambda\right) &= \lambda \mathbb{P}(T \leq t) = \lambda \mathbb{E} \mathbf{1}_{\{T \leq t\}} \leq \mathbb{E} X_T \mathbf{1}_{\{T \leq t\}} \\ &= \mathbb{E} X_{t \wedge T} \mathbf{1}_{\{T \leq t\}} = \mathbb{E} X_{t \wedge T}^+ \mathbf{1}_{\{T \leq t\}} \leq \mathbb{E} X_{t \wedge T}^+ \leq \mathbb{E} X_t^+ \end{aligned}$$

Hence, we completed proof, by transferring original question to new stopping time. $\square$

The above theorem is to consider the tail probability, and next one is for moment:

Theorem 2.1.9. Let $X = \{X_t, t \geq 0\}$ be submartingale, then $\forall p > 1$:

$$\mathbb{E}\left(\sup_{0 \leq s \leq t} |X_s|\right)^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}|X_t|^p$$

Proof. Similar to previous inequality, here we need more dedicate tail probability estimation, where we first define $Y = \sup_{0 \leq s \leq t} |X_s|$, then:

$$\begin{aligned} \lambda \mathbb{P}(Y \geq \lambda) &\leq \mathbb{E}|X_{T \wedge t}| \mathbf{1}_{\{T \leq t\}} = \mathbb{E}|X_{T \wedge t}| - \mathbb{E}|X_{T \wedge t}| \mathbf{1}_{\{T > t\}} \\ &\leq \mathbb{E}|X_t| - \mathbb{E}|X_t| \mathbf{1}_{\{T > t\}} = \mathbb{E}|X_t| - \mathbb{E}|X_t| \mathbf{1}_{\{Y < \lambda\}} = \mathbb{E}|X_t| \mathbf{1}_{\{Y \geq \lambda\}} \end{aligned}$$

Then, using basic calculus to expand the moment:

$$\begin{aligned} \mathbb{E} Y^p &= \int_0^\infty p \lambda^{p-1} \mathbb{P}(Y \geq \lambda) d\lambda \leq \int_0^\infty p \lambda^{p-2} \mathbb{E}|X_t| \mathbf{1}_{\{Y \geq \lambda\}} d\lambda \\ &= \mathbb{E}|X_t| \int_0^\infty p \lambda^{p-2} \mathbf{1}_{\{Y \geq \lambda\}} d\lambda = \mathbb{E}|X_t| \int_0^Y p \lambda^{p-2} d\lambda \\ &= \frac{p}{p-1} \mathbb{E}|X_t| Y^{p-1} \leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{1/p} (\mathbb{E} Y^p)^{(p-1)/p} \end{aligned}$$

The last inequality is due to holder's inequality, then we arrange terms to get:

$$(\mathbb{E} Y^p)^{1/p} \leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{1/p}$$

For unbounded case, we stopped the process by $Y \wedge k$, and then sending k to infinity, hence we could easily completed the proof. $\square$

2.2 The Doob-Meyer Decomposition

Recall the discretised version of Doob Decomposition:

$$\underbrace{X_n}_{\text{Submartingale}} = \underbrace{M_n}_{\text{Martingale}} + \underbrace{A_n}_{\text{Predictable Increasing}}$$

The intuition is simple, from above construction:

$$\mathbb{E}(M_{n+1} | \mathcal{F}_n) = M_n \implies \mathbb{E}(X_{n+1} - A_{n+1} | \mathcal{F}_n) = X_n - A_n$$

Using the predictability of $\{A_n, n \in \mathbb{N}\}$, i.e. $A_n \in \mathcal{F}_{n-1}$:

$$\mathbb{E}(X_{n+1} | \mathcal{F}_n) - A_{n+1} = X_n + A_n \implies A_{n+1} - A_n = \mathbb{E}(X_{n+1} | \mathcal{F}_n) - X_n \geq 0$$

Since $\{X_n, n \in \mathbb{N}\}$ is submartingale, then in general, it is increasing, hence A_n is to capture the additional increasing parts, and martingale can be seen as "white noise", by iterating, and by default to set $A_0 = 0$, we have:

$$A_n = \sum_{k=0}^{n-1} [\mathbb{E}(X_{k+1} | \mathcal{F}_k) - X_k]$$

Hence, this subsection is to delve into such decomposition with continuous setting:

$$\underbrace{X_t}_{\text{Submartingale}} = \underbrace{M_t}_{\text{Martingale}} + \underbrace{A_t}_{\substack{\text{Predictable} \\ \text{Increasing}}}$$

, where assuming $\{X_t, t \geq 0\}$ is **right-continuous** submartingale. First is to consider increasing process in continuous setting:

Definition 2.2.1. An adapted process $A = \{A_t, t \geq 0\}$ is called **increasing** if:

- (a) $A_0 = 0$ a.s.
- (b) $t \mapsto A_t$ is non-decreasing and right-continuous for ω -a.e.
- (c) $\mathbb{E}A_t < \infty, \forall t \geq 0$.

Remark 2.2.1. Note that for integrability condition is a little different from (c):

$$\mathbb{E}A_\infty = \mathbb{E} \lim_{t \rightarrow \infty} A_t < \infty$$

Besides increasing, other than predictable, we introduce **natural**:

Definition 2.2.2. An increasing process $A = \{A_t, t \geq 0\}$ is called **natural** if $\forall$ bounded, right continuous martingale, $M = \{M_t, t \geq 0\}$, then $\forall t \geq 0$:

$$\mathbb{E} \int_0^t M_s dA_s = \mathbb{E} \int_0^t M_{s-} dA_s$$

Remark 2.2.2. It is easy to see, if $s \mapsto A_s$ is a.s. continuous, then A is natural, since $M_s - M_{s-} \neq 0$ only for jump points, for a.s. continuity, there are only finite of them.

It seems wired to propose this definition, by thinking from discretised version:

$$\begin{aligned} A_n \in \mathcal{F}_{n-1} &\iff \text{predictable} \\ A_t \in \mathcal{F}_{t-} &\iff \text{"predictable"} \end{aligned}$$

Hence, we first check its discrete-time version:

Definition 2.2.3. An adapted and increasing sequence, $A = \{A_n, n \in \mathbb{N}\}$ is called **natural** if $\forall$ bounded martingale, $M = \{M_n, n \in \mathbb{N}\}$, then, $\forall n \in \mathbb{N}$:

$$\mathbb{E}(M_n A_n) = \mathbb{E} \left[\sum_{k=1}^n M_{k-1} (A_k - A_{k-1}) \right]$$

Then we are eager to see the connection between **natural** and **predictable**:

Proposition 2.2.1. *For an increasing sequence, A is predictable iff it is natural.*

Proof. The proof is direct, mainly by dedicated algebra:

$\Rightarrow$ To prove, it is natural, it is same to show for every martingale M :

$$\mathbb{E}(M_n A_n) = \mathbb{E} \left[\sum_{k=1}^n M_{k-1} (A_k - A_{k-1}) \right] = \sum_{k=1}^n [\mathbb{E} M_{k-1} A_k - \mathbb{E} M_{k-1} A_{k-1}]$$

Now we could rearrange things to put minus term to left hand side:

$$\sum_{k=1}^n \mathbb{E} M_{k-1} A_{k-1} + \mathbb{E} M_n A_n = \sum_{k=1}^n \mathbb{E} M_{k-1} A_k$$

Since $A_0 = 0$ and $M_n A_n$ serves as final term of first summation, then:

$$\sum_{k=1}^n \mathbb{E} M_k A_k = \sum_{k=1}^n \mathbb{E} M_{k-1} A_{k-1} \Rightarrow \sum_{k=1}^n \mathbb{E} (M_k - M_{k-1}) A_k = 0$$

From this equivalent statement, we plug in the predictability, if $A_n \in \mathcal{F}_{n-1}$:

$$\mathbb{E} A_n (M_n - M_{n-1}) = \mathbb{E} [A_n \mathbb{E} (M_n - M_{n-1} | \mathcal{F}_{n-1})] = 0$$

Then each term of the summation is zero, we completed this direction.

$\Leftarrow$ Assume A is natural, and $\forall$ bounded martingale M , from above, then:

$$\mathbb{E} A_n (M_n - M_{n-1}) = 0, \forall n \geq 1$$

Then goal is to show predictability, i.e. $A_n \in \mathcal{F}_{n-1}$, or $A_n = \mathbb{E}(A_n | \mathcal{F}_{n-1})$ a.s., now since A is natural, then for every bounded M , we notice that:

$$\begin{aligned} \mathbb{E} M_n [A_n - \mathbb{E}(A_n | \mathcal{F}_{n-1})] &= \mathbb{E} A_n M_{n-1} - \mathbb{E} [M_n \mathbb{E}(A_n | \mathcal{F}_{n-1})] \\ &= \mathbb{E} M_{n-1} [A_n - \mathbb{E}(A_n | \mathcal{F}_{n-1})] + \mathbb{E} [\mathbb{E}(A_n | \mathcal{F}_{n-1}) (M_n - M_{n-1})] \\ &= \underbrace{\mathbb{E} M_{n-1} A_n - \mathbb{E} \mathbb{E}(A_n M_{n-1} | \mathcal{F}_{n-1})}_{=0} + \mathbb{E} \left[\underbrace{\mathbb{E}(A_n | \mathcal{F}_{n-1}) \mathbb{E}(M_n - M_{n-1} | \mathcal{F}_{n-1})}_{=0} \right] = 0 \end{aligned}$$

Hence, by the arbitrariness, for each fixed n , we choose:

$$\xi_n = \text{sgn}(A_n - \mathbb{E}(A_n | \mathcal{F}_{n-1}))$$

, and define our martingale as:

$$M_k = \begin{cases} \mathbb{E}(\xi_n | \mathcal{F}_k) & 0 \leq k \leq n-1 \\ \xi_n & k \geq n \end{cases}$$

, which is bounded, and martingale, from this well-designed martingale, we know:

$$0 = \mathbb{E}M_n[A_n - \mathbb{E}(A_n | \mathcal{F}_{n-1})] = \mathbb{E}|A_n - \mathbb{E}(A_n | \mathcal{F}_{n-1})|$$

, above gives us:

$$A_n = \mathbb{E}(A_n | \mathcal{F}_{n-1}) \text{ a.s.} \implies A_n \in \mathcal{F}_{n-1}$$

Hence, we completed the proof by showing both directions. □

Then next thing is to bridge the discrete one with continuous one:

$$\mathbb{E}(M_n A_n) = \mathbb{E} \left[\sum_{k=1}^n M_{k-1} (A_k - A_{k-1}) \right] \iff \mathbb{E} \int_0^t M_s dA_s = \mathbb{E} \int_0^t M_{s-} dA_s$$

Hence, use the traditional discretisation method, by partition, $\Pi = \{t_0, t_1, \dots, t_n\} \subseteq [0, t]$ with $0 = t_0 < t_1 < t_2 < \dots < t_n = t$ and by setting $\|\Pi\| = \max_{1 \leq k \leq n} |t_k - t_{k-1}|$:

$$\int_0^t M_s dA_s = \lim_{\|\Pi\| \rightarrow 0} \int_0^t M_s^\Pi dA_s$$

, where by right continuous, we chose our discrete martingale to be:

$$M_s^\Pi = \sum_{k=1}^n M_{t_k} \mathbf{1}_{(t_{k-1}, t_k]}(s)$$

Then, by this setup, for each Π , we could see:

$$\begin{aligned} \mathbb{E} \int_0^t M_s^\Pi dA_s &= \mathbb{E} \left[\sum_{k=1}^n M_{t_k} (A_{t_k} - A_{t_{k-1}}) \right] = \mathbb{E} M_{t_n} A_{t_n} + \sum_{k=1}^{n-1} \mathbb{E} M_{t_k} A_{t_k} - \sum_{k=2}^n \mathbb{E} M_{t_k} A_{t_{k-1}} \\ &= \mathbb{E} M_t A_t + \sum_{k=1}^{n-1} \mathbb{E} M_{t_k} A_{t_k} - \sum_{k=1}^{n-1} \mathbb{E} M_{t_{k+1}} A_{t_k} \\ &= \mathbb{E} M_t A_t - \sum_{k=1}^{n-1} \mathbb{E} \left[A_{t_k} \underbrace{\mathbb{E}(M_{t_{k+1}} - M_{t_k} | \mathcal{F}_{t_k})}_{=0} \right] = \mathbb{E} M_t A_t \end{aligned}$$

Hence, we see the connection between discrete version and continuous one, next:

Definition 2.2.4. Let $S = \{ \text{stopping times } T \text{ with } T < \infty \text{ a.s.} \}$, we say a right-continuous $X_t, t \geq 0$ is of **class D** if $\{X_T, T \in S\}$ is u.i., furthermore, to fix $a > 0$, let $S_a = \{ \text{stopping times } T \text{ with } T \leq a \text{ a.s.} \}$, then $\{X_t, t \geq 0\}$ is of **class DL** if $\{X_T, T \in S_a\}$ is u.i. for every $a > 0$.

Then Doobs and Meyer found membership of class DL is sufficient for decomposition:

Theorem 2.2.2. (Doob-Meyer Decomposition)

If $\{X_t, t \geq 0\}$ is of **class DL**, and a submartingale, then it admits, $\forall t \geq 0$:

$$X_t = M_t + A_t$$

, with $\{M_t, t \geq 0\}$ is martingale, $\{A_t, t \geq 0\}$ is increasing process. In particular, if we restrict $\{A_t, t \geq 0\}$ to be natural, then such decomposition is unique.

Proof. Let us only consider the special cases, where we admit A is natural:

(1) For uniqueness, we assume it admits two different decomposition:

$$X_t = \begin{cases} M_t + A_t \\ M'_t + A'_t \end{cases} \implies B_t := M_t - M'_t = A'_t - A_t$$

, where $\{B_t, t \geq 0\}$ is martingale and of bounded variation, also natural, from the definition and exercise before, $\forall$ bounded continuous martingale $\{\xi_t, t \geq 0\}$:

$$\mathbb{E}\xi_t(A_t - A'_t) = \mathbb{E} \int_0^t \xi_{s-} (dA_s - dA'_s) = \mathbb{E} \int_0^t \xi_{t-} dB_s = \lim_{n \rightarrow \infty} \mathbb{E} \sum_{j=1}^{m_n} \xi_{t_{j-1}^{(n)}} (B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}})$$

, where $\Pi_n = \{t_0^{(n)}, t_1^{(n)}, \dots, t_{m_n}^{(n)}\} \subseteq [0, t]$, for each term:

$$\mathbb{E}\xi_{t_{j-1}^{(n)}} (B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}) = \mathbb{E} \left[\xi_{t_{j-1}^{(n)}} \underbrace{\mathbb{E} (B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}} \mid \mathcal{F}_{t_{j-1}^{(n)}})}_{=0} \right] = 0$$

, then by similar trick, we obtained:

$$\mathbb{E} |A_t - A'_t| = 0 \implies A_t = A'_t \text{ a.s. } \forall t > 0 \implies A_t = A'_t \text{ a.s. } \forall t \in \mathbb{Q}^+$$

Lastly, by right-continuity, we achieved $A_t = A'_t$ a.s. $\forall t \geq 0$.

(2) For existence, fix $a > 0$, it suffices to construct the decomposition on $[0, a]$, consider:

$$Y_t := X_t - \mathbb{E}(X_a \mid \mathcal{F}_t) \leq 0$$

Then $\{Y_t, 0 \leq t \leq a\}$ is right-continuous submartingale. Consider dyadic partition:

$$\Pi_n = \left\{ t_0^{(n)}, t_1^{(n)}, \dots, t_{2^n}^{(n)} \right\}, \quad t_j^{(n)} = \frac{j}{2^n} a, \quad 0 \leq j \leq 2^n$$

For each $n \geq 1$, the discrete-time submartingale admits the Doob decomposition

$$Y_{t_j^{(n)}} = M_{t_j^{(n)}}^{(n)} + A_{t_j^{(n)}}^{(n)}, \quad 0 \leq j \leq 2^n$$

where $A_0^{(n)} = 0$, and for $1 \leq j \leq 2^n$,

$$A_{t_j^{(n)}}^{(n)} = \sum_{k=0}^{j-1} \mathbb{E} \left(Y_{t_{k+1}^{(n)}} - Y_{t_k^{(n)}} \mid \mathcal{F}_{t_k^{(n)}} \right)$$

In particular, $\{A_{t_j^{(n)}}^{(n)}\}_{0 \leq j \leq 2^n}$ is predictable and increasing, notice:

$$Y_a = X_a - \mathbb{E}(X_a \mid \mathcal{F}_a) = 0 \implies 0 = Y_a = M_a^{(n)} + A_a^{(n)}$$

Hence, we may rewrite this into:

$$M_{t_j^{(n)}}^{(n)} = \mathbb{E}(M_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}}) = -\mathbb{E}(A_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}}) \implies Y_{t_j^{(n)}} = A_{t_j^{(n)}}^{(n)} - \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}})$$

We now claim that the family $\{A_a^{(n)}\}_{n \geq 1}$ is u.i. (you may verify). Hence, by the Dunford-Pettis compactness criterion, there exist an integrable random variable A_a and a subsequence $\{A_a^{(n_k)}\}_{k \geq 1}$ such that $A_a^{(n_k)} \xrightarrow{L^1} A_a$, i.e.

$$\lim_{k \rightarrow \infty} \mathbb{E}(\xi A_a^{(n_k)}) = \mathbb{E}(\xi A_a)$$

for every bounded random variable ξ . To simplify notation, we relabel this subsequence again by $\{A_a^{(n)}\}$, now define

$$A_t := Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t), \quad 0 \leq t \leq a$$

where we choose its right-continuous modification. Likewise, for each n , define

$$A_t^{(n)} := Y_t + \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_t), \quad 0 \leq t \leq a$$

again with the right-continuous version. Then, for every $t \in \Pi_n$, this agrees with the discrete compensator constructed above, let

$$\mathbb{T} := \bigcup_{n=1}^{\infty} \Pi_n$$

By the weak L^1 stability of conditional expectation, for every $t \in \mathbb{T}$ and every bounded random variable ξ , we have

$$\lim_{n \rightarrow \infty} \mathbb{E}(\xi A_t^{(n)}) = \mathbb{E}(\xi A_t)$$

Now take $s, t \in \mathbb{T}$ with $0 \leq s < t \leq a$, and let ξ be bounded and nonnegative. For all large n , both s, t belong to Π_n , hence

$$\mathbb{E}[\xi(A_t - A_s)] = \lim_{n \rightarrow \infty} \mathbb{E}[\xi(A_t^{(n)} - A_s^{(n)})] \geq 0$$

because $\{A_u^{(n)}\}_{u \in \Pi_n}$ is increasing. Choosing $\xi = \mathbf{1}_{\{A_s > A_t\}}$, we obtain

$$A_s \leq A_t \quad \text{a.s.}$$

Since $\mathbb{T}$ is countable and by right-continuity, it is then nondecreasing on the whole interval $[0, a]$. Therefore, $\{A_t, 0 \leq t \leq a\}$ is an increasing process. Next:

$$Y_t = A_t - \mathbb{E}(A_a \mid \mathcal{F}_t) = X_t - \mathbb{E}(X_a \mid \mathcal{F}_t) \implies X_t = A_t + \mathbb{E}(X_a - A_a \mid \mathcal{F}_t)$$

Define $M_t := \mathbb{E}(X_a - A_a \mid \mathcal{F}_t)$, then it is a right-continuous martingale, and

$$X_t = M_t + A_t$$

It remains to show that A is natural. Let $\{\xi_t, 0 \leq t \leq a\}$ be any bounded right-continuous martingale. For each n , since $\{A_{t_j^{(n)}}^{(n)}\}_{0 \leq j \leq 2^n}$ is predictable on Π_n :

$$\begin{aligned} \mathbb{E}(\xi_a A_a^{(n)}) &= \mathbb{E} \left[\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} \left(A_{t_j^{(n)}}^{(n)} - A_{t_{j-1}^{(n)}}^{(n)} \right) \right] = \mathbb{E} \left[\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} \left(Y_{t_j^{(n)}} - Y_{t_{j-1}^{(n)}} \right) \right] \\ &= \mathbb{E} \left[\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} \left(A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}} \right) \right] \end{aligned}$$

Letting $n \rightarrow \infty$, the left-hand side converges to $\mathbb{E}(\xi_a A_a)$ by weak convergence:

$$\mathbb{E}(\xi_a A_a) = \mathbb{E} \int_0^a \xi_s dA_s = \mathbb{E} \int_0^a \xi_{s-} dA_s$$

, so A is natural. Hence, on $[0, a]$, we have constructed

$$X_t = M_t + A_t, \quad 0 \leq t \leq a,$$

, with M a right-continuous martingale, and A a natural increasing process.

Then we completed this long proof. □

2.3 Square-Integrable Martingale

After a complicated derivation about Doob-Meyer decomposition, we had to know what it can do, then it comes to our approach. Here firstly propose a definition:

Definition 2.3.1. Let $X = \{X_t, t \geq 0\}$ be a right-continuous martingale, if:

(a) X is square-integrable, i.e.:

$$\mathbb{E} X_t^2 < \infty, \forall t \geq 0$$

(b) $X_0 = 0$

, then we say $X \in \mathcal{M}_2$.

Notice that, if $X \in \mathcal{M}_2$, then $Y := X^2 := \{X_t^2, t \geq 0\}$ is a submartingale, to make it meaningful, we require it to be integrable, which explains why we are interested into square-integrable. Since it is square integrable, then it is clearly in DL, then it admits Doob-Meyer decomposition:

$$Y_t = M_t + A_t$$

Furthermore, if we restrict all the sample path of X to be continuous, noted as $X \in \mathcal{M}_2^c$, then from last subsection, M_t and A_t be continuous. Empowered by this, we define:

Definition 2.3.2. $\forall X \in \mathcal{M}_2$, the quadratic variation of X is defined by:

$$\langle X \rangle_t := A_t$$

Remark 2.3.1. By theorem ??, $\langle X \rangle_t$ is the unique, adapted, natural, increasing process, such that $\langle X \rangle_0 = 0$, moreover, $X_t^2 - \langle X \rangle_t$ is martingale.

Now, if we take, $X, Y \in \mathcal{M}_2$, then:

$$\begin{cases} (X_t + Y_t)^2 - \langle X + Y \rangle_t \\ (X_t - Y_t)^2 - \langle X - Y \rangle_t \end{cases} \text{ are both martingale.}$$

, then we could do subtraction of these two:

$$4X_t Y_t - (\langle X + Y \rangle_t - \langle X - Y \rangle_t) \text{ is also martingale.}$$

Inspired by this, we could define the **cross variation**:

Definition 2.3.3. For $X, Y \in \mathcal{M}_2$, we define the **cross variation**:

$$\langle X, Y \rangle_t := \frac{1}{4} (\langle X + Y \rangle_t - \langle X - Y \rangle_t)$$

, such that $X_t Y_t - \langle X, Y \rangle_t$ is a martingale.

Notice that this definition recovers our quadratic variation, since:

$$\langle X, X \rangle_t = \frac{1}{4} \langle 2X \rangle_t = \frac{1}{4} \cdot 4 \cdot A_t = \langle X \rangle_t$$

But more commonly, quadratic or generally, p-th variation is defined by:

Definition 2.3.4. The **p-th variation** of X over partition:

$$\Pi = \{t_0, t_1, \dots, t_m\} \subseteq [0, t]$$

is defined to be:

$$V_t^{(p)}(\Pi) = \sum_{k=1}^m |X_{t_k} - X_{t_{k-1}}|^p$$

Hence, we want to justify, under class of $\mathcal{M}_2^c$, it is "equivalent":

Theorem 2.3.1. Let $X \in \mathcal{M}_2^c$, for partition, Π , of $[0, t]$, we have:

$$\lim_{\|\Pi\| \rightarrow 0} V_t^{(2)}(\Pi) = \langle X \rangle_t, \text{ in probability}$$

, i.e., $\forall \varepsilon > 0, \eta > 0, \exists \delta > 0$, s.t.:

$$\|\Pi\| < \delta \implies \mathbf{P} \left(\left| V_t^{(2)}(\Pi) - \langle X \rangle_t \right| > \varepsilon \right) < \eta$$

Proof. To see it is "equal" in probability, we consider L^2 distance:

$$\begin{aligned} \mathbb{E} \left(V_t^{(2)}(\Pi) - \langle X \rangle_t \right)^2 &= \mathbb{E} \left[\sum_{k=1}^m |X_{t_k} - X_{t_{k-1}}|^2 - \langle X \rangle_t \right]^2 \\ &= \mathbb{E} \left\{ \sum_{k=1}^m \left[|X_{t_k} - X_{t_{k-1}}|^2 - (\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}) \right] \right\}^2 \end{aligned}$$

Hence we need to consider them separately:

- (1) Firstly, for centered terms, notice that, for $t_j < t_{i-1}$:

$$\mathbb{E} \left[(X_{t_i} - X_{t_{i-1}})^2 \mid \mathcal{F}_{t_j} \right] = \mathbb{E} (X_{t_i}^2 \mid \mathcal{F}_{t_j}) - 2\mathbb{E} (X_{t_i} X_{t_{i-1}} \mid \mathcal{F}_{t_j}) + \mathbb{E} (X_{t_{i-1}}^2 \mid \mathcal{F}_{t_j})$$

Furthermore, if we focus on cross term, we may stick in a conditional expectation:

$$\begin{aligned} \mathbb{E} (X_{t_i} X_{t_{i-1}} \mid \mathcal{F}_{t_j}) &= \mathbb{E} [\mathbb{E} (X_{t_i} X_{t_{i-1}} \mid \mathcal{F}_{t_{i-1}}) \mid \mathcal{F}_{t_j}] \\ &= \mathbb{E} [X_{t_{i-1}} \mathbb{E} (X_{t_i} \mid \mathcal{F}_{t_{i-1}}) \mid \mathcal{F}_{t_j}] = \mathbb{E} (X_{t_{i-1}}^2 \mid \mathcal{F}_{t_j}) \end{aligned}$$

Therefore, we could see the cross term is gone, since:

$$\mathbb{E} \left[(X_{t_k} - X_{t_{k-1}})^2 \mid \mathcal{F}_{t_j} \right] = \mathbb{E} (X_{t_k}^2 - X_{t_{k-1}}^2 \mid \mathcal{F}_{t_j}) = \mathbb{E} (\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}} \mid \mathcal{F}_{t_j})$$

By symmetry, another parts also works.

- (2) Secondly, for centered term, we use inequality to divide and conquer:

$$\begin{aligned} & \sum_{k=1}^m \mathbb{E} \left[\left| X_{t_k} - X_{t_{k-1}} \right|^2 - (\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}) \right]^2 \\ & \leq \sum_{k=1}^m \left[2\mathbb{E} (X_{t_k} - X_{t_{k-1}})^4 + 2\mathbb{E} (\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}})^2 \right] \\ & = 2 \mathbb{E} V_t^4(\Pi) + 2 \sum_{k=1}^m \mathbb{E} (\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}})^2 \\ & \leq 2 \mathbb{E} V_t^4(\Pi) + 2 \mathbb{E} \left[\sup_{1 \leq k \leq m} |\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}| \cdot \langle X \rangle_t \right] \end{aligned}$$

, where we only used a simple inequality, $(a - b)^2 \leq 2a^2 + 2b^2$. Then next step is to prove these two terms go to zero:

- (i) For blue term, we provide a lemma without proof:

Lemma. Let $X \in \mathcal{M}_2^c$ satisfy $|X_s| \leq K < \infty$, $\forall s \leq t$, then:

$$\mathbb{E} \left[V_t^{(2)}(\Pi) \right]^2 \leq 6K^4$$

Assuming this, you should verify, we can see:

$$\mathbb{E} V_t^{(4)}(\Pi) \leq \mathbb{E} \left(\sum_{k=1}^m |X_{t_k} - X_{t_{k-1}}|^2 \cdot m_t(X, \|\Pi\|) \right)$$

, where $m_t(X, \delta)$ is, the δ -distance, defined by:

$$m_t(X, \delta) = \sup_{0 \leq r < s \leq t} \{|X_r - X_s| : |s - r| \leq \delta\}$$

Then, under the above lemma condition, if $X \in \mathcal{M}_2^c$ and $|X_s| \leq K, \forall 0 \leq s \leq t$ by bounded convergence:

$$\mathbb{E} m_t(X, \|\Pi\|) \xrightarrow{\|\Pi\| \rightarrow 0} 0$$

Now, using Cauchy-Schwartz, to separate and control them:

$$\mathbb{E} V_t^{(4)}(\Pi) \leq \left\{ \mathbb{E} \left[V_t^{(2)}(\Pi) \right]^2 \right\}^{1/2} [\mathbb{E} m_t^4(X, \|\Pi\|)]^{1/2} \leq \sqrt{6} K^2 \cdot [\mathbb{E} m_t^4(X, \|\Pi\|)]^{1/2} \rightarrow 0$$

(a) For greened term, also introduce new lemma omitted proof:

Lemma. Let $X \in \mathcal{M}_2^c$ satisfy $\langle X \rangle_s \leq K < \infty, \forall 0 \leq s \leq t$, then:

$$\mathbb{E} \left[\sup_{1 \leq k \leq m} |\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}| \cdot \langle X \rangle_t \right] \rightarrow 0$$

This can be done by uniform continuity.

Lastly, we need to deal with general $X \in \mathcal{M}_2^c$, i.e. unbounded, then we use the technique, "localisation", to define, $\forall n \geq 1$:

$$T_n = \inf \{t \geq 0 : |X_t| \geq n \text{ or } \langle X \rangle_t \geq n\}$$

Then, considered localised process, $X_t^{(n)} = X_{t \wedge T_n}$ is bounded martingale, also as the following squared process:

$$\{X_{t \wedge T_n}^2 - \langle X \rangle_{t \wedge T_n}, t \geq 0\}$$

, where it is done by the uniqueness of $\langle X^{(n)} \rangle_t$, then we get:

$$\langle X^{(n)} \rangle_t = \langle X \rangle_{t \wedge T_n}$$

For this localised process, from above discussion we know:

$$\mathbb{E} \left[\sum_{k=1}^m |X_{t_k \wedge T_n} - X_{t_{k-1} \wedge T_n}|^2 - \langle X \rangle_{t \wedge T_n} \right]^2 \xrightarrow{\|\Pi\| \rightarrow 0} 0$$

Unfortunately, we cannot directly send $n \rightarrow \infty$, since there are several variables in control, however, which can be done by ε - δ language, now, $\forall \varepsilon > 0, \forall \eta > 0$, we can always choose a large enough fixed $t > 0$ s.t.

$$\mathbb{P}(T_n < t) \leq \frac{\eta}{2}$$

Besides, $\exists \delta > 0$, s.t. $\|\Pi\| \leq \delta$, here we can choose $\|\Pi\|$ small enough s.t.:

$$\mathbb{E} \left[\sum_{k=1}^m |X_{t_k \wedge T_n} - X_{t_{k-1} \wedge T_n}|^2 - \langle X \rangle_{t \wedge T_n} \right]^2 \leq \varepsilon^2 \cdot \frac{\eta}{2}$$

Then, by simple event separation and Markov Inequality:

$$\begin{aligned} \mathbb{P} \left(\left| V_t^{(2)}(\Pi) - \langle X \rangle_t \right| > \varepsilon \right) &= \mathbb{P}(T_n < t) + \mathbb{P} \left(\left| V_{t \wedge T_n}^{(2)}(\Pi) - \langle X \rangle_{t \wedge T_n} \right| > \varepsilon \right) \\ &\leq \frac{\eta}{2} + \frac{1}{\varepsilon} \mathbb{E} \left[\sum_{k=1}^m |X_{t_k \wedge T_n} - X_{t_{k-1} \wedge T_n}|^2 - \langle X \rangle_{t \wedge T_n} \right]^2 \leq \eta \end{aligned}$$

Hence, we completed the proof by using ε - δ language to bypass. $\square$

Since the power 2 gives you an exact convergence, then if the power changes a little:

Corollary 2.3.2. *Same setting as before, $\forall q > 2$:*

$$V_t^{(q)} \xrightarrow{p.} 0$$

, moreover, on $\{\langle X \rangle_t \geq 0\}$, $\forall q < 2$:

$$V_t^{(q)} \xrightarrow{p.} \infty$$

The conclusion to be drawn from above is that for continuous, square-integrable martingales, quadratic variation is the "right" variation to study. All variations of higher order are zero, and except in trivial cases, where the martingale is a.s. constant. Thus, the sample paths of continuous, square-integrable martingales are quite different from "ordinary" continuous martingales. Being of unbounded first variation, they cannot be differentiable, nor is it possible to define integral of form:

$$\int_0^t Y_s dX_s$$

, w.r.t $X \in \mathcal{M}_2^c$ in a pathwise Lebesgue-Stieltjes sense.

Chapter 3

Stochastic Integration

In the last section, we focus on continuous square-integrable martingale class:

$$\mathcal{M}_2 = \{X : \mathbb{E}X_t^2 < \infty, \forall t \geq 0\}$$

Before, we move into integration, we delve into this space, by restricting it first:

$$\mathcal{M}^2 := \left\{ X : \sup_{t \geq 0} \mathbb{E}X_t^2 < \infty \right\}$$

, clearly this space is smaller, since it implies continuous, square-integrable class.

3.1 The L^2 Theory of Martingales

Beside further, we will see it is a **Hilbert Space**, moreover it may be easy for us to study the process that starts at zero (otherwise, we may subtract the initial value):

$$\mathcal{M}_0^2 = \{X \in \mathcal{M}^2 : X_0 = 0\}$$

Theorem 3.1.1. *The space $\mathcal{M}^2$ is same as Hilbert space, $L^2(\mathcal{F}_\infty)$, with bijection:*

$$\begin{aligned} \{M_t, t \geq 0\} \in \mathcal{M}^2 &\mapsto M_\infty := \lim_{t \rightarrow \infty} M_t \in L^2(\mathcal{F}_\infty) \\ M_\infty \in L^2(\mathcal{F}_\infty) &\mapsto M_t := \mathbb{E}(M_\infty \mid \mathcal{F}_t) \in \mathcal{M}^2 \end{aligned}$$

Proof. omitted with intention. □

Remark 3.1.1. The live of space for backward is provided by:

$$\mathbb{E}X_t^2 = \mathbb{E}[\mathbb{E}(X_\infty^2 \mid \mathcal{F}_t)] \leq \mathbb{E}[\mathbb{E}(X_\infty^2 \mid \mathcal{F}_t)] = \mathbb{E}X_\infty^2 < \infty$$

, then recall the Doob's L^2 inequality:

$$\mathbb{E} \sup_{0 \leq s \leq t} |X_s|^2 \leq 4\mathbb{E}X_t^2$$

, sending $t \rightarrow \infty$ to see:

$$\mathbb{E} \sup_{t \geq 0} X_s^2 \leq 4\mathbb{E}X_\infty^2$$

Next interesting things about $\mathcal{M}^2$ space is that it is **complete**, i.e. for any cauchy sequence $\{M^{(n)}\} \in \mathcal{M}^2$, we have $\{M_\infty^{(n)}\} \in L^2(\mathcal{F}_\infty)$ is also cauchy sequence:

Theorem 3.1.2. *The space $\mathcal{M}^2$ is complete.*

Proof. We only need to show, if $\exists M_\infty \in L^2(\mathcal{F}_\infty)$, s.t.:

$$M_\infty^{(n)} \xrightarrow{L^2} M_\infty$$

, with definition, $M_t = \mathbb{E}(M_\infty \mid \mathcal{F}_t)$, then:

$$\sup_{t \geq 0} |M_t^{(n)} - M_t| \xrightarrow{L^2} 0$$

, however, above is easy to show as we indicated in remark 3.1.1:

$$\begin{aligned} \mathbb{E} \sup_{t \geq 0} |M_t^{(n)} - M_t| &= \mathbb{E} \sup_{t \geq 0} \mathbb{E}(M_\infty^{(n)} - M_\infty \mid \mathcal{F}_t)^2 \\ &\leq 4 \mathbb{E}(M_\infty^{(n)} - M_\infty)^2 \longrightarrow 0 \end{aligned}$$

Therefore, we completed the proof. □

More interesting, we focus on the continuous part of this space:

Corollary 3.1.3. *The space, $c\mathcal{M}^2$, of continuous martingale in $\mathcal{M}^2$ is closed in the topology of $\mathcal{M}^2$.*

Proof. Notice, if $M^{(n)} \rightarrow M$, for $M^{(n)} \in c\mathcal{M}^2$, then:

$$\sup_{t \geq 0} |M_t^{(n)} - M_t| \xrightarrow{L^2} 0$$

, therefore, we could pick a subsequence, i.e. $\exists \{n_k, k \geq 0\}$, s.t.:

$$\sup_{t \geq 0} |M_t^{(n_k)} - M_t| \xrightarrow{a.s.} 0$$

, since each one is continuous, we reached M_t is continuous. □

Then next thing is to think about cross variation:

Definition 3.1.1. For $X, Y \in \mathcal{M}^2$, we define:

$$\langle X, Y \rangle := \mathbb{E} X_\infty Y_\infty$$

To connect previous intuition, we define weakly orthogonal.

Definition 3.1.2. $\forall M, N \in \mathcal{M}_0^2$, we say M and N are weakly orthogonal, if:

$$\mathbb{E} M_\infty N_\infty = 0$$

Then correspondingly, we have strongly orthogonal:

Definition 3.1.3. $\forall M, N \in \mathcal{M}_0^2$, M and N are strongly orthogonal, if, $\forall$ s.t. T :

$$\mathbb{E} M_T N_T = 0$$

To better understand these orthogonality, we present following lemma:

Lemma 3.1.4. M and N are strongly orthogonal iff MN is u.i.

Proof. Here, we omit the proof, but introduce a new lemma for verifying u.i. □

Lemma 3.1.5. For any progressively measurable process, s.t. $\forall$ s.t. T , if:

(a) $\mathbb{E} |M_T| < \infty$;

(b) $\mathbb{E} M_T = 0$.

, then M is u.i. martingale.

As procedure in functional analysis or topology, we are interested into:

Definition 3.1.4. (Stable subspace of $\mathcal{M}_0^2$)

We say $\mathcal{N}_0 \subseteq \mathcal{M}_0^2$ is a stable subspace if:

(a) $\mathcal{N}_0$ is closed subspace;

(b) $\forall N \in \mathcal{N}_0$, $\forall$ s.t., T we have:

$$N^T \in \mathcal{N}_0$$

, where N^T should be understood as: $N_t^T = N_{t \wedge T}$.

Above definition sounds astonishing, here we present example:

Example 3.1.1. $c\mathcal{M}_0^2$ is stable subspace.

Empowered by this new tool, we may re-represent our old results:

Theorem 3.1.6. Let $\mathcal{N}_0$ be a stable subspace of $\mathcal{M}_0^2$, let $\mathcal{N}_0^\perp$ be the set of $Z \in \mathcal{M}_0^2$, which are weakly orthogonal to every element of $\mathcal{N}_0$, i.e.:

$$\mathcal{N}_0^\perp = \{Z \in \mathcal{M}_0^2 : \langle Z, N \rangle = 0, \forall N \in \mathcal{N}_0\}$$

, then $\mathcal{N}_0^\perp$ is a stable subspace of $\mathcal{M}_0^2$, and $\forall Z \in \mathcal{N}_0^\perp$, we have Z and N are strongly orthogonal for all $N \in \mathcal{N}_0$, finally, $\forall M \in \mathcal{M}_0^2$, it admits:

$$M = N + Z$$

Proof. We only show it is stable, the decomposition part is just Doob-Meyer's result.

(a) Firstly, it is easy to see, if $Z_n \xrightarrow{L^1} Z$, then:

$$\mathbb{E}Z_n N = 0 \implies \mathbb{E}Z N = 0$$

(b) Next, $\forall Z \in \mathcal{N}_0^\perp$, then $\forall N \in \mathcal{N}_0$:

$$\mathbb{E}Z_\infty N_\infty = 0$$

, we are left to show, $\forall$ s.t. T , we have $Z^T \in \mathcal{N}_0^\perp$, which is due to:

$$\begin{aligned} \mathbb{E}Z_\infty^T N_\infty &= \mathbb{E}Z_T N_\infty = \mathbb{E}[Z_T \mathbb{E}(N_\infty | \mathcal{F}_T)] = \mathbb{E}Z_T N_T \\ &= \mathbb{E}[\mathbb{E}(Z_\infty | \mathcal{F}_T) N_T] = \mathbb{E}Z_\infty N_T = \mathbb{E}Z_\infty N_\infty^T = 0 \end{aligned}$$

, last step is by $N^T \in \mathcal{N}_0$.

Hence, it is stable, we completed the proof. $\square$

Lastly, we consider the discontinuous part, or the complement of continuous:

Definition 3.1.5. Define the discountinuous space as:

$$d\mathcal{M}_0^2 = (c\mathcal{M}_0^2)^\perp$$

Then $\forall M \in \mathcal{M}_0^2$, we write:

$$M = M^c + M^d$$

, with $M^c \in c\mathcal{M}_0^2$ and $M^d \in d\mathcal{M}_0^2$ and $M^c \cdot M^d$ is u.i, by $\mathbb{E}M_T^c M_T^d = 0$, and jump part:

Definition 3.1.6. Define the jump as:

$$\Delta f(t) = f(t) - f(t-)$$

And Meyer develop following theorem to understand this by new representation:

Theorem 3.1.7. Let $M \in \mathcal{M}_0^2$, then $\exists$ unique increasing process, $[M]$, null at 0, s.t.:

(a) $M^2 - [M]$ is u.i.;

(b) $\Delta[M] = (\Delta M)^2$.

Proof. Omitted with intention. $\square$

Remark 3.1.2. Intuitively, it is to say the quadratic variation process jumps at the same point as based-off martingale, and the intensity is its squared.

As similarly as before, for $M, N \in \mathcal{M}_0^2$, let:

$$[M, N] := \frac{1}{4} ([M + N] - [M - N])$$

Then Kunita and Watanabe studied the jump of this related to based-off martingale:

Theorem 3.1.8. *The process $[M, N]$ is the unique FV process null at 0 s.t.:*

- (a) $MN - [M, N]$ is u.i.;
- (b) $\Delta[M, N] = \Delta M \Delta N$.

Proof. Omitted with intention. □

Remark 3.1.3. Intuitively, it is to say the cross variation jump iff two processes both jump.

3.2 Elementary Previsible Integrands

After those preparation, we restrict ourself into this class, let, $M \in \mathcal{M}_0^2$, now for two s.t.s $S \leq T$, and some bounded $Z \in \mathcal{F}_S$, if $H_t = Z \mathbf{1}_{(S, T]}(t)$, then we could define :

$$\int_0^t H_s dM_s := (H \cdot M)(t) = Z(M_T - M_S)$$

$$\left\{ = \int_0^t \mathbf{1}_{(a, b]} dF(s) = F(t \wedge b) - F(t \wedge a) \right\}$$

As we discussion at motivation, we want to show $(H \cdot M) \in \mathcal{M}_0^2$:

- (1) Firstly we will prove $\forall$ s.t. R :

- (a) $\mathbb{E} |(H \cdot M)_R| < \infty$:

Proof. Let $K > 0$, s.t. $|Z| \leq K < \infty$, then:

$$\mathbb{E} |(H \cdot M)_R| \leq K [\mathbb{E} |M_{T \wedge R}| + \mathbb{E} |M_{S \wedge R}|] \leq 2K \mathbb{E} |M_\infty| < \infty$$

, where it is due to OST:

$$\mathbb{E} |M_{T \wedge R}| = \mathbb{E} [\mathbb{E} (M_\infty | \mathcal{F}_{T \wedge R})] \leq \mathbb{E} [\mathbb{E} (|M_\infty| | \mathcal{F}_{T \wedge R})] = \mathbb{E} |M_\infty|$$

Therefore, it is simply finished. □

- (b) $\mathbb{E} (H \cdot M)_R = 0$:

Proof. To prove above, it is equivalent to show:

$$\mathbb{E} Z (M_{T \wedge R} - M_{S \wedge R}) = 0$$

Considering replace it with indicator function since it is the basis of this space:

$$\text{For } A \in \mathcal{F}_S, \text{ if } Z = \mathbf{1}_A \implies \mathbb{E} \mathbf{1}_A (M_{T \wedge R} - M_{S \wedge R}) = \mathbb{E} \mathbf{1}_A (M_{T_A \wedge R} - M_{S_A \wedge R}) = 0$$

, where we could define:

$$S_A := \begin{cases} S, & \text{on } A \\ \infty, & \text{on } A^c \end{cases}$$

Then we completed the proof by specifying the indicator function. □

Then it is to say $(H \cdot M)$ is u.i. martingale.

(2) Lastly, we remain to show it is L^2 -bounded, which is easy to see, assume $|Z| \leq K$:

$$\mathbb{E} (H \cdot M)_t^2 = \mathbb{E} \left[Z^2 (M_{t \wedge T} - M_{t \wedge S})^2 \right] \leq 2K^2 \mathbb{E} M_{t \wedge T}^2 + 2K^2 \mathbb{E} M_{t \wedge S}^2$$

Since, $M \in \mathcal{M}_0^2$, then M^2 is submartingale, by OST:

$$\mathbb{E} (H \cdot M)_t^2 \leq 4K^2 \mathbb{E} M_t^2 \leq 4K^2 \sup_{u \geq 0} \mathbb{E} M_u^2 < \infty \implies \sup_{t \geq 0} \mathbb{E} (H \cdot M)_t^2 < \infty$$

Then we showed $(H \cdot M) \in \mathcal{M}_0^2$, to extend this result, we have following lemma:

Lemma 3.2.1. *Let $S \leq T$ be two s.t., if $Z \in b\mathcal{F}_S$ and M is u.i. martingale, define:*

$$H_t = Z \mathbf{1}_{(S, T]}(t)$$

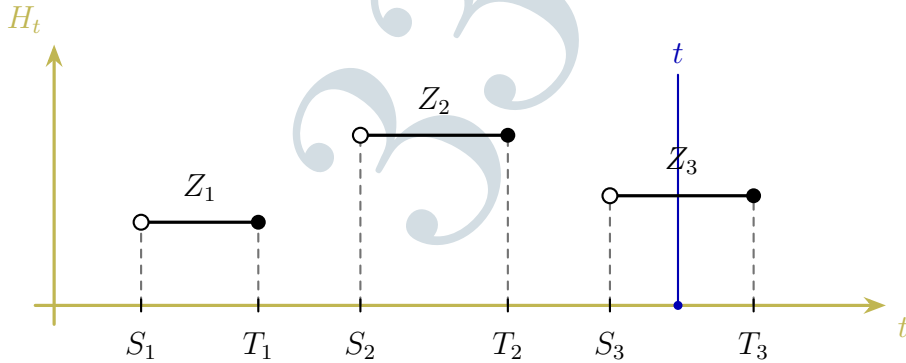
, then $(H \cdot M)$ is u.i. martingale.

Proof. Follow from above discussion. □

Next, a natural idea is to extend our result to linear combination, considering:

$$H_t = \sum_{i=1}^n Z_i \mathbf{1}_{(S_i, T_i]}(t)$$

, where each $Z_i \in b\mathcal{F}_{S_i}$, and a series s.t.s, s.t.: $S_1 \leq T_1 \leq S_2 \leq \dots$ as illustration below:



The left endpoint is not included, so Z_i is already known at time S_i .

Corollary 3.2.2. *Let $\{S_i, i \in [n]\}$, $\{T_i, i \in [n]\}$ be two series of s.t.s, s.t:*

$$S_i \leq T_i, i \in [n]$$

, if $Z_i \in b\mathcal{F}_{S_i}$ and M is u.i. martingale, define:

$$H_t = \sum_{i=1}^n Z_i \mathbf{1}_{(S_i, T_i]}(t)$$

, then $(H \cdot M)$ is also u.i. martingale.

Proof. Directly follow from lemma 3.2.1. □

Therefore, a follow-up question is that, what is the largest class that inherit such property:

Definition 3.2.1. A previsible σ -algebra $\mathcal{P}$ on $(0, \infty) \times \Omega$ is defined to be the smallest σ -algebra s.t. every adapted L-process is $\mathcal{P}$ -measurable.

Remark 3.2.1. The so called **L-process** is just what we used before, the L.C.R.L process, and correspondingly, we say $\overline{\{X_t, t \geq 0\}}$ is previsible if it is $\mathcal{P}$ -measurable.

Lemma 3.2.3. Suppose S and T are two s.t.s with $S \leq T < \infty$, let $Z \in b\mathcal{F}_S$ and:

$$H = Z\mathbf{1}_{(S, T]}$$

, then H defined above is previsible.

Proof. Here, $\forall B \in \mathcal{B}(\mathbb{R})$, and we assume $0 \notin B$, then $\forall t > 0$:

$$\begin{aligned} \{H_t \in B\} &= \{Z \in B\} \cap \{S < t \leq T\} \\ &= \{Z \in B\} \cap \{S < t\} \cap \{t \leq T\} \\ &= \bigcup_{n=1}^{\infty} \left(\underbrace{\{Z \in B\} \cap \{S \leq t - 1/n\}}_{\in \mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t} \right) \cap \underbrace{\{T \geq t\}}_{=\{T < t\}^c \in \mathcal{F}_t} \in \mathcal{F}_t \end{aligned}$$

Hence, we completed the proof. □

Now, we want to see the connection of simple process and previsible process:

Definition 3.2.2. Define the σ -algebra of bounded simple process:

$$b\mathcal{E} = \left\{ \sum_{i=1}^n Z_i \mathbf{1}_{(S_i, T_i]} : n \geq 1, Z_i \in b\mathcal{F}_{S_i}, S_1 \leq T_1 \leq S_2 \leq \dots \right\}$$

And then we claim that there are the same after completion:

Claim 3.2.4. $\sigma(b\mathcal{E}) = \mathcal{P}$

Proof. The forward is simple, and backward conclusion can be donw by using that fact:

$$H_t = \lim_{k \uparrow \infty} \lim_{n \uparrow \infty} \sum_{i=2}^{nk} H_{\frac{i-1}{n}} \mathbf{1}_{(\frac{i-1}{n}, \frac{i}{n}]}(t)$$

We leave the details to the reader. □

3.3 The L^2 -Itô Integral

After discussion with integrands, we now relook at the L^2 theory:

$$\mathcal{M}_0^2 \longleftrightarrow L^2(\mathcal{F}_\infty)$$

In the Itô integral, the integrand is allowed to remember the past, but not to foresee the future. This non-anticipative nature is exactly what predictability formalises.

Lemma 3.3.1. Let $H \in b\mathcal{E}$, and $M \in \mathcal{M}_0^2$, then:

$$(H \cdot M) \in \mathcal{M}_0^2$$

, and more specific, if $H = \sum_{i=1}^n Z_{i-1} \mathbf{1}_{(T_{i-1}, T_i]}$, then:

$$\mathbb{E} (H \cdot M)_\infty^2 = \sum_{i=1}^n \mathbb{E} \left[Z_{i-1}^2 (M_{T_i} - M_{T_{i-1}})^2 \right]$$

Proof. The first result have been showed in the preparation of section 3.2, now since:

$$(H \cdot M) \text{ is u.i. martingale} \implies (H \cdot M)_t \xrightarrow{L^1} (H \cdot M)_\infty := \sum_{i=1}^n Z_{i-1} \Delta_i$$

, where we denote, $\Delta_i = M_{T_i} - M_{T_{i-1}}$, now notice that:

$$\mathbb{E} (H \cdot M)_\infty^2 = \mathbb{E} \left[\sum_{i=1}^n Z_{i-1}^2 \Delta_i^2 \right] + 2\mathbb{E} \left[\sum_{i < j} Z_{i-1} \Delta_i Z_{j-1} \Delta_j \right]$$

, then it remains to show the cross terms is vanished, obviously, for $i \leq j-1 < j$:

$$\mathbb{E} [Z_{i-1} \Delta_i Z_{j-1} \Delta_j] = \mathbb{E} [Z_{i-1} \Delta_i \cdot Z_{j-1} \mathbb{E} (\Delta_j \mid \mathcal{F}_{T_{j-1}})] = 0$$

Lastly, by Doob's inequality, we also have:

$$\mathbb{E} \sup_{t \geq 0} (H \cdot M)_t^2 \leq 4\mathbb{E} (H \cdot M)_\infty^2 < \infty$$

, hence $(H \cdot M) \in \mathcal{M}_0^2$, and we completed the proof. □

Next, recall that, if $M \in \mathcal{M}_0^2$, then $\exists$ quadratic variation, $[M]$, s.t.:

$$M^2 - [M] \text{ is u.i. martingale.}$$

Then, using this fact, notice that:

$$\begin{aligned} \mathbb{E} \left[(M_{T_i} - M_{T_{i-1}})^2 \mid \mathcal{F}_{T_{i-1}} \right] &= \mathbb{E} [M_{T_i}^2 \mid \mathcal{F}_{T_{i-1}}] + M_{T_{i-1}}^2 - 2\mathbb{E} [M_{T_i} M_{T_{i-1}} \mid \mathcal{F}_{T_{i-1}}] \\ &= \mathbb{E} [M_{T_i}^2 \mid \mathcal{F}_{T_{i-1}}] + M_{T_{i-1}}^2 - 2M_{T_{i-1}} \mathbb{E} [M_{T_i} \mid \mathcal{F}_{T_{i-1}}] \\ &= \mathbb{E} [M_{T_i}^2 \mid \mathcal{F}_{T_{i-1}}] + M_{T_{i-1}}^2 - 2M_{T_{i-1}}^2 \\ &= \mathbb{E} [M_{T_i}^2 - M_{T_{i-1}}^2 \mid \mathcal{F}_{T_{i-1}}] = \mathbb{E} [[M]_{T_i} - [M]_{T_{i-1}} \mid \mathcal{F}_{T_{i-1}}] \end{aligned}$$

Hence, by this additional information, we may rewrite:

$$\begin{aligned} \mathbb{E} (H \cdot M)_\infty^2 &= \sum_{i=1}^n \mathbb{E} \left[Z_{i-1}^2 (M_{T_i} - M_{T_{i-1}})^2 \right] \\ &= \sum_{i=1}^n \mathbb{E} \left[Z_{i-1}^2 ([M]_{T_i} - [M]_{T_{i-1}}) \right] \xrightarrow{n \rightarrow \infty} \mathbb{E} \int_0^\infty H_s^2 d[M]_s \end{aligned}$$

For any previsible process, H , we now define the norm, $\forall$ fixed $M \in \mathcal{M}_0^2$:

$$\|H\|_M = \left(\mathbb{E} \int_0^\infty H_s^2 d[M]_s \right)^{1/2}$$

Then the space:

$$L^2(M) := \{\text{previsible } H \text{ with } \|H\|_M < \infty\}$$

, (strictly speaking, this is of course a space of certain equivalence classes) is a Hilbert space, containing $b\mathcal{E}$. Hence, stochastic integral, $(H \cdot M)$, is already defined for $H \in b\mathcal{E}$:

$$\begin{aligned} I : L^2(M) \cap b\mathcal{E} &\mapsto L^2(\mathcal{F}_\infty) \\ H &\mapsto (H \cdot M)_\infty \end{aligned}$$

, is an isometry:

$$\|H\|_M = \|I(H)\|$$

, then it extends uniquely to $L^2(M) \cap b\mathcal{E}$. We summarise into following theorem:

Theorem 3.3.2. (*Kunita and Watanabe*)

Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$, then the Itô integral, $(H \cdot M)_t$ is the image of H under the extension of $I : L^2(M) \mapsto L^2(\mathcal{F}_\infty)$.

Proof. Follow above discussion. □

3.4 Stopping and Localisation

Firstly, we start with our familiar friends:

Definition 3.4.1. Let T be a s.t. and H be any process, define stopped process:

$$H\mathbf{1}_{(0,T]}(t) = \begin{cases} H_t, & 0 < t \leq T \\ 0, & t > T \end{cases}$$

And in some of cases, we may want it to stop but not vanish, then:

Definition 3.4.2. Let T be a s.t. and X be any process, define localised process:

$$X^T(t) = \begin{cases} X(t), & 0 \leq t \leq T \\ X(T), & t > T \end{cases}$$

Then, if we want to extend the result to localised process, notice:

$$\begin{aligned} (H \cdot M)_t^T &= [Z_i (M_{T_i \wedge t} - M_{T_{i-1} \wedge t})]^T = (Z_i \mathbf{1}_{(T_{i-1} \wedge T, T_i \wedge T]} \cdot M^T)_t \\ &= (H \mathbf{1}_{(0,T]} \cdot M^T)_t = Z_i (M_{T_i \wedge T \wedge t} - M_{T_{i-1} \wedge T \wedge t}) \end{aligned}$$

The next result provides the key to extending the stochastic integral by localisation:

Theorem 3.4.1. Fix $M \in \mathcal{M}_0^2$, $\forall H \in L^2(M)$ and any s.t. T , we have:

- (a) $(H \cdot M)^T = (H \mathbf{1}_{(0,T]} \cdot M) = (H \cdot M^T)$,
- (b) $(H \cdot M)_t^2 - \int_0^t H_s^2 d[M]_s$ is u.i. martingale,
- (c) If $H, K \in b\mathcal{P}$, then $(HK) \cdot M = H \cdot (K \cdot M) = K \cdot (H \cdot M)$,
- (d) $\Delta(H \cdot M) = H \Delta M$,
- (e) If M is continuous, then $(H \cdot M)$ is also continuous.

Proof. Firstly, we prove the result for $H \in b\mathcal{E}$, and then extend it to $L^2(M)$ by the isometry. Recall that if $H^n \rightarrow H$ in $L^2(M)$, then:

$$\mathbb{E} \sup_{t \geq 0} |(H^n \cdot M)_t - (H \cdot M)_t|^2 \leq 4\mathbb{E} [(H^n - H) \cdot M]_\infty^2 = 4\|H^n - H\|_M^2 \rightarrow 0.$$

Hence, after passing to a subsequence if necessary, the convergence is uniform in t a.s.

- (a) We first assume that, two s.t.s $U \leq V$, and $Z \in b\mathcal{F}_S$, considering:

$$H = Z \mathbf{1}_{(U,V]}$$

Then by the definition of stochastic integral for elementary previsible processes,

$$(H \cdot M)_t = Z (M_{t \wedge V} - M_{t \wedge U}) \implies (H \cdot M)_t^T = Z (M_{t \wedge V \wedge T} - M_{t \wedge U \wedge T})$$

Secondly, we could see:

$$H \mathbf{1}_{(0,T]}(t) = Z \mathbf{1}_{(U \wedge T, V \wedge T]}(t) \implies (H \mathbf{1}_{(0,T]} \cdot M) = Z (M_{t \wedge V \wedge T} - M_{t \wedge U \wedge T})$$

Lastly, the left one is also have same structure:

$$(H \cdot M^T)_t = Z (M_{V \wedge t}^T - M_{U \wedge t}^T) = Z (M_{t \wedge V \wedge T} - M_{t \wedge U \wedge T})$$

By linearity, the same conclusion holds for every $H \in b\mathcal{E}$. Hence the whole space.

- (b) Firstly, notice that:

$$\mathbb{E} \sup_{t \geq 0} (H \cdot M)_t^2 \leq 4\mathbb{E} \int_0^\infty H_s^2 d[M]_s < \infty \implies \mathbb{E} \sup_{t \geq 0} \left| (H \cdot M)_t^2 - \int_0^t H_s^2 d[M]_s \right| < \infty$$

, then $N_t := (H \cdot M)_t^2 - \int_0^t H_s^2 d[M]_s$ is u.i., it remains to show $\forall$ s.t. T :

$$\begin{aligned} \mathbb{E} N_T &= \mathbb{E} (H \cdot M)_T^2 - \mathbb{E} \int_0^T H_s^2 d[M]_s \\ &= \mathbb{E} (H \mathbf{1}_{(0,T]} \cdot M)_\infty^2 - \mathbb{E} \int_0^T H_s^2 d[M]_s \\ &= \mathbb{E} \int_0^\infty (H \mathbf{1}_{(0,T]}(s))^2 d[M]_s - \mathbb{E} \int_0^T H_s^2 d[M]_s = 0 \end{aligned}$$

Hence, by lemma 3.1.5, we completed (b), moreover, recall that, if $M \in \mathcal{M}_0^2$, then

$$M^2 - [M] \text{ is martingale} \implies [H \cdot M]_t = \int_0^t H_s^2 d[M]_s.$$

- (c) We first prove the identity for $H, K \in b\mathcal{E}$, considering $H = Z\mathbf{1}_{(S,T]}$, $K = Y\mathbf{1}_{(U,V]}$, then HK is again a finite linear combination of stochastic intervals, and on each such interval the increment of $K \cdot M$ is just K times the increment of M . Hence,

$$H \cdot (K \cdot M) = (HK) \cdot M.$$

By linearity, this holds for all $H, K \in b\mathcal{E}$. Similarly,

$$K \cdot (H \cdot M) = (HK) \cdot M.$$

Now let $H, K \in b\mathcal{P}$. Since H, K are bounded, all the integrals below are well-defined. Choose $H^n, K^n \in b\mathcal{E}$ such that:

$$H^n \xrightarrow{L^2} H, K^n \xrightarrow{L^2} K \implies H^n K^n \xrightarrow{L^2} HK$$

, and therefore:

$$\|H^n - H\|_{K \cdot M}^2 = \mathbb{E} \int_0^\infty (H_s^n - H_s)^2 K_s^2 d[M]_s \rightarrow 0$$

Passing to the limit, then we completed the (c).

- (d) For $H \in b\mathcal{E}$:

$$\Delta(H \cdot M)_t = \sum_{i=1}^n Z_i \mathbf{1}_{\{S_i < t \leq T_i\}} \Delta M_t = H_t \Delta M_t$$

For general $H \in L^2(M)$, take $H^n \in b\mathcal{E}$ with:

$$\|H^n - H\|_M \rightarrow 0 \implies \mathbb{E} \sum_{s \geq 0} (H_s^n - H_s)^2 (\Delta M_s)^2 \leq \mathbb{E} \int_0^\infty (H_s^n - H_s)^2 d[M]_s \rightarrow 0$$

Thus, passing to a subsequence, the jump identity passes to the limit, and we get

$$\Delta(H \cdot M) = H \Delta M$$

- (e) Follow from (d).

Therefore, all statements are proved. □

Thanks to the theorem 3.4.1's (a), we can extend:

Lemma 3.4.2. *If $H \in b\mathcal{P}$, and $M \in \mathcal{M}_{0,loc}^2$, then:*

$$(H \cdot M) \in \mathcal{M}_{0,loc}^2$$

, moreover, if $H \in b\mathcal{P}$, and $M \in c\mathcal{M}_{0,loc}^2$, then:

$$(H \cdot M) \in c\mathcal{M}_{0,loc}^2$$

Proof. Follow the result from theorem 3.4.1's (a). □

Suppose that $H \in b\mathcal{P}, M \in \mathcal{M}_{0,loc}^2$, and that M has paths of finite variation. We certainly want $(H \cdot M)$ to agree with the pathwise Stieltjes integral, and indeed it does.

3.5 Integration w.r.t. Local Martingales

3.6 Integration w.r.t. Continuous Semimartingales

3.7 The Itô's Formula

The Itô integral is defined by a limiting procedure far too clumsy to provide an effective means of handling such integrals in practice. Here, we shall develop the rules by which it can be manipulated, commonly called the stochastic calculus. As before, we start with stochastic calculus for bounded continuous processes, and build up via localisation.

Lemma 3.7.1. *Let $M \in c\mathcal{M}_0$ be bounded, and let V be a continuous FV₀, then:*

$$\begin{aligned} M_t^2 &= \int_0^t 2M_s dM_s + [M]_t \\ M_t V_t &= \int_0^t M_s dV_s + \int_0^t V_s dM_s \end{aligned}$$

Proof. The proof is straightforward, by simple algebraic manipulation:

$$\begin{aligned} M_t^2 &= (M_t - M_0)^2 = \left[\sum_{k=1}^{2^n} M_{t_k^n}^2 - M_{t_{k-1}^n}^2 \right] \\ &= \sum_{k=1}^{2^n} \left(M_{t_k^n} - M_{t_{k-1}^n} \right)^2 + 2 \sum_{k=1}^{2^n} \left[M_{t_{k-1}^n} \left(M_{t_k^n} - M_{t_{k-1}^n} \right) \right] \\ &\xrightarrow{n \rightarrow \infty} [M]_t + 2 \int_0^t M_s dM_s \end{aligned}$$

Similarly, for another one, we may see:

$$\begin{aligned} M_t V_t &= M_{t_{2^n}^n} V_{t_{2^n}^n} - M_0 V_0 = \sum_{k=1}^{2^n} \left[M_{t_k^n} V_{t_k^n} - M_{t_{k-1}^n} V_{t_{k-1}^n} \right] \\ &= \sum_{k=1}^{2^n} \left[M_{t_k^n} V_{t_k^n} - M_{t_k^n} V_{t_{k-1}^n} + M_{t_k^n} V_{t_{k-1}^n} - M_{t_{k-1}^n} V_{t_{k-1}^n} \right] \\ &= \sum_{k=1}^{2^n} M_{t_k^n} \left(V_{t_k^n} - V_{t_{k-1}^n} \right) + \sum_{k=1}^{2^n} V_{t_{k-1}^n} \left(M_{t_k^n} - M_{t_{k-1}^n} \right) \\ &\xrightarrow{n \rightarrow \infty} \int_0^t M_s dV_s + \int_0^t V_s dM_s \end{aligned}$$

Hence, we completed the proof, and this provides us a powerful tool for I.B.P. □

The next result is the cornerstone of stochastic calculus:

Theorem 3.7.2. *(Integration by parts formula)*

Let X and Y be continuous semimartingales, then:

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t$$

Proof. Since $X, Y \in b\mathcal{P}$, then the integral is well defined, w.l.o.g., we may assume, $X_0 = Y_0 = 0$. In deed, if it is not and assume we have the above result, notice:

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \tilde{X}_t \tilde{Y}_t + X_0 \tilde{Y}_t + Y_0 \tilde{X}_t \\ &= \int_0^t \tilde{X}_s d\tilde{Y}_s + \int_0^t \tilde{Y}_s d\tilde{X}_s + [\tilde{X}, \tilde{Y}]_t + X_0 \tilde{Y}_t + Y_0 \tilde{X}_t \\ &= \int_0^t \tilde{X}_s dY_s + \int_0^t \tilde{Y}_s dX_s + [X, Y]_t + X_0 \int_0^t dY_s + Y_0 \int_0^t dX_s \\ &= \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t \end{aligned}$$

Then assume they all start from null, if $X = M + A, Y = N + B$, and:

$$M, N \in c\mathcal{M}_{0,\text{loc}}, A, B \in \text{FV}_0 \implies \begin{cases} M_t N_t = \int_0^t M_s dN_s + \int_0^t N_s dM_s + [M, N]_t \\ A_t B_t = \int_0^t A_s dB_s + \int_0^t B_s dA_s \end{cases}$$

Then simple algebra gives us plus lemma 3.7.1:

$$\begin{aligned} X_t Y_t &= (M_t + A_t) \times (N_t + B_t) = M_t N_t + A_t B_t + M_t B_t + N_t A_t \\ &= \int_0^t (M_s + A_s) d(N_s + B_s) + \int_0^t (N_s + B_s) d(M_s + A_s) + [X, Y]_t \\ &= \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t \end{aligned}$$

Hence, we completed the proof. $\square$

Just as in the case of finite-variation stochastic calculus, where we derived the change of variables formula (Itô's formula) from the integration-by-parts formula, so here we repeat the exercise (and the proof is essentially the same!) to derive the powerful Itô's formula for continuous semimartingales.

Theorem 3.7.3. (Itô's formula)

Let $f : \mathbb{R}^d \mapsto \mathbb{R}$ be C^2 , and let $X_t = (X_t^1, \dots, X_t^d)$ be a continuous semimartingale in $\mathbb{R}^d$, (that is, each X^i is a continuous semimartingales), then:

$$f(X_t) - f(X_0) = \int_0^t \sum_i D_i f(X_s) dX_s^i + \frac{1}{2} \int_0^t \sum_{i,j} D_{i,j} f(X_s) d[X^i, X^j]_s$$

Proof. Firstly, we notice the below corollary:

Corollary 3.7.4. Under the same condition above, $f(X)$ is continuous semimartin-

gale, if $X = X_0 + M + A$ is the canonical decomposition of X , then:

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t \sum_i D_i f(X_s) dM_s^i \\ &\quad + \left\{ \int_0^t \sum_i D_i f(X_s) dA_s^i + \frac{1}{2} \int_0^t \sum_{i,j} D_{i,j} f(X_s) d[M^i, M^j]_s \right\} \end{aligned}$$

, then if we consider two functions: $F_t := f(X_t)$, $G_t := g(X_t)$, by definition:

$$\begin{aligned} [F, G]_t &= [F^{\text{cnt mart}}, G^{\text{cnt mart}}]_t \\ &= \left[\int_0^t f'(X_s) dM, \int_0^t g'(X_s) dM \right] = \int_0^t f'(X_s) g'(X_s) d[M]_s \end{aligned}$$

Now, using the I.B.P, (theorem 3.7.2):

$$\begin{aligned} F_t G_t - F_0 G_0 &= \int_0^t F_s dG_s + \int_0^t G_s dF_s + \int_0^t F'_s G'_s d[M]_s \\ &= \int_0^t \{F_s G'_s + F'_s G_s\} dX_s \\ &\quad + \frac{1}{2} \int_0^t \{F_s G''_s + 2F'_s G'_s + F''_s G_s\} d[M]_s \end{aligned}$$

This is exactly the Itô's formula for product fg , and clearly $f(x) = x$ satisfies Itô's formula, which gives the starting points of this induction, thus all polynomials hold.

Lemma 3.7.5. *Let $f \in C^2(\mathbb{R}^d)$. For every compact set $K \subset \mathbb{R}^d$, there exists a sequence of polynomials $\{p_n\}_{n \geq 1}$ such that*

$$p_n \rightarrow f, \quad D_i p_n \rightarrow D_i f, \quad D_{ij} p_n \rightarrow D_{ij} f$$

uniformly on K , for every $1 \leq i, j \leq d$.

Inspired by above lemma, define, for any each $N > 0$:

$$U_N = \inf \{t \geq 0 : |X_t| + [M]_t > N\}$$

, then $U_N \uparrow \infty$, so it suffice to prove Itô's formula holds for $\forall t \leq U_N$, note that:

$$p_n(X_t) = p_n(X_0) + \int_0^t \sum_i D_i p_n(X_s) dX_s^i + \frac{1}{2} \int_0^t \sum_{i,j} D_{i,j} p_n(X_s) d[X^i, X^j]_s$$

, for $0 \leq t \leq U_N$, by letting $n \rightarrow \infty$, we will prove the limits are:

$$f(X_t) = f(X_0) + \int_0^t \sum_i D_i f(X_s) dX_s^i + \frac{1}{2} \int_0^t \sum_{i,j} D_{i,j} f(X_s) d[X^i, X^j]_s$$

(1) The first convergence is simple, since from the lemma:

$$p_n \xrightarrow{n \rightarrow \infty} f, \forall 0 \leq t \leq U_N$$

(2) For the continuous semimartingale term, considering its decomposition:

$$\int_0^t \sum_i D_i p_n(X_s) dX_s^i = \int_0^t \sum_i D_i p_n(X_s) dM_s^i + \int_0^t \sum_i D_i p_n(X_s) dA_s^i$$

(a) For FV term, it is simple since:

$$\begin{aligned} & \left| \int_0^t \sum_i D_i p_n(X_s) dA_s^i - \int_0^t \sum_i D_i f(X_s) dA_s^i \right| \\ & \leq d \cdot \sup_i \|D_i p_n - D_i f\| \cdot A_t \xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

(b) For continuous martingale term, since we restrict $t \leq U_N$:

$$\begin{aligned} & \mathbb{E} \sup_{s \leq t} \left| \int_0^t \sum_i D_i p_n(X_s) dM_s^i - \int_0^t \sum_i D_i f(X_s) dM_s^i \right|^2 \\ & \leq 4 \mathbb{E} \int_0^t \left(\sum_i D_i p_n(X_s) - \sum_i D_i f(X_s) \right)^2 d[M]_s \\ & \leq d \cdot \sup_{1 \leq i \leq d} \|D_i p_n - D_i f\|^2 \cdot \mathbb{E}[M]_t \xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

, since $\mathbb{E}[M]_t = \mathbb{E}[M]_{t \wedge U_N} \leq N$, taking subsequence, we have a.s. convergence.

(3) Using same way to demonstrate, by noticing, under $t \leq U_N$:

$$\sum_{i,j} [X^i, X^j]_t \text{ is bounded.}$$

Hence, we completed the proof of Itô's formula. $\square$

Remark 3.7.1. We can regard theorem 3.7.3 as a Taylor series expansion where we set:

$$\begin{aligned} dX^i dX^j &:= d[X^i, X^j] \\ dX^i dX^j dX^k &:= 0 \end{aligned}$$

We shall make heavy use of this type of differential notation. Particularly important is the fact that if X is a continuous 1-dimensional semimartingale and V is a continuous 1-dimensional finite-variation process, then:

$$dX dV = 0$$

Hence, if X_1, X_2, Y_1 and Y_2 are 1-dimensional continuous semimartingales such that both $X_1 - X_2$, and $Y_1 - Y_2$ are of finite variation, then:

$$dX_1 dY_1 = dX_2 dY_2$$

If X and Y are continuous semimartingales in $\mathbb{R}$, and H, K are previsible $\mathbb{R}$ -valued processes, then:

$$d(H \cdot X) = H dX$$

, therefore:

$$d[H \cdot X, K \cdot Y] = d(H \cdot X) d(K \cdot Y) = HK dX dY = HK d[X, Y]$$

Chapter 4

Beyond Itô Formula

4.1 Lévy's Theorem

Many of the most important and useful results of probability, originally proved by very complicated methods, become elegant and simple when we use Itô's formula. We begin with what is everyone's favourite proof, the Kunita-Watanabe proof of Lévy's theorem.

Theorem 4.1.1. *Let $\{B_t^i, t \geq 0\}$ be in $c\mathcal{M}_{0,loc}$, for $i = 1, 2, \dots, n$, if $\forall i, j$, we have:*

$$[B^i, B^j]_t = \begin{cases} t, & i = j \\ 0, & i \neq j \end{cases}$$

Then, $B_t = (B_t^1, \dots, B_t^n)$ is a Brownian motion on $\mathbb{R}^n$ with distribution $\mathcal{N}(0, I_n)$.

Proof. Recall each distribution is captured entirely by its characteristic function, then we will show, $\forall s < t$, and also $\forall \theta \in \mathbb{R}^n$, the following holds:

$$\mathbb{E} [e^{i\theta(B_t - B_s)} | \mathcal{F}_s] = \exp \left[-\frac{1}{2} (t - s) |\theta|^2 \right]$$

The structure is clear, then we define a function: $f : \mathbb{R}^{n+1} \mapsto \mathbb{C}$ as:

$$f(x, t) = e^{i\theta \cdot x + \frac{1}{2} |\theta|^2 t}$$

Hence, we can apply Itô formula to:

$$f(B_t, t) := f(X_t)$$

, where $X_t := (B_t^1, \dots, B_t^n, t)$, then we expand that into:

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t \sum_{i=1}^n D_i f(X_s) dB_s^i + \int_0^t D_{n+1} f(X_s) ds \\ &\quad + \frac{1}{2} \int_0^t \sum_{1 \leq i, j \leq n} D_{i,j} f(X_s) d[B^i, B^j]_s + \frac{1}{2} \int_0^t \sum_{1 \leq k \leq n} D_{k,n+1} f(X_s) d[B^k, s]_s \end{aligned}$$

, then using the information of quadratic variation:

$$d[B^k, s]_s = 0, [B^i, B^j]_t = \begin{cases} t, & i = j \\ 0, & i \neq j \end{cases}$$

And also by the derivative of $f(x, t)$, we reduce to:

$$\begin{aligned}
f(\mathbf{B}_t, t) - f(\mathbf{B}_0, 0) &= \int_0^t \sum_i \mathbf{D}_i f(\mathbf{B}_s, s) d\mathbf{B}_s^i + \int_0^t \frac{\partial}{\partial s} f(\mathbf{B}_s, s) ds \\
&\quad + \frac{1}{2} \int_0^t \sum_{1 \leq i, j \leq n} \mathbf{D}_{i,j} f(\mathbf{B}_s) d[\mathbf{B}^i, \mathbf{B}^j]_s \\
&= c\mathcal{M}_{0, \text{loc}} + \int_0^t \left(e^{i\theta \mathbf{B}_s + \frac{1}{2}|\theta|^2 s} \cdot \frac{1}{2} |\theta|^2 \right) ds \\
&\quad + \int_0^t \frac{1}{2} \left[\sum_{j=1}^n e^{i\theta \mathbf{B}_s + \frac{1}{2}|\theta|^2 s} \cdot (i\theta_j)^2 \right] ds = c\mathcal{M}_{0, \text{loc}}
\end{aligned}$$

Then we may conclude:

$$M_t^\theta := f(\mathbf{B}_t, t) = e^{i\theta \mathbf{B}_t + \frac{1}{2}|\theta|^2 t} = 1 + c\mathcal{M}_{0, \text{loc}} \in c\mathcal{M}_{\text{loc}}$$

Besides, we know: $|M_t^\theta| = e^{\frac{1}{2}|\theta|^2 t} < \infty$, by dominated convergence theorem:

$$\mathbb{E}(M_t^\theta \mid \mathcal{F}_s) = M_s^\theta \implies \mathbb{E}[e^{i\theta(\mathbf{B}_t - \mathbf{B}_s)} \mid \mathcal{F}_s] = e^{-\frac{1}{2}|\theta|^2(t-s)}$$

Then it tells us, $\mathbf{B}_t - \mathbf{B}_s \sim \mathcal{N}(0, (t-s)I_n)$ and independent of $\mathcal{F}_s$. □

Remark 4.1.1. Lévy's theorem is the fundamental martingale characterization result, and thus it can be seen as motivation for the martingale-problem approach to diffusion theory.

4.2 $c\mathcal{M}_{0, \text{loc}}$ as time-changes of BM

A striking application of Lévy's theorem shows that, modulo time-transformations, Brownian motion is the most general continuous local martingale. More specifically, let $M \in c\mathcal{M}_{0, \text{loc}}$, and regard $[M]_t$ as the intrinsic clock carried by M . Then M has delusions of grandeur: it thinks it is a Brownian motion!

Theorem 4.2.1. (*Dubins and Schwarz*)

Let $M \in c\mathcal{M}_{0, \text{loc}}$, such that: $[M]_t \uparrow \infty$ as $t \rightarrow \infty$, then, $\forall t \geq 0$:

$$M_t = \mathbf{B}_{[M]_t}$$

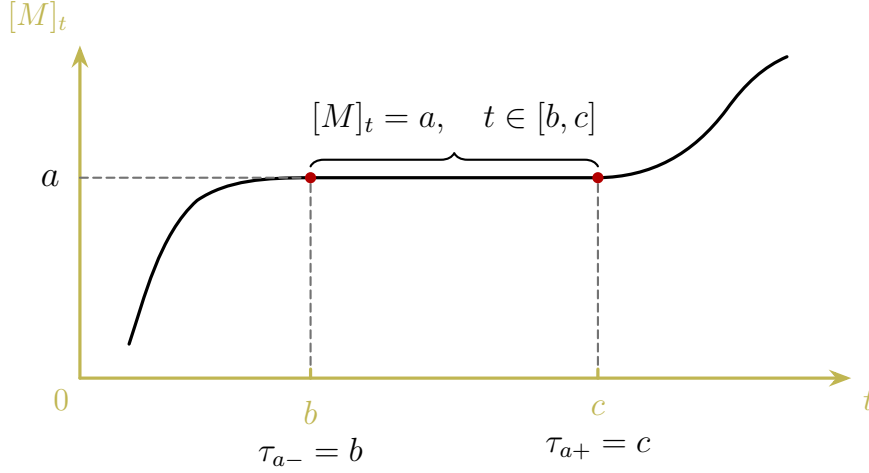
, where $\mathbf{B}$ is some B.M. More general, define stopping time:

$$\tau_t = \inf \{u \geq 0 : [M]_u > t\}$$

Then, $\mathbf{B}_t := M_{\tau_t}$ is a B.M w.r.t $\mathcal{G}_t := \mathcal{F}_{\tau_t}$.

Proof. (1) Firstly, for adaptivity, we know M is right-continuous, then:

$$M_{\tau_t} \in \mathcal{F}_{\tau_t} \implies \mathbf{B}_t := M_{\tau_t} \in \mathcal{F}_{\tau_t} = \mathcal{G}_t$$



- (2) Next, we want to show: $t \mapsto B_t$ defined above is continuous, as we can see from graph above, it suffices to show, $\forall q \in \mathbb{Q}^+$, define:

$$S_q = \inf \left\{ t > q : [M]_t > [M]_q \right\}$$

that $M_t = M_q, \forall t \in [q, S_q)$, note that $S_q \geq q$, then by OST:

$$\mathbb{E} \left[M_{S_q}^2 - [M]_{S_q} \mid \mathcal{F}_q \right] = M_q^2 - [M]_q$$

Since, on this interval, the constancy of $[M]$, i.e. $[M]_{S_q} = [M]_q$:

$$0 = \mathbb{E} \left[M_{S_q}^2 - M_q^2 \mid \mathcal{F}_q \right] = \mathbb{E} \left[M_{S_q}^2 - 2M_{S_q}M_q + M_q^2 \mid \mathcal{F}_q \right] = \mathbb{E} \left[(M_{S_q} - M_q)^2 \mid \mathcal{F}_q \right]$$

Then taking another expectation on both side gives us equivalence in L^2 :

$$\mathbb{E} \left\{ \mathbb{E} \left[(M_{S_q} - M_q)^2 \mid \mathcal{F}_q \right] \right\} = \mathbb{E} (M_{S_q} - M_q)^2 = 0 \implies M_{S_q} = M_q$$

- (3) Lastly, we want to apply Lévy's theorem, then it remains to show:

$$B_t \text{ and } B_t^2 - t \in c\mathcal{M}_{0,\text{loc}} \implies [B]_t = t \implies B \text{ is B.M.}$$

Here, we define $T_n = \inf \{ t \geq 0 : |M_t| > n \}$, and $U_n = [M]_{T_n}$, it is easy to show:

$$\tau_{t \wedge U_n} = T_n \wedge \tau_t$$

, (verify by yourself) then, we denote following:

$$B_{t \wedge U_n} = B_t^{U_n} = M_{\tau_{t \wedge U_n}} = M_{T_n \wedge \tau_t} = M_{\tau_t}^{T_n}$$

And by OST, we still have:

$$\mathbb{E} [B_t^{U_n} \mid \mathcal{F}_{\tau_s}] = \mathbb{E} [M_{\tau_t}^{T_n} \mid \mathcal{F}_{\tau_s}] = M_{\tau_s}^{T_n} = B_s^{U_n}$$

, hence in order to show $B \in c\mathcal{M}_{0,\text{loc}}$, it remains to show U_n is $\{\mathcal{G}_t\}$ -s.t.:

$$\{U_n \leq t\} = \{[M]_{T_n} \leq t\} = \{[M]_{T_n} \leq [M]_{\tau_t}\} = \{T_n \leq \tau_t\} \in \mathcal{F}_{\tau_t} = \mathcal{G}_t$$

Finally, notice:

$$[M]_{\tau_t \wedge T_n} = [M]_{\tau_t \wedge U_n} = t \wedge U_n$$

, then using the fact that $M \in c\mathcal{M}_{0,\text{loc}}$:

$$\begin{aligned} \mathbb{E} \left[(\mathbb{B}_t^{U_n})^2 - t \wedge U_n \mid \mathcal{G}_s \right] &= \mathbb{E} \left[(M_{\tau_t}^{T_n})^2 - [M]_{\tau_t}^{T_n} \mid \mathcal{F}_{\tau_s} \right] \\ &= (M_{\tau_s}^{T_n})^2 - [M]_{\tau_s}^{T_n} = (\mathbb{B}_s^{U_n})^2 - s \wedge U_n \end{aligned}$$

Therefore, $(\mathbb{B}_t^2 - t)^{U_n}$ is a martingale.

For special case, we notice that:

$$M_t = M_{\tau_{[M]_t}} = \mathbb{B}_{[M]_t}$$

, hence we completed the proof. $\square$

From the condition, we require that $[M]_\infty = \infty$, but what if not? In this case, $[M]_\infty < \infty$:

$$\tau_t = \infty, \forall t \geq [M]_\infty$$

, then it is to say, $\mathbb{B}_t = M_\infty, \forall t \geq [M]_\infty$, which is clearly not a B.M, so how to fix it? The natural idea is that, we may piece together another B.M. at the end, such that:

$$\mathbb{B}_t = \begin{cases} M_{\tau_t}, & t < [M]_\infty \\ M_\infty + W_{(t-[M]_\infty)}, & t \geq [M]_\infty \end{cases}$$

, where W is a s.B.M. independent of M . But whether the limit exists?

Lemma 4.2.2. *On $\{[M]_\infty < \infty\}$, the limit:*

$$M_\infty := \lim_{t \uparrow \infty} M_t \exists \text{ a.s.}$$

Proof. Define:

$$S_k = \inf \{t \geq 0 : [M]_t > k\}$$

, then on $\{[M]_\infty < \infty\}$,

$$S_k = \infty, \forall k > [M]_\infty$$

Since, M^{S_k} is a L^2 -bounded martingale, then:

$$\lim_{t \uparrow \infty} M_t^{S_k} \exists \text{ a.s.}$$

Now, we send $k \uparrow \infty$, to get our result. $\square$

Lastly, by enlargement of the probability space $(\Omega, \mathcal{F}, \mathbb{P})$:

$$\begin{aligned} \tilde{\Omega} &= \Omega \otimes \Omega^* \\ \tilde{\mathcal{F}} &= \mathcal{F} \otimes \mathcal{F}^* \\ \tilde{\mathbb{P}} &= \mathbb{P} \otimes \mathbb{P}^* \end{aligned}$$

, where on $(\Omega^*, \mathcal{F}^*, \mathbb{P}^*)$, we have W is B.M. independent of M , and on this enriched probability space, we fixed with the general case, to conclude:

Theorem 4.2.3. Let $M \in c\mathcal{M}_{0,loc}$, then $\exists B.M., B$, s.t.:

$$M_t = B_{[M]_t}, \forall t \geq 0$$

Proof. Follow above discussion. □

4.3 Exponential semimartingales

The next application of Itô's formula is of central importance because it yields a rare example of a stochastic differential equation with an explicit solution, we can discuss it outside the framework of the general theory of stochastic differential equations.

Theorem 4.3.1. (Doléans's Theorem)

Let X be a continuous semimartingale with $X_0 = 0$. Suppose $Z_0 \in \mathcal{F}_0$, then there exists a unique continuous semimartingale Z s.t.:

$$Z_t = Z_0 + \int_0^t Z_s dX_s$$

, the solution is:

$$Z_t = Z_0 \exp \left(X_t - \frac{1}{2} [X]_t \right)$$

Remark 4.3.1. Before we move into the proof, we need to clarify that even though we have not touched SDE theory, but we could be motivated to see why it has such form:

$$dZ_t = Z_t dX_t \approx dy = y dx \implies y = ce^x$$

, and in order to proceed, we may make following notation:

$$\mathcal{E}(X)_t := \exp \left(X_t - \frac{1}{2} [X]_t \right)$$

, is called the exponential semimartingale of X .

Proof. Considering its exponential semimartingale as:

$$\mathcal{E}(X)_t = \exp \left(X_t - \frac{1}{2} [X]_t \right) = f(X_t, [X]_t)$$

Then we can apply Itô's formula to $f(x, y) = e^{x - \frac{1}{2}y}$:

$$\begin{aligned} d\mathcal{E}(X)_t &= df(X_t, [X]_t) = \partial_X f(X_t, [X]_t) dX_t + \partial_{[X]} f(X_t, [X]_t) d[X]_t + \frac{1}{2} \partial_{XX} f(X_t, [X]_t) d[X]_t \\ &= \mathcal{E}(X)_t dX_t + \left(-\frac{1}{2} \right) \mathcal{E}(X)_t d[X]_t + \frac{1}{2} \mathcal{E}(X)_t d[X]_t = \mathcal{E}(X)_t dX_t \end{aligned}$$

Hence, $\mathcal{E}(X)_t$ is a solution to:

$$dZ_t = Z_t dX_t$$

Next is to show the uniqueness, assume Z is another solution, we will show:

$$Z_t = Z_0 \mathcal{E}(X)_t$$

Since we have have: $dZ_t = Z_t dX_t$, then we define:

$$Y_t = \exp \left(-X_t + \frac{1}{2} [X]_t \right) = (\mathcal{E}(X)_t)^{-1}$$

It suffices to prove,

$$d(Z_t Y_t) = 0 \implies Z_t Y_t = Z_0 Y_0 = Z_0$$

, which is simple to show by noticing:

$$dY_t = (-1) Y_t dX_t + \frac{1}{2} Y_t d[X]_t + \frac{1}{2} (-1)^2 Y_t d[X]_t = -Y_t dX_t + Y_t d[X]_t$$

Lastly, since I.B.P:

$$\begin{aligned} d(Z_t Y_t) &= Z_t dY_t + Y_t dZ_t + d[Y_t, Z_t] \\ &= Z_t (-Y_t dX_t + Y_t d[X]_t) + Y_t (Z_t dX_t) + (-Y_t Z_t) d[X]_t = 0 \end{aligned}$$

Then we completed the proof. □

Remark 4.3.2. The last step, we used the result from homework:

$$d[H \cdot X, K \cdot Y]_t = HK d[X, Y]_t$$

In general, let $M \in c\mathcal{M}_{0, \text{loc}}$, then:

$$\mathcal{E}(\theta M)_t = \exp \left(\theta M_t - \frac{1}{2} \theta^2 [M]_t \right) \in c\mathcal{M}_{\text{loc}}$$

Moreover, by Fatou's Lemma, we can see it is also a supermartingale:

$$\begin{aligned} \mathbb{E}[\mathcal{E}(\theta M)_t \mid \mathcal{F}_s] &= \mathbb{E} \left[\lim_{n \rightarrow \infty} \mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s \right] \\ &\leq \liminf_{n \rightarrow \infty} \mathbb{E} \left[\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s \right] = \liminf_{n \rightarrow \infty} \mathcal{E}(\theta M)_s^{T_n} = \mathcal{E}(\theta M)_s \end{aligned}$$

, with bounded expectation: $\mathbb{E} \mathcal{E}(M)_t \leq 1$. For more information, we want to estimate it:

Theorem 4.3.2. *If for some $t > 0$, $\exists K_t > 0$, s.t.:*

$$[M]_t < K_t \text{ a.s.}$$

, then for this $t > 0$, and $\forall y \geq 0$:

$$\mathbb{P} \left(\max_{s \leq t} M_s > y \right) \leq \exp \left(-\frac{y^2}{2K_t} \right)$$

Proof. By using the fact $\mathcal{E}(\theta M)$ is supermartingale, and Doob's inequality:

$$\begin{aligned} \mathbb{P}\left(\max_{s \leq t} M_s > y\right) &= \mathbb{P}\left(\max_{s \leq t} \mathcal{E}(\theta M)_s > \exp\left(\theta y - \frac{1}{2}\theta^2 K_t\right)\right) \\ &\leq \frac{\mathbb{E}\mathcal{E}(\theta M)_t}{\exp\left(\theta y - \frac{1}{2}\theta^2 K_t\right)} \leq \exp\left(-\theta y + \frac{1}{2}\theta^2 K_t\right) \leq \exp\left(-\frac{y^2}{2K_t}\right) \end{aligned}$$

The last step is to pick θ so that we minimise the exponent. □

And if we restrict the condition, we may gain more powerful result:

Theorem 4.3.3. Suppose $\forall t > 0, \exists K_t > 0$, s.t.:

$$[M]_t < K_t \text{ a.s.}$$

, then $\mathcal{E}(\theta M)$ is a true martingale.

Proof. Firstly, we define:

$$Z_t = \max_{s \leq t} M_s$$

, and by theorem 4.3.2, notice that, $\forall \theta > 0$:

$$\begin{aligned} \mathbb{E}e^{\theta Z_t} &= 1 + \mathbb{E} \int_0^{Z_t} \theta e^{\theta y} dy = 1 + \mathbb{E} \int_0^\infty \mathbf{1}_{\{Z_t > y\}} \theta e^{\theta y} dy \\ &= 1 + \int_0^\infty \theta e^{\theta y} \mathbb{P}(Z_t > y) dy \leq 1 + \int_0^\infty \theta e^{\theta y} e^{-\frac{y^2}{2K_t}} dy < \infty \end{aligned}$$

Then using the fact $\mathcal{E}(\theta M)_t \in c\mathcal{M}_{\text{loc}}$, we get $\exists T_n \uparrow \infty$, s.t., $\forall s < t$:

$$\mathbb{E}\left[\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s\right] = \mathcal{E}(\theta M)_s^{T_n}$$

Lastly, since $\mathcal{E}(\theta M)_t^{T_n} \leq e^{\theta Z_t}$ and $\mathbb{E}e^{\theta Z_t} < \infty$, using B.C.T we completed the proof. □

As natural extension, we can check the special case on B.M.

Corollary 4.3.4. Let B be B.M. in $\mathbb{R}^n$, let H be bounded previsible process, then:

$$\exp\left(\int_0^t H dB - \frac{1}{2} \int_0^t |H_s|^2 ds\right)$$

, is a true martingale.

Proof. Directly follow from theorem 4.3.3. □

Remark 4.3.3. If we take, $M = \int_0^t H dB$, then:

$$[M]_t = \int_0^t |H_s|^2 ds < K_t$$

Corollary 4.3.5. Let $M \in c\mathcal{M}_{0,loc}$, then $\forall t > 0, y > 0, K > 0$:

$$\mathbb{P} \left(\max_{s \leq t} M_s \geq y, [M]_t \leq K \right) \leq \exp \left(-\frac{y^2}{2K} \right)$$

This also implies:

$$\mathbb{P} \left(\max_{s \leq t} |M_s| \geq y, [M]_t \leq K \right) \leq 2 \exp \left(-\frac{y^2}{2K} \right)$$

Proof. Let $T = \inf \{u : [M]_u \geq K\}$, then:

$$\mathbb{P} \left(\max_{s \leq t} M_s \geq y, [M]_t \leq K \right) \leq \mathbb{P} \left(\max_{s \leq t} M_s^T \geq y \right) \leq \exp \left(-\frac{y^2}{2K} \right)$$

The last step it apply theorem 4.3.2 to M^T . □

Lastly, we enclose this section by another theorem without proof:

Theorem 4.3.6. (Novikov's Theorem)

Let $M \in c\mathcal{M}_{loc}$, if $\forall t > 0, \mathbb{E} e^{\frac{1}{2}[M]_t} < \infty$, then $\mathcal{E}(M)_t$ is true martingale.

Proof. See p.198 of KS. □

4.4 Burkholder-Davis-Gundy Inequalities

Martingale inequalities of diverse and wondrous forms have been the subject of intensive investigation, and it would be impossible to attempt a detailed coverage. But B-D-G inequality worth its own place, and firstly, we need to introduce one concept:

Definition 4.4.1. We say a function $F : \mathbb{R}^+ \mapsto \mathbb{R}^+$ is moderate if:

- (a) it is continuous,
- (b) it is increasing,
- (c) $F(x) = 0 \Leftrightarrow x = 0$,
- (d) $\forall \alpha > 1$:

$$\sup_{x>0} \frac{F(\alpha x)}{F(x)} < \infty$$

The most obvious examples of moderate F are the powers:

$$F(x) = x^p, \forall p > 0$$

, and in some of cases, we may want to use logarithm to control, then:

$$F(x) = x^p \log(x \vee 1), \forall p > 0$$

Now, we can introduce the B-D-G inequality:

Theorem 4.4.1. (Burkholder-Davis-Gundy Inequalities)

Let F be moderate, then $\exists$ universal constants, $c_F \leq C_F < \infty$, s.t., $\forall M \in c\mathcal{M}_{0,loc}$:

$$c_F \mathbb{E}F\left([M]_{\infty}^{\frac{1}{2}}\right) \leq \mathbb{E}F(M_{\infty}^*) \leq C_F \mathbb{E}F\left([M]_{\infty}^{\frac{1}{2}}\right)$$

, where $M_t^* = \sup_{s \leq t} |M_s|$. Generally, fix $t > 0$, let $\widetilde{M}_s^t = M_{s \wedge t}$, then:

$$c_F \mathbb{E}F\left([M]_t^{\frac{1}{2}}\right) \leq \mathbb{E}F(M_t^*) \leq C_F \mathbb{E}F\left([M]_t^{\frac{1}{2}}\right)$$

Remark 4.4.1. For the case $F(x) = x^p$, Gettoor and Sharpe give a nice proof using stochastic calculus. We follow Burkholder's proof which relies on the good λ inequality.

Then we postpone the proof of B-D-G inequalities, and introduce an auxiliary lemma:

Lemma 4.4.2. ("good" λ inequality)

Let X and Y be nonnegative r.v.s, suppose $\exists \beta > 1$, s.t. $\forall \lambda > 0, \delta > 0$:

$$\mathbb{P}(X > \beta\lambda, Y \leq \delta\lambda) \leq \psi(\delta) \mathbb{P}(X > \lambda)$$

, where $\psi(\delta) \downarrow 0$ as $\delta \downarrow 0$. Then for each moderate F , $\exists C = C(\beta, \psi, F) > 0$, s.t.:

$$\mathbb{E}F(X) \leq C \mathbb{E}F(Y)$$

Proof. W.l.o.g., we may assume $\mathbb{E}F(Y) < \infty$, similarly we assume the good λ inequality holds, then it also holds true, if we replace X with $X \wedge a$, $\forall a > 0$:

(1) If $a \leq \beta\lambda$, then:

$$X \wedge a \leq a \leq \beta\lambda \implies \mathbb{P}(X > \beta\lambda, Y \leq \delta\lambda) = 0$$

(2) If $a > \beta\lambda > \lambda$, then:

$$\begin{aligned} X \wedge a > \beta\lambda &\Leftrightarrow X > \beta\lambda &\implies \mathbb{P}(X \wedge a > \beta\lambda, Y \leq \delta\lambda) &= \mathbb{P}(X > \beta\lambda, Y \leq \delta\lambda) \\ X \wedge a > \lambda &\Leftrightarrow X > \lambda &\implies \mathbb{P}(X \wedge a > \lambda) &= \mathbb{P}(X > \lambda) \end{aligned}$$

Hence, we may also assume $\mathbb{E}F(X) < \infty$, since if not, we send $a \uparrow \infty$, and:

$$\mathbb{E}F(X) = \lim_{a \uparrow \infty} \mathbb{E}F(X \wedge a)$$

Now, notice the moderate property, then:

$$\sup_{x>0} \frac{F(x)}{F(x/\beta)} < \infty \implies \exists \gamma > 0, \text{ s.t. } F(x) \leq \frac{1}{\gamma} F(x/\beta), \forall x > 0$$

Now, we can integrate on both side:

$$\int F(d\lambda) \psi(\delta) \mathbb{P}(X > \lambda) \geq \int F(d\lambda) \mathbb{P}(X > \beta\lambda, Y \leq \delta\lambda)$$

Now, we may check one by one:

(a) For LHS:

$$\begin{aligned}
\int F(d\lambda) \psi(\delta) \mathbf{P}(X > \lambda) &= \psi(\delta) \int F(d\lambda) \mathbf{P}(X > \lambda) \\
&= \psi(\delta) \int F(d\lambda) \mathbf{E} \mathbf{1}_{\{X > \lambda\}} \\
&= \psi(\delta) \mathbf{E} \int F(d\lambda) \mathbf{1}_{\{X > \lambda\}} \\
&= \psi(\delta) \mathbf{E} \int_0^X F(d\lambda) = \psi(\delta) \mathbf{E} F(X)
\end{aligned}$$

(b) For RHS:

$$\begin{aligned}
\int F(d\lambda) \mathbf{P}(X > \beta\lambda, Y \leq \delta\lambda) &= \int F(d\lambda) \mathbf{E} \mathbf{1}_{\{\frac{Y}{\delta} \leq \lambda \leq \frac{X}{\beta}\}} \\
&= \mathbf{E}[F(X/\beta) - F(Y/\delta)] \geq \mathbf{E} \gamma F(X) - \mathbf{E} F(Y/\delta)
\end{aligned}$$

Now, since $\psi(\delta)$ can be arbitrarily small, then let $\delta > 0$ be small s.t.:

$$\psi(\delta) < \frac{\gamma}{2} \implies \gamma - \psi(\delta) \geq \frac{\gamma}{2} > 0$$

Lastly, we use the same trick to find a parameter for Y , i.e. using:

$$\sup_{y>0} \frac{F(y/\delta)}{F(y)} < \infty \implies \exists \mu > 0, \text{ s.t. } \mu F(y) \geq F(y/\delta), \forall y > 0$$

Hence, we can combine the two side together:

$$\mu \mathbf{E} F(Y) \geq \mathbf{E} F(Y/\delta) \geq (\gamma - \psi(\delta)) \mathbf{E} F(X) \geq \frac{\gamma}{2} \mathbf{E} F(X)$$

Then, we completed the proof, by solving out $C := 2\mu/\gamma$. □

Using lemma 4.4.2, we can proceed:

Proof. of theorem 4.4.1.

The key idea is to use lemma 4.4.2, define $\tau = \inf \{u \geq 0 : |M_u| > \lambda\}$, notice that:

$$\begin{aligned}
\mathbf{P}(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \leq \delta\lambda) &= \mathbf{P}(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \leq \delta\lambda, \tau < \infty) \\
&= \mathbf{E} \left[\mathbf{1}_{\{\tau < \infty\}} \mathbf{P}(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \leq \delta\lambda \mid \mathcal{F}_\tau) \right]
\end{aligned}$$

To better use the gambler's ruin, we define:

$$N_t = (M_{t+\tau} - M_\tau)^2 - ([M]_{t+\tau} - [M]_\tau) \in c\mathcal{M}_{0,\text{loc}}$$

Besides, on $\{M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \leq \delta\lambda\}$, we have $\exists \tilde{t} > 0$, s.t.:

$$M_{\tilde{t}+\tau} \geq \beta\lambda \implies N_{\tilde{t}} = (M_{\tilde{t}+\tau} \pm \lambda)^2 - ([M]_{\tilde{t}+\tau} - [M]_{\tilde{t}}) \geq [(\beta - 1)\lambda]^2 - (\delta\lambda)^2$$

Moreover, since $N_t \geq -\delta^2 \lambda^2, \forall t \geq 0$, hence N_t reaches $[(\beta - 1) \lambda]^2 - (\delta \lambda)^2$ before $-\delta^2 \lambda^2$:

$$\begin{aligned} \mathbf{P} \left(M_{\infty}^* > \beta \lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta \lambda \mid \mathcal{F}_{\tau} \right) &\leq \mathbf{P} \left(\tau_{[(\beta-1)\lambda]^2 - (\delta\lambda)^2} - \tau_{-\delta^2\lambda^2} \right) \\ &\leq \frac{(\delta\lambda)^2}{[(\beta-1)\lambda]^2} = \frac{\delta^2}{(\beta-1)^2} \end{aligned}$$

, here we take, $0 < \delta < \beta - 1$, therefore:

$$\mathbf{P} \left(M_{\infty}^* > \beta \lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta \lambda \right) \leq \frac{\delta^2}{(\beta-1)^2} \mathbf{P}(\tau < \infty) = \frac{\delta^2}{(\beta-1)^2} \mathbf{P}(M_{\infty}^* > \lambda)$$

For another part, we similarly define:

$$\tilde{N}_t = [M]_{t+\tilde{\tau}} - [M]_{\tilde{\tau}} - (M_{t+\tilde{\tau}} - M_{\tilde{\tau}})^2$$

, with: $\tilde{\tau} = \int \left\{ u \geq 0 : [M]_u^{\frac{1}{2}} > \lambda \right\}$. Then by lemma 4.4.2 we completed the proof. $\square$

4.5 Semimartingale local time

The applications of Itô's formula we have seen so far have been valid for continuous semimartingales with values in $\mathbb{R}^n$ for any $n \geq 1$. For 1-dimensional continuous semimartingales, we have the added great advantage of having the concept of local time. Itô's formula and local time together supply simple and beautiful existence, uniqueness, and comparison results for 1-dimensional SDEs, and the combination of local time and Itô excursion theory can be applied very effectively to 1-dimensional diffusions.

For a B.M. $\{B_t, t \geq 0\}$, we consider its local time at 0 as:

$$\ell_t^0 = \int_0^t \delta_0(B_s) ds$$

, where δ_0 is the Dirac delta function, s.t., $\forall$ function f :

$$\int f(x) \delta_0(x) dx = f(0)$$

, which is actually a distribution, here we may imagine a r.v. ξ , s.t. $\mathbf{P}(\xi = 0) = 1$, then:

$$\mathbf{E}f(\xi) = \int f(x) \delta_0(x) dx = f(0)$$

Since, clearly, this is increasing, then it can induce a measure as:

$$\mu([a, b)) = \ell_b^0 - \ell_a^0$$

, and its support is, for sure, that:

$$\text{Supp}(\mu) = \{t \geq 0 : B_t = 0\}$$

And generally, we call the **level set**: $\{t \geq 0 : B_t = x\}$, and its local time:

$$\ell_t^x = \int_0^t \delta_x(B_s) ds$$

To be rigourous about this concept, we propose the Tanaka's formula:

Theorem 4.5.1. (Tanaka's formula)

Let X be a continuous semimartingale, then $\exists$ continuous increasing adapted process, $\{\ell_t, t \geq 0\}$, such that:

$$|X_t| - |X_0| = \int_0^t \text{sgn}(X_s) dX_s + \ell_t$$

, where $\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}$, then ℓ is called the **local time** of X at 0, and:

$$\int_0^t \mathbf{1}_{\{X_s \neq 0\}} d\ell_s = 0$$

Remark 4.5.1. We postpone the poof, and check some interesting points:

(a) Intuitively, consider the function, $f(x) = |x|$, then basic calculus gives us:

$$f'(x) = \text{sgn}(x)$$

, at $x \neq 0$, and if we are able to do another derivative, then:

$$f''(x) \stackrel{?}{=} 2\delta_0(x)$$

since this is not legal in general, then we call it, **distributional derivative**:

Definition 4.5.1. We say $\frac{d}{dx}T(t)$ is the **distributional derivative** of $T(x)$, if $\forall$ smooth function, $\phi(x)$, with $\lim_{|x| \rightarrow \infty} \phi(x) = 0$, then:

$$\int \left(\frac{d}{dx}T(x) \right) \phi(x) dx = - \int \phi'(x) T(x) dx$$

It is easy to see, it is the same as I.B.P, with the test function, ϕ , be null at the boundary. Then we also can see why the distribution derivative of $\text{sgn}(x)$:

$$\begin{aligned} 2\phi(0) &= \int_{\mathbb{R}} 2\delta_0(x) \phi(x) dx = - \int_{\mathbb{R}} \phi'(x) \text{sgn}(x) dx \\ &= - \int_0^{\infty} \phi'(x) dx + \int_{-\infty}^0 \phi'(x) dx = 2\phi(0) \end{aligned}$$

(b) If we do a simple transformation, we recover the "level set" process:

$$|X_t - x| - |X_0 - x| = \int_0^t \text{sgn}(X_s - x) dX_s + \ell_t^x$$

(c) Another important observation is that:

$$\begin{aligned} |X_t| &= X_t^+ + X_t^- \\ X_t &= X_t^+ - X_t^- \end{aligned} \implies 2X_t^+ = |X_t| + X_t$$

Then, we may use theorem 4.5.1 to expand $|X_t|$:

$$2X_t^+ = |X_t| + X_t = |X_0| + \underbrace{\int_0^t \mathbf{1}_{\{X_s > 0\}} dX_s - \int_0^t \mathbf{1}_{\{X_s \leq 0\}} dX_s}_{|X_t|} + \underbrace{\ell_t + \int_0^t \mathbf{1} \cdot dX_s}_{X_t} + X_0$$

Then if we rearrange the terms, we gain the expansion of X_t^+ :

$$X_t^+ - X_0^+ = \int_0^t \mathbf{1}_{\{X_s > 0\}} dX_s + \frac{1}{2} \ell_t$$

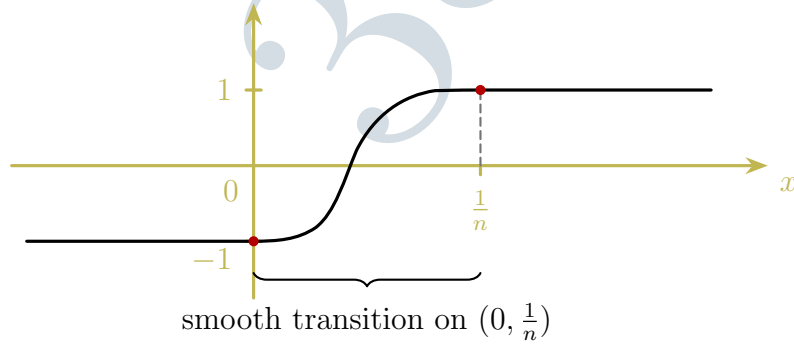
Similarly, by $2X_t^- = |X_t| - X_t$, we also have:

$$X_t^- - X_0^- = - \int_0^t \mathbf{1}_{\{X_s \leq 0\}} dX_s + \frac{1}{2} \ell_t$$

Proof. Motivated by the observation that, if we can apply Itô's formula to $f(x) = |x|$:

$$\begin{aligned} |X_t| &= |X_0| + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s \\ &= |X_0| + \int_0^t \operatorname{sgn}(X_s) dX_s + \int_0^t \delta_0(X_s) d[X]_s \end{aligned}$$

And for B.M. $d[X]_s$ becomes ds , and above gives more general definition of local time, but the issue is we cannot do so, as the absolute function is not second differentiable. Then our idea is to find a series of function, whose properties is good enough, so that we can interchange the limit and integral. Starting from $\operatorname{sgn}(\cdot)$, considering:



$$\rho(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases} \implies \varphi_n(x) = \begin{cases} -1, & x \leq 0 \\ -1 + 2 \frac{\rho(x)}{\rho(x) + \rho(1/n - x)}, & 0 < x < 1/n \\ 1, & x \geq 1/n \end{cases}$$

Clearly, $\varphi_n \in C^\infty(\mathbb{R})$ and $\operatorname{sgn}(x) = \lim_{n \rightarrow \infty} \varphi_n(x)$, and then we could define:

$$f_n(x) = \begin{cases} -x, & x \leq 0 \\ \int_0^x \varphi_n(t) dt, & x > 0 \end{cases} \implies f'_n(x) = \varphi_n(x), |x| = \lim_{n \rightarrow \infty} f_n(x)$$

Besides, it is also uniformly bounded, since:

$$|f(x) - f_n(x)| = \left| \int_0^{|x|} [1 - \varphi_n(t)] dt \right| \leq \int_0^{|x|} |1 - \varphi_n(t)| dt \leq \int_0^{1/n} |1 - \varphi_n(t)| dt \leq 2 \cdot \frac{1}{n}$$

Now we shall apply the Itô formula to f_n to gain:

$$f_n(X_t) = f_n(X_0) + \int_0^t f'_n(X_s) dX_s + \frac{1}{2} \int_0^t f''_n(X_s) d[X]_s$$

Here, we know from the construction that:

$$f(X_0) = \lim_{n \rightarrow \infty} f_n(X_0), \text{ and } f(X_t) = \lim_{n \rightarrow \infty} f_n(X_t)$$

Then it suffice to show the convergence of first derivative term, since if it holds:

$$\frac{1}{2} \int_0^t f''_n(X_s) d[X]_s =: C_t^n \xrightarrow{\text{a.s.}} \ell_t := |X_t| - |X_0| - \int_0^t \text{sgn}(X_s) dX_s$$

, as the other terms already converged, and for first derivative term, notice:

$$\int_0^t f'_n(X_s) dX_s = \int_0^t f'_n(X_s) dM_s + \int_0^t f'_n(X_s) dA_s$$

Then we shall consider them one by one:

(1) For the FV term, notice that

$$\begin{aligned} \left| \int_0^t [f'(X_s) - f'_n(X_s)] dA_s \right| &= \left| \int_0^t [\text{sgn}(X_s) - \varphi_n(X_s)] dA_s \right| \\ &\leq \int_0^t |\text{sgn}(X_s) - \varphi_n(X_s)| \cdot |dA_s| \end{aligned}$$

Then since, $\varphi_n(x) \uparrow \text{sgn}(x)$, $\forall x$, by D.C.T:

$$\int_0^t \varphi_n(X_s) dA_s \xrightarrow{\text{a.s.}} \int_0^t \text{sgn}(X_s) dA_s$$

, uniformly in t .

(2) For martingale term, we need to bound it first, define:

$$T_k = \inf \{t \geq 0 : |M_t| \geq k\} \implies M^{T_k} \text{ is bounded}$$

Then we may assume M is bounded, hence:

$$\left\| \int_0^\infty [\text{sgn}(X_s) - \varphi_n(X_s)] dM_s \right\|_2^2 = \mathbb{E} \int_0^\infty [\text{sgn}(X_s) - \varphi_n(X_s)]^2 d[M]_s \longrightarrow 0$$

Then by Doob's inequality, we also have:

$$\mathbb{E} \sup_{t \leq T} \left| \int_0^\infty [\text{sgn}(X_s) - \varphi_n(X_s)] dM_s \right|^2 \longrightarrow 0$$

Hence, from L^2 convergence, we may have a subsequence so that:

$$\sup_{t \geq 0} \int_0^t [\text{sgn}(X_s) - \varphi_n(X_s)] dM_s \xrightarrow{\text{a.s.}} 0$$

Lastly, we remain to show the support, since we see C_t^n as distribution function, then

$$dC_s^n = \frac{1}{2} f_n''(X_s) d[X]_s$$

Therefore, $\forall \gamma > 0$, if $1/n < \gamma$, then:

$$|X_s| > \gamma > \frac{1}{n} \implies f_n''(X_s) = 0 \implies \int_0^t \mathbf{1}_{\{|X_s| > \gamma\}} dC_s^n = 0$$

Hence, we split into smaller interval and send it to infinity, i.e.:

$$\int_0^t \mathbf{1}_{\{|X_s| > \gamma\}} d\ell_s \leq \lim_{n \rightarrow \infty} \int_0^t \mathbf{1}_{\{|X_s| > \gamma/2\}} dC_s^n = 0$$

Then this completes the proof since γ can be arbitrary. $\square$

4.6 Stratonovich Calculus

All above we are discussing is Itô calculus, it is powerful, and there are lots of results derived from it, the last section aims to remind reader that there are not only one way to define the stochastic integral, and the key is how the point is picked.

Definition 4.6.1. Let X and Y be continuous semimartingale, then the Stratonovich integral is defined by:

$$\int_0^t Y_s \partial X_s := \int_0^t Y_s dX_s + \frac{1}{2} [X, Y]_t$$

Remark 4.6.1. Throughout this section, we always use:

d to denote the Itô differential

∂ to denote the Stratonovich differential

The intuition of why we define like this is that, recall:

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t \\ &= \underbrace{\int_0^t X_s dY_s + \frac{1}{2} [X, Y]_t}_{\int_0^t X_s \partial Y_s} + \underbrace{\int_0^t Y_s dX_s + \frac{1}{2} [X, Y]_t}_{\int_0^t Y_s \partial X_s} \end{aligned}$$

Then since the symmetry of cross variation, we may give each term one half cross variation to make the integral "clean", benefiting from that, the differential form is also clean:

Theorem 4.6.1. (Stratonovich formula)

Let $f \in C^3(\mathbb{R})$, then:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) \partial X_s$$

Proof. By Itô formula:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s$$

From the definition of Stratonovich integral:

$$\int_0^t f'(X_s) \partial X_s = \int_0^t f'(X_s) dX_s + \frac{1}{2} [f'(X), X]_t$$

And to see the cross variation of f' , we apply Itô formula to f' :

$$df'(X_t) = f''(X_t) dX_t + \frac{1}{2} f'''(X_t) d[X]_t \implies [f'(X), X]_t = \int_0^t f''(X_s) d[X]_s$$

Therefore, we recovered the term and also completed the proof. $\square$

From another point of view, the Riemann-Sum approximation, we could see the difference more intuitive, it is how to choose the point in the integrand:

Lemma 4.6.2. *Let c be a locally bounded adapted L -process, and X be a continuous semimartingale. Then, with $t_i^n = t \wedge \frac{i}{2^n}$:*

$$I^n(t) := \sum_{i=1}^{\infty} c(t_{i-1}^n) (X_{t_i^n} - X_{t_{i-1}^n}) \longrightarrow \int_0^t c(s) dX_s =: I(t)$$

, the convergence is uniform in probability on compacts, i.e. $\forall K > 0$,

$$\sup_{t \leq K} |I^n(t) - I(t)| \xrightarrow{P} 0$$

Proof. Let $c^n(t) = c(t_{i-1}^n)$ on $(t_{i-1}^n, t_i^n]$, then $\forall t, \omega$:

$$c^n(t, \omega) \longrightarrow c(t, \omega)$$

Also,

$$I^n(t) = \int_0^t c^n(s) dX_s$$

Consider the canonical decomposition, $X_t = X_0 + M_t + A_t$, for each ω :

$$\int_0^t c^n(s) dA_s \longrightarrow \int_0^t c(s) dA_s$$

, which is guaranteed by classical Lebesgue integration, it remains to show:

$$\int_0^t c^n(s) dM_s \longrightarrow \int_0^t c(s) dM_s$$

, which is equivalent: c is bounded, and $M \in \mathcal{M}_0^2$. Then we completed the proof. $\square$

Lemma 4.6.3. *Let X, Y be two continuous semimartingale, define:*

$$S^n(t) = \sum_{i=1}^{\infty} \frac{1}{2} \left(Y_{t_i^n} + Y_{t_{i-1}^n} \right) \cdot \left(X_{t_i^n} - X_{t_{i-1}^n} \right)$$

, and $S(t) = \int_0^t Y_s \partial X_s$, then $\forall K > 0$:

$$\sup_{t \leq K} |S^n(t) - S(t)| \xrightarrow{p} 0$$

Proof. Notice the following rearrangement:

$$\begin{aligned} S^n(t) &= \sum_{i=1}^{\infty} \frac{1}{2} \left(Y_{t_i^n} + Y_{t_{i-1}^n} \right) \cdot \left(X_{t_i^n} - X_{t_{i-1}^n} \right) \\ &= \sum_{i=1}^{\infty} \left[Y_{t_{i-1}^n} + \frac{1}{2} \left(Y_{t_i^n} - Y_{t_{i-1}^n} \right) \right] \cdot \left(X_{t_i^n} - X_{t_{i-1}^n} \right) \\ &= \sum_{i=1}^{\infty} Y_{t_{i-1}^n} \cdot \left(X_{t_i^n} - X_{t_{i-1}^n} \right) + \frac{1}{2} \sum_{i=1}^{\infty} \Delta Y_{t_i^n} \Delta X_{t_i^n} \end{aligned}$$

, where $\Delta Y_{t_i^n} = Y_{t_i^n} - Y_{t_{i-1}^n}$, and $\Delta X_{t_i^n} = X_{t_i^n} - X_{t_{i-1}^n}$. It remains to show:

$$\sum_{i=1}^{\infty} \Delta Y_{t_i^n} \Delta X_{t_i^n} \longrightarrow [X, Y]_t = \frac{1}{4} ([X + Y]_t - [X - Y]_t)$$

Then this is equivalent to show:

$$\sum_{i=1}^{\infty} (\Delta X_{t_i^n})^2 = \sum_{i=1}^{\infty} (\Delta M_{t_i^n} + \Delta A_{t_i^n})^2 \longrightarrow [X]_t = [M]_t$$

Other left as exercise for reader. □

The two lemmas just given are clearly relevant to the problem of simulating processes described by stochastic differential equations, which we will see later.

Chapter 5

Stochastic Differential Equations

After all the above setup, we may touch the theory of SDEs,

$$dX_t = \underbrace{b(t, X_t)}_{\text{drift}} dt + \underbrace{\sigma(t, X_t)}_{\text{diffusion}} dW_t \quad (5.1)$$

This is convenient way to put, but rigorously, what above means is:

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$$

In higher dimension, say $\mathbb{R}^d$, Eq. (5.1), should be read as:

$$d \begin{bmatrix} X_t^{(1)} \\ \vdots \\ X_t^{(d)} \end{bmatrix}_{d \times 1} = \begin{bmatrix} b_1 \\ \vdots \\ b_d \end{bmatrix}_{d \times 1} (t, X_t) dt + \underbrace{\begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1r} \\ \vdots & \ddots & \vdots \\ \sigma_{d1} & \cdots & \sigma_{dr} \end{bmatrix}_{d \times r}}_{\text{dispersion matrix}} \begin{bmatrix} dW_t^{(1)} \\ \vdots \\ dW_t^{(r)} \end{bmatrix}_{r \times 1}$$

Moreover, we call $[a_{ij}]$ the diffusion matrix with:

$$a_{ij} = [\sigma \cdot \sigma^\top]_{ij} = \sum_{k=1}^r \sigma_{ik} \sigma_{kj}^\top = \sum_{k=1}^r \sigma_{ik} \sigma_{jk}$$

Then it has appealing property since:

$$d[X^{(i)}, X^{(j)}]_t = a_{ij} dt$$

5.1 Strong Solutions

A natural question would be, what is the solution?

Definition 5.1.1. A **strong solution** of the SDE, on a given probability space, $(\Omega, \mathcal{F}, \mathbb{P})$ and a given, B.M., $\{W_t, t \geq 0\}$ and its natural filtration, $\{\mathcal{F}_t^W, t \geq 0\}$, with initial condition, ξ , is a process $\{X_t, t \geq 0\}$ with continuous path and:

- (a) $X_t \in \mathcal{F}_t$ with $\{\mathcal{F}_t, t \geq 0\}$ being the **augmented filtration** of the B.M.'s filtration, $\{\mathcal{F}_t^W, t \geq 0\}$,

(b) $\mathbb{P}(X_0 = \xi) = 1$,

(c) For all i, j, t :

$$\mathbb{P}\left(\int_0^t |b_i(s, X_s)| + \sigma_{ij}^2(s, X_s) ds < \infty\right) = 1$$

(d) The process, $\{X_t, t \geq 0\}$ follows:

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$$

As usual, after we defined what is solution, we are curious about the uniqueness:

Definition 5.1.2. Given $b(t, x)$ and $\sigma(t, x)$. Suppose $\forall$ B.M. $\{W_t, t \geq 0\}$ on some $(\Omega, \mathcal{F}, \mathbb{P})$ and ξ is the initial condition, and $\{\mathcal{F}_t, t \geq 0\}$ is the augmented filtration, then we say a solution has **strong uniqueness**, if X and $\tilde{X}$ are two strong solutions w.r.t. the same W , then:

$$\mathbb{P}(X_t = \tilde{X}_t, \forall t \geq 0) = 1$$

And easy example would be:

$$dX_t = dW_t \implies X_t = \xi_0 + W_t$$

An somewhat complicated one would be:

Example 5.1.1. Consider following SDE, the strong uniqueness holds:

$$dX_t = b(t, X_t) dt + dW_t$$

, where $b : [0, \infty) \times \mathbb{R} \mapsto \mathbb{R}$ is bounded Borel-measurable, and $\forall x \leq y$, we have:

$$b(t, x) \geq b(t, y), \forall t \geq 0$$

Proof. $\forall$ B.M. on $(\Omega, \mathcal{F}, \mathbb{P})$, $\forall \xi$, if $X^{(1)}$ and $X^{(2)}$ are two strong solutions, then:

$$X_t^{(i)} = \xi + \int_0^t b(s, X_s^{(i)}) ds + W_t, \forall i = 1, 2$$

, then since they are on the same B.M., hence denote $\Delta_t = X_t^{(1)} - X_t^{(2)}$:

$$d\Delta_t = dX_t^{(1)} - dX_t^{(2)} = [b(t, X_t^{(1)}) - b(t, X_t^{(2)})] dt$$

Since we want to show $\Delta_t = 0$, we may apply Itô formula to Δ_t^2 :

$$\begin{aligned} \Delta_t^2 &= 2 \int_0^t \Delta_s d\Delta_s + \frac{1}{2} \int_0^t 2d[\Delta]_t \\ &= 2 \int_0^t (X_s^{(1)} - X_s^{(2)}) [b(s, X_s^{(1)}) - b(s, X_s^{(2)})] ds \leq 0 \end{aligned}$$

The last step from the monotonicity, hence we have:

$$\mathbb{P}(\Delta_t = 0, \forall t \geq 0) = 1$$

Thus, we reached the strong uniqueness. $\square$

However, above is a well-designed example, then follow-up question would be a general theory for uniqueness, before that, we shall introduce a lemma:

Lemma 5.1.1. (Gronwall's Inequality)

Suppose a continuous function $g(t)$ satisfies:

$$0 \leq g(t) \leq \alpha(t) + \beta \int_0^t g(s) ds$$

, for all $t \in [0, T]$ with $\beta \geq 0$ and $\alpha : [0, T] \mapsto \mathbb{R}$ integrable, then, $\forall t \leq T$:

$$g(t) \leq \alpha(t) + \beta \int_0^t \alpha(s) e^{\beta(t-s)} ds$$

Proof. Clearly, this is an integral inequality, we denote:

$$I(t) := \int_0^t g(s) ds \implies I'(t) = g(t) \leq \alpha(t) + \beta I(t)$$

Hence, we may rewrite above into, and it reminds us with the integrating factor:

$$I'(t) - \beta I(t) \leq \alpha(t) \implies [e^{-\beta t} I(t)]' = [I'(t) - \beta I(t)] e^{-\beta t} \leq \alpha(t) e^{-\beta t}$$

Integrating from both side, we reached:

$$e^{-\beta t} I(t) - 0 = \int_0^t [e^{-\beta s} I(s)]' ds \leq \int_0^t \alpha(s) e^{-\beta s} ds$$

Lastly, normalised the constant, and reuse previous inequality to get:

$$g(t) \leq \alpha(t) + \beta I(t) \leq \beta \int_0^t \alpha(s) e^{\beta(t-s)} ds$$

Hence we completed the proof. $\square$

Empowered by this estimation tool, we may present the Itô Uniqueness theorem:

Theorem 5.1.2. (Itô Uniqueness Theorem)

Suppose $b(t, x)$ and $\sigma(t, x)$ are locally Lipschitz in the space variable, i.e. $\forall n \geq 1, \exists K_n > 0$, s.t. $\forall t \geq 0, \|x\|, \|y\| \leq n$, we have:

$$\|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\| \leq K_n \|x - y\|$$

, then the strong uniqueness holds.

Proof. Given a B.M., W , and a probability space, $(\Omega, \mathcal{F}, \mathbb{P})$ also a initial condition, ξ , if X and $\tilde{X}$ are two strong solutions w.r.t. the W given, then define:

$$\begin{aligned}\tau_n &= \inf \{t \geq 0 : |X_t| \geq n\} \\ \tilde{\tau}_n &= \inf \{t \geq 0 : |\tilde{X}_t| \geq n\} \implies S_n := \tau_n \wedge \tilde{\tau}_n\end{aligned}$$

Then, $\lim_{n \rightarrow \infty} S_n = \infty$ is $\mathbb{P}$ -a.s., and is bounded on S_n , hence notice that:

$$\begin{aligned}X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n} &= \int_0^{t \wedge S_n} [b(s, X_s) - b(s, \tilde{X}_s)] ds \\ &\quad + \int_0^{t \wedge S_n} [\sigma(s, X_s) - \sigma(s, \tilde{X}_s)] dW_s\end{aligned}$$

Using AM-GM, i.e. $(a + b)^2 \leq 2a^2 + 2b^2$, we get:

$$\begin{aligned}\mathbb{E} \|X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n}\|^2 &\leq 2 \sum_{i=1}^d \mathbb{E} \left[\int_0^{t \wedge S_n} [b_i(s, X_s) - b_i(s, \tilde{X}_s)] ds \right]^2 \\ &\quad + 2 \sum_{i=1}^d \mathbb{E} \left[\sum_{j=1}^r \int_0^{t \wedge S_n} [\sigma_{ij}(s, X_s) - \sigma_{ij}(s, \tilde{X}_s)] dW_s^{(j)} \right]^2 \\ &\leq 2t \mathbb{E} \int_0^{t \wedge S_n} \|b(s, X_s) - b(s, \tilde{X}_s)\|^2 ds \\ &\quad + 2 \mathbb{E} \int_0^{t \wedge S_n} \|\sigma(s, X_s) - \sigma(s, \tilde{X}_s)\|^2 ds \\ &\leq 2(T+1)K_n^2 \int_0^t \mathbb{E} \|X_{s \wedge S_n} - \tilde{X}_{s \wedge S_n}\|^2 ds\end{aligned}$$

, where, for drift term, we use Cachy-Schwarz, and diffusion terms, we use the independent increment of B.M. to simplify, lastly to use Gronwall's inequality:

$$\mathbb{E} \|X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n}\|^2 \leq 0 \implies X_{t \wedge S_n} = \tilde{X}_{t \wedge S_n}, \text{ a.s., } \forall t \geq 0$$

By the (right) continuity, to get:

$$X_{t \wedge S_n} = \tilde{X}_{t \wedge S_n}, \forall t \geq 0, \text{ a.s.}$$

, then let $S_n \uparrow \infty$ to conclude, $X_t = \tilde{X}_t, \forall t \geq 0$, a.s. □

Naturally, we are going to talk about uniqueness, let us first see a toy example,

$$dX_t = X_t^2 dt, X_0 = 1 \implies X_t = 1 + \int_0^t X_s^2 ds$$

, we may see that, the process start from 1 (> 0), and each increment is related to X_t^2 , then intuitively, it will explode rapidly, this can be seen from its exact solution:

$$X_t = \begin{cases} \frac{1}{1-t}, & 0 \leq t < 1 \\ \infty, & t \geq 1 \end{cases}$$

, hence we need a universal theorem to ignore such cases:

Theorem 5.1.3. Suppose $\exists K > 0$, s.t. $\forall t \geq 0, \forall x, y \in \mathbb{R}^d$:

$$\begin{aligned}\|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\| &\leq K \|x - y\| \\ \|b(t, x)\| + \|\sigma(t, x)\| &\leq K(1 + \|x\|)\end{aligned}$$

On $(\Omega, \mathcal{F}, \mathbf{P})$, let ξ be r.v. and r -dimensional B.M. $W = \{W_t, t \geq 0\}$ is independent of ξ , also $\mathbf{E} \|\xi\|^2 < \infty$. Let $\{\mathcal{F}_t, t \geq 0\}$ be the augmented filtration, then $\exists$ continuous adapted process, $X = \{X_t, t \geq 0\}$, which is a strong solution w.r.t. W and $X_0 = \xi$ a.s. Moreover, $\forall T > 0, \exists C = C(K, T) > 0$, s.t. $\forall 0 \leq t \leq T$:

$$\mathbf{E} \|X_t\|^2 \leq C(1 + \mathbf{E} \|\xi\|^2) e^{Ct}$$

Proof. The idea is to use picard iteration, define, $X_t^{(0)} := \xi, \forall t \geq 0$, then $\forall k \geq 1$:

$$X_t^{(k)} = \xi + \int_0^t b(s, X_s^{(k-1)}) ds + \int_0^t \sigma(s, X_s^{(k-1)}) dW_s$$

Hence, (left as exercise, you should verify) $\forall T > 0, \exists C = C(K, T) > 0$, s.t.:

$$\mathbf{E} \|X_t^{(k)}\|^2 \leq C(1 + \mathbf{E} \|\xi\|^2) e^{Ct}, \forall t \leq T, K \geq 0$$

Now, for any $k \geq 0$, notice:

$$\begin{aligned}X_t^{(k+1)} - X_t^{(k)} &= \int_0^t [b(s, X_s^{(k+1)}) - b(s, X_s^{(k)})] ds \\ &\quad + \int_0^t [\sigma(s, X_s^{(k+1)}) - \sigma(s, X_s^{(k)})] dW_s\end{aligned}$$

We may separate and check each terms:

(1) For drift term, denote:

$$\begin{aligned}\mathbf{E} \sup_{s \leq t} \|D_s\|^2 &= \mathbf{E} \left\| \sup_{s \leq t} \int_0^s [b(u, X_u^{(k+1)}) - b(u, X_u^{(k)})] du \right\|^2 \\ &\leq \mathbf{E} \left[t \cdot \int_0^t \|b(u, X_u^{(k+1)}) - b(u, X_u^{(k)})\|^2 du \right] \\ &\leq tK^2 \int_0^t \mathbf{E} \|X_u^{(k+1)} - X_u^{(k)}\|^2 du\end{aligned}$$

(2) For diffusion term, denote:

$$M_t = \int_0^t [\sigma(s, X_s^{(k+1)}) - \sigma(s, X_s^{(k)})] dW_s$$

Then its quadratic variation:

$$[M]_t = \int_0^t \|\sigma(s, X_s^{(k+1)}) - \sigma(s, X_s^{(k)})\|^2 ds \leq \int_0^t K^2 \|X_s^{(k+1)} - X_s^{(k)}\|^2 ds$$

Hence, using Doob's maximal inequality we can also bound:

$$\mathbf{E} \sup_{s \leq t} \|M_s\|^2 \leq 4\mathbf{E} [M]_t \leq 4K^2 \int_0^t \mathbf{E} \|X_u^{(k+1)} - X_u^{(k)}\|^2 du$$

Then to combine these two inequality, we land in:

$$\begin{aligned} \mathbb{E} \sup_{s \leq t} \|X_s^{(k+1)} - X_s^{(k)}\|^2 &\leq 2\mathbb{E} \sup_{s \leq t} \|D_s\|^2 + 2\mathbb{E} \sup_{s \leq t} \|M_s\|^2 \\ &\leq L \int_0^t \mathbb{E} \|X_u^{(k+1)} - X_u^{(k)}\|^2 du \end{aligned}$$

, where if you work detailedly, we denote $L := 2K^2(T+4)$, now when $k = 1$:

$$\mathbb{E} \|X_t^{(1)} - X_t^{(0)}\|^2 \leq 2\mathbb{E} \|X_t^{(1)}\|^2 + 2\mathbb{E} \|X_t^{(0)}\|^2 \leq C^*$$

, where C^* definitely exist (since bounded) and is constant. Hence:

$$\mathbb{E} \sup_{0 \leq s \leq t} \|X_s^{(2)} - X_s^{(1)}\|^2 \leq L \int_0^t C^* ds = C^* \cdot (tL)$$

And consequently, we can see:

$$\mathbb{E} \sup_{0 \leq s \leq t} \|X_s^{(3)} - X_s^{(2)}\|^2 \leq L \int_0^t C^* L ds = C^* \cdot \frac{1}{2} (tL)^2$$

, then by induction, we can show that, $\forall k \geq 1$:

$$\mathbb{E} \sup_{0 \leq s \leq t} \|X_s^{(k+1)} - X_s^{(k)}\|^2 \leq C^* \frac{1}{k!} (tL)^k$$

Then, by Chebyshev's inequality, we could get:

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} \|X_s^{(k+1)} - X_s^{(k)}\|^2 > \frac{1}{2^{k+1}} \right) \leq 4^{k+1} C^* \frac{1}{k!} (LT)^k$$

By Borel-Cantalli Lemma, It is same to put:

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} \|X_s^{(k+1)} - X_s^{(k)}\|^2 > \frac{1}{2^{k+1}}, \text{ i.o.} \right) = 0$$

Hence, P-a.s. $\exists N(\omega) < \infty$, s.t., $\forall k \geq N$:

$$\sup_{t \leq T} \|X_t^{(k+1)} - X_t^{(k)}\| \leq \frac{1}{2^{k+1}} \xrightarrow{\forall m \geq 1} \sup_{t \leq T} \|X_t^{k+m} - X_t^k\| \leq \sum_{j=k}^{k+m-1} \frac{1}{2^{j+1}} \leq \frac{1}{2^k}$$

, thus $\{X_t^{(k)}(\omega), 0 \leq t \leq T\}$ is a Cauchy sequence in $\|\cdot\|_{\sup}$. Then $\exists$ a limit, X_t , s.t.:

$$\lim_{k \rightarrow \infty} \sup_{t \leq T} \|X_t^{(k)} - X_t\| = 0$$

, since the arbitrary of T , we can extend it to ∞ , next notice:

$$\mathbb{E} \|X_t\|^2 = \mathbb{E} \lim_{k \rightarrow \infty} \|X_t^{(k)}\|^2 \leq \lim_{k \rightarrow \infty} \mathbb{E} \|X_t^{(k)}\|^2 \leq C (1 + \|\xi\|^2) e^{Ct}$$

Moreover, since:

$$\begin{aligned} b(s, X_s^{(k)}) &\xrightarrow{a.s.} b(s, X_s) \\ \sigma(s, X_s^{(k)}) &\xrightarrow{L^2} \sigma(s, X_s) \end{aligned} \implies X_t^{(k+1)} \xrightarrow{a.s.} X_t := \xi + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$$

Hence, we completed the proof. $\square$

As closure, for the proof of strong uniqueness, the condition is too restricted, then we may wonder, what is the near necessary and sufficient condition:

Proposition 5.1.4. (Yamada & Watanabe)

Let $d = 1$, consider the following SDE:

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_s$$

, assume that $\exists$ constant, $K > 0$ and $h : [0, \infty) \mapsto [0, \infty)$ is strictly increasing function with $h(0) = 0$, such that:

$$\begin{aligned} |b(t, x) - b(t, y)| &\leq K |x - y| \\ |\sigma(t, x) - \sigma(t, y)| &\leq h(|x - y|) \end{aligned}$$

holds for all $t > 0$ and $x, y \in \mathbb{R}$, with:

$$\int_0^\varepsilon h^{-2}(u) du = \infty$$

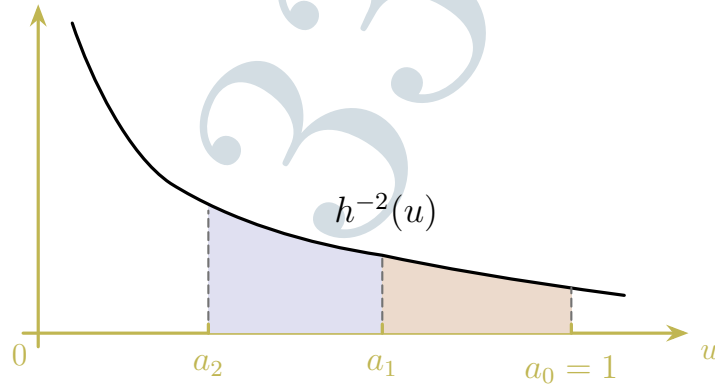
, $\forall \varepsilon > 0$, then the strong uniqueness holds.

Remark 5.1.1. The importance of Y-W (1971) is that, it gives us a critical point, i.e.:

$$h(u) = u^\alpha, \alpha \geq 1/2$$

, holds. But $\alpha < 1/2$ the strong uniqueness fails.

Proof. The idea is to construct a C^2 function, $\psi_n(x)$ s.t. $\psi_n(x) \uparrow |x|$.



Let $1 = a_0 > a_1 > a_2 > \dots > a_n \downarrow 0$, s.t. (showed as above figure):

$$\int_{a_n}^{a_{n-1}} h^{-2}(u) du = n$$

Then, there exists a **continuous**, $x \mapsto \rho_n(x) < 2/n \cdot h^{-2}(x)$, s.t.:

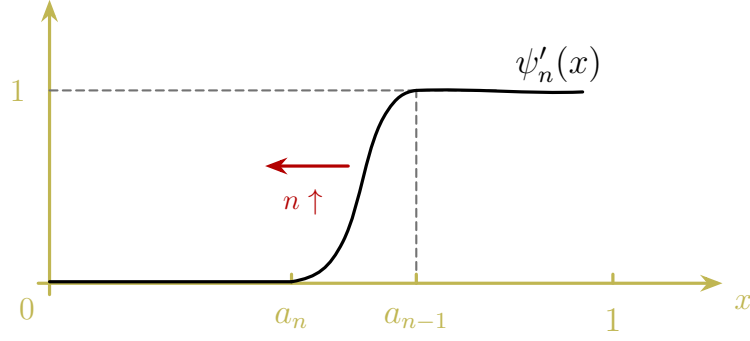
$$\int_{a_n}^{a_{n-1}} \rho_n(x) dx = 1$$

And $\rho_n(x) = 0, \forall x \notin (a_n, a_{n-1})$, then it is also true that:

$$\int_0^\infty \rho_n(x) dx = 1$$

Next step, we denote, $\forall x \geq 0$:

$$\psi'_n(x) := \int_0^x \rho_n(y) dy$$



As showed above, if we set it as odd function, as n increase, this will approach to derivative of $|x|$, hence, last construction is to define, $\psi_n(x) = \psi_n(-x), \forall x < 0$, and $\forall x \geq 0$:

$$\psi_n(x) = \int_0^x \psi'_n(y) dy$$

One can check that: $\psi_n(x) \uparrow |x|, \forall x \in \mathbb{R}$, besides, we may summarise the properties:

- (1) $\forall x \in \mathbb{R}, \psi_n(x) \rightarrow |x|$,
- (2) $\forall x \in \mathbb{R}, |\psi'_n(x)| \leq 1$,
- (3) $\forall x \in \mathbb{R}, |\psi''_n(x)| \leq \frac{2}{n} h^{-2}(x)$.

Now, suppose $X^{(1)}$ and $X^{(2)}$ are two strong solutions with, $X_0^{(1)} = X_0^{(2)}$ a.s., define:

$$\tau_N = \inf \left\{ t \geq 0 : \int_0^t \sigma^2(s, X_s^{(1)}) ds \geq N \text{ or } \int_0^t \sigma^2(s, X_s^{(2)}) ds \geq N \right\}$$

Since $\tau_N \uparrow \infty$, and $\mathbb{E} \int_0^{t \wedge \tau_N} \sigma^2(s, X_s^{(i)}) ds \leq N$, then we shall assume finiteness, i.e.:

$$\mathbb{E} \int_0^t \sigma^2(s, X_s^{(i)}) ds < \infty$$

Define $\Delta_t := X_t^{(1)} - X_t^{(2)}$, then apply Itô's formula to $\psi_n(\Delta_t)$ and taking expectation:

$$\begin{aligned} \mathbb{E} \psi_n(\Delta_t) &= \mathbb{E} \int_0^t \psi'_n(\Delta_s) [b(s, X_s^{(1)}) - b(s, X_s^{(2)})] ds \\ &\quad + \frac{1}{2} \mathbb{E} \int_0^t \psi''_n(x) [\sigma(s, X_s^{(1)}) - \sigma(s, X_s^{(2)})]^2 ds \\ &\leq K \int_0^t \mathbb{E} |\Delta_s| ds + \mathbb{E} \int_0^t \frac{1}{2} \cdot \frac{2}{n} h^{-2}(|\Delta_s|) \cdot [h(|\Delta_s|)]^2 ds \\ &= K \int_0^t \mathbb{E} |\Delta_s| ds + \frac{1}{n} \cdot t \end{aligned}$$

Let $n \rightarrow \infty$ to get:

$$\mathbb{E} |\Delta_t| \leq K \int_0^t \mathbb{E} |\Delta_s| ds \implies X_t^{(1)} = X_t^{(2)} \text{ a.s. }, \forall t \geq 0$$

The last step is from Gronwall, and by continuity we completed the proof. $\square$

Remark 5.1.2. Above proof is the original Yamada-Watanabe's proof, and it starts a new way even for later SPDE, and the we shall present the second proof from well-known Legall's genius observation, but firstly we need a auxiliary lemma:

Lemma 5.1.5. If $Y = X - X'$ and $h : [0, \infty) \mapsto [0, \infty)$ is increasing s.t., $\forall \varepsilon > 0$:

$$\int_0^\varepsilon h^{-2}(x)dx = \infty$$

, then $\ell_t^0(Y) = 0$ if, $\forall t > 0$:

$$\int_0^t h^{-2}(Y_s) \mathbf{1}_{\{Y_s > 0\}} d[Y]_s < \infty \text{ a.s.}$$

, here $\ell_t^0(Y)$ is the local time of Y at 0.

Proof. Recall the local time from tanaka formula:

$$|Y_t| = |Y_0| + \int_0^t \text{sgn}(Y_s) dY_s + \ell_t^0(Y)$$

And from the derivation, the local time is somewhat defined like:

$$\ell_t^a \approx \int_0^t \delta_a(Y_s) d[Y]_s$$

Then we can see that $\forall$ function $f : \mathbb{R} \mapsto \mathbb{R}$:

$$\begin{aligned} \int_0^t f(Y_s) d[Y]_s &= \int_0^t d[Y]_s \int_{-\infty}^{\infty} f(a) \delta_a(Y_s) da \\ &= \int_{-\infty}^{\infty} f(a) da \int_0^t \delta_a(Y_s) d[Y]_s = \int_{-\infty}^{\infty} f(a) \ell_t^a(Y) da \end{aligned}$$

Here the ℓ_t^a is called occupation density, then we may rewrite:

$$\int_0^t h^{-2}(Y_s) \mathbf{1}_{\{Y_s > 0\}} d[Y]_s = \int_0^\infty h^{-2}(a) \ell_t^a(Y) da$$

If $\ell_t^0(Y) > 0$, then by the right-continuity of $a \mapsto \ell_t^a(Y)$, we have:

$$\ell_t^a(Y) \geq \varepsilon_0 \implies \int_0^\infty h^{-2}(a) \ell_t^a(Y) da \geq \int_0^{\varepsilon_0} h^{-2}(a) \varepsilon_0 da = \infty$$

, for some $\varepsilon_0 > 0$ and $0 < a < \varepsilon_0$. Thus it is a contradiction, then we finished. $\square$

Empowered by this lemma, we are ready to check:

$$\begin{aligned} \int_0^t h^{-2}(\Delta_s) \mathbf{1}_{\{\Delta_s > 0\}} d[\Delta]_s &= \int_0^t h^{-2}(\Delta_s) \mathbf{1}_{\{\Delta_s > 0\}} [\sigma(s, X_s^{(1)}) - \sigma(s, X_s^{(2)})]^2 ds \\ &\leq \int_0^t h^{-2}(\Delta_s) \mathbf{1}_{\{\Delta_s > 0\}} h^2(\Delta_s) d \leq t < \infty \text{ a.s.} \end{aligned}$$

Hence, we have $\ell_t^0(Y) = 0$ a.s., thus by tanaka formula:

$$|\Delta_t| = \int_0^t \text{sgn}(\Delta_s) (b(s, X_s^{(1)}) - b(s, X_s^{(2)})) ds + \int_0^t \text{sgn}(\Delta_s) (\sigma(s, X_s^{(1)}) - \sigma(s, X_s^{(2)})) dW_s$$

Then if we take expectation on both side, the martingale term will eliminate, i.e.:

$$\mathbb{E} |\Delta_t| \leq K \int_0^t \mathbb{E} |\Delta_s| ds$$

Lastly, by the Gronwall's inequality, Legall completed the proof. We can see that Legall carefully find the key point of Y-M's proof is to eliminate the diffused term, and this can be done by asking the local time at 0 to be 0.

Remark 5.1.3. For S(O)DE, we discussed the problem that:

$$dX_t = X_t^\alpha dW_t$$

, if $\alpha \geq 1/2$, then it has strong uniqueness. A similar version for SPDE:

$$\frac{\partial}{\partial t} X(t, x) = \frac{\Delta}{2} X(t, x) + X^\alpha(t, x) \dot{W}(t, x)$$

Edwin Perkins solved the case where $\alpha > 3/4$, the strong uniqueness holds, and other parts remain as a big open question.

Lastly, we close this chapter by introducing comparison principle:

Theorem 5.1.6. (Comparison Principle)

Given $(\Omega, \mathcal{F}, \mathbb{P})$ and a B.M. $\{W_t, t \geq 0\}$ with its natural filtration. If two continuous, adapted process, $X^{(1)}$ and $X^{(2)}$ satisfy that:

$$\begin{cases} X_t^{(1)} = X_0^{(1)} + \int_0^t b_1(s, X_s^{(1)}) ds + \int_0^t \sigma(s, X_s^{(1)}) dW_s \\ X_t^{(2)} = X_0^{(2)} + \int_0^t b_2(s, X_s^{(2)}) ds + \int_0^t \sigma(s, X_s^{(2)}) dW_s \end{cases}$$

Assume the followings hold:

- (a) $\sigma(t, x)$ and $b_i(t, x)$ are continuous, real-valued functions on $[0, \infty) \times \mathbb{R}$,
- (b) The same σ , holds: $|\sigma(t, x) - \sigma(t, y)| \leq h(|x - y|)$, with:

$$\int_{0^+} h^{-2}(x) dx = \infty$$

- (c) The initial condition, $X_0^{(1)} \leq X_0^{(2)}$,
- (d) The drift term, $b_1(t, x) \leq b_2(t, x)$,
- (e) Either $b_1(t, x)$ or $b_2(t, x)$ satisfy:

$$|b_i(t, x) - b_i(t, y)| \leq K|x - y|$$

Then $\mathbb{P}(X_t^{(1)} \leq X_t^{(2)}, \forall t \geq 0) = 1$

Proof. Assume the finiteness of σ as usual, and also:

$$|b_1(t, x) - b_1(t, y)| \leq K|x - y|$$

Let $\psi_n(x)$ as before, then define $\varphi_n(x) = \psi_n(x)\mathbf{1}_{(0,\infty)}(x) \rightarrow x^+$, apply Itô formula:

$$\begin{aligned} \mathbb{E}\varphi_n(\Delta_t) - \frac{t}{n} &\leq \mathbb{E} \int_0^t \varphi'_n(\Delta_s) [b_1(s, X_s^{(1)}) - b_2(s, X_s^{(2)})] ds \\ &= \mathbb{E} \int_0^t \varphi'_n(\Delta_s) \left[b_1(s, X_s^{(1)}) - b_1(s, X_s^{(2)}) + \underbrace{b_1(s, X_s^{(2)}) - b_2(s, X_s^{(2)})}_{\leq 0} \right] ds \\ &\leq \mathbb{E} \int_0^t K |\Delta_s| \varphi'_n(\Delta_s) ds = \mathbb{E} \int_0^t K \Delta_s^+ \varphi'_n(\Delta_s^+) ds \leq \mathbb{E} \int_0^t K \Delta_s^+ ds \end{aligned}$$

Sending $n \rightarrow \infty$, we reached:

$$\mathbb{E}\Delta_t^+ \leq \int_0^t \mathbb{E}\Delta_s^+ ds \implies \mathbb{E}\Delta_t^+ = 0 \implies \mathbb{P}(X_t^{(1)} \leq X_t^{(2)}) = 1, \text{ a.s.}$$

Last step is from Gronwall, by continuity, we completed the proof. $\square$

5.2 Weak Solutions

Definition 5.2.1. A weak solution of SDE is a triple, constructed process with corresponding B.M. (X, W) , on a given space $(\Omega, \mathcal{F}, \mathbb{P})$, w.r.t. $\{\mathcal{F}_t, t \geq 0\}$, where,

- (a) $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, and $\{\mathcal{F}_t, t \geq 0\}$ is filtration, also $\mathcal{F}_t \in \mathcal{F}$,
- (b) $X = \{X_t, t \geq 0\}$ is continuous and adapted to $\{\mathcal{F}_t, t \geq 0\}$,
- (c) $W = \{W_t, t \geq 0\}$ is an r -dimensional B.M. adapted to $\{\mathcal{F}_t, t \geq 0\}$,
- (d) $\mathbb{P}\left(\int_0^t [|b_j(s, X_s)| + \sigma_{ij}^2(s, X_s)] ds < \infty, \forall t \geq 0\right) = 1$,
- (e) The process satisfies the integral equation:

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$$

Similar to strong uniqueness, some call it pathwise uniqueness:

Definition 5.2.2. Suppose whenever $(X, W), (\Omega, \mathcal{F}, \mathbb{P}), \{\mathcal{F}_t, t \geq 0\}$ and another triple, $(\tilde{X}, \tilde{W}), (\Omega, \mathcal{F}, \mathbb{P}), \{\tilde{\mathcal{F}}_t, t \geq 0\}$ are two weak solutions with $X_0 = \tilde{X}_0$ a.s. if:

$$\mathbb{P}(X_t = \tilde{X}_t, \forall t \geq 0) = 1$$

, then the pathwise uniqueness holds.

Another one is weaker one called weak uniqueness or uniqueness in law:

Definition 5.2.3. If for any two weak solutions with $(X, W), (\Omega, \mathcal{F}, \mathbb{P}), \{\mathcal{F}_t, t \geq 0\}$, and another triple, $(\tilde{X}, \tilde{W}), (\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}}), \{\tilde{\mathcal{F}}_t, t \geq 0\}$ with same initial distribution,

i.e. $\forall B \in \mathcal{B}(\mathbb{R}^d)$, $P(X_0 \in B) = \tilde{P}(\tilde{X}_0 \in B)$, we get X and $\tilde{X}$ have the same law,
i.e. $\forall t_0 \leq t_1 \leq \dots \leq t_n$:

$$P(X_{t_0} \in B_0, \dots, X_{t_n} \in B_n) = \tilde{P}(\tilde{X}_{t_0} \in B_0, \dots, \tilde{X}_{t_n} \in B_n)$$

, for all $B_i \in \mathcal{B}(\mathbb{R}^d)$, then weak uniqueness holds.

The reason why we call it is a "weak" solution, is clearly the constraint is less, to see that, some of the SDEs, they have weak solution(uniqueness) but no strong solution(uniqueness), then it comes to famous example, **Tanaka SDE**:

$$X_t = \int_0^t \text{sgn}(X_s) dW_s$$

, with sgn function defined as:

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}$$

- **Weak Uniqueness:** If (X, W) , $(\Omega, \mathcal{F}, P)$, and $\{\mathcal{F}_t, t \geq 0\}$ is a weak solution, then the process $\{X_t, t \geq 0\}$ is a continuous, square-integrable martingale, with:

$$[X]_t = \int_0^t \text{sgn}^2(X_s) ds = \int_0^t 1 \cdot ds = t$$

, then by Lévy criterion, X is a B.M, hence every finite dimensional distribution is the same, then it is easy to see the weak uniqueness to hold.

- **Pathwise Uniqueness:** However, it is easy to verify (left as exercise), that $(X, -W)$, $(\Omega, \mathcal{F}, P)$, and $\{\mathcal{F}_t, t \geq 0\}$ is also a weak solution, then:

$$P(X_t \neq -X_t, \text{ a.e. } t \geq 0) = P(X_t \neq 0, \text{ a.e. } t \geq 0) = 1$$

, then the pathwise uniqueness fails.

- **Weak Existence:** By construction, given $(\Omega, \mathcal{F}, P)$ and a B.M., $X = \{X_t, t \geq 0\}$, with its natural filtration, $\{\mathcal{F}_t^X, t \geq 0\}$. Let $\tilde{\mathcal{F}}_t^X$ be the augmentation of $\mathcal{F}_t^X$, assume that $P(X_0 = 0) = 1$, we could define:

$$W_t := \int_0^t \text{sgn}(X_s) dX_s$$

, thus the above construction is also a B.M. w.r.t. $\tilde{\mathcal{F}}_t^X$, last to see integral equation:

$$X_t = \int_0^t \text{sgn}(X_s) dW_s = \int_0^t \text{sgn}^2(X_s) dX_s = X_t$$

Hence, $X, W, (\Omega, \mathcal{F}, P), \{\tilde{\mathcal{F}}_t^X, t \geq 0\}$ is a weak solution.

- **No Strong Solution:** Given $(\Omega, \mathcal{F}, \mathbf{P})$ and a B.M., W , if X is strong solution:

$$X_t \in \mathcal{F}_t^W \subseteq \tilde{\mathcal{F}}_t^W \implies \mathcal{F}_t^X \subseteq \tilde{\mathcal{F}}_t^W$$

, where $\tilde{\mathcal{F}}_t^W$ is the augmentation, then by X is B.M, from Tanaka's formula:

$$\begin{aligned} |X_t| &= |X_0| + \int_0^t \operatorname{sgn}(X_s) dX_s + \ell_t \\ &= \int_0^t \operatorname{sgn}^2(X_s) dW_s + \ell_t = W_t + \ell_t \end{aligned}$$

Recall that, the local time, can be written as:

$$\ell_t = \int_0^t \delta_0(X_s) ds = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \left| \{0 \leq s \leq t : |X_s| \leq \varepsilon\} \right|$$

The last step is from:

$$\int_0^t \mathbf{1}_{\{|X_s| \leq \varepsilon\}} ds = \int \ell_t^y \mathbf{1}_{\{|y| \leq \varepsilon\}} dy \approx 2\varepsilon \ell_t$$

Hence, rewrite our result into:

$$W_t = |X_t| - \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \left| \{0 \leq s \leq t : |X_s| \leq \varepsilon\} \right| \in \mathcal{F}_t^{|X|}$$

This is to say:

$$\tilde{\mathcal{F}}_t^W \subseteq \tilde{\mathcal{F}}_t^{|X|} \implies \tilde{\mathcal{F}}_t^X \subseteq \tilde{\mathcal{F}}_t^{|X|}$$

, which is a contradiction, then we showed there is no strong solution.

Remark 5.2.1. We can see the strong uniqueness is important, as we will see later:

$$\text{Strong Uniqueness} \implies \text{Weak Uniqueness}$$

$$\text{Weak Existence} + \text{Pathwise Uniqueness} \implies \text{Strong Existence}$$

Lastly, in this section, we consider the construction of weak solution, but before that we need to study the **Girsanov Theorem**: Given $(\Omega, \mathcal{F}, \mathbf{P})$ and a d-dimensional B.M. $W = \{W_t, t \geq 0\}$ with $\mathbf{P}(W_0 = 0) = 1$. Let $X = \{X_t, t \geq 0\}$ be a d-dimensional measurable, adapted process w.r.t. $\{\mathcal{F}_t, t \geq 0\}$, such that, $\forall T \geq 0, \forall 1 \leq i \leq d$:

$$\mathbf{P} \left(\int_0^T (X_t^{(i)})^2 dt < \infty \right) = 1$$

, then the following integral is well-defined:

$$\int_0^t X_s^{(i)} dW_s^{(i)}$$

, now set:

$$Z_t(X) = \exp \left(\int_0^t X_s dW_s - \frac{1}{2} \int_0^t \|X_s\|^2 ds \right)$$

Assume $Z(X)$ is a martingale, i.e. $\forall t \geq 0$:

$$\mathbb{E}Z_t(X) = 1$$

Define a probability measure, $\tilde{\mathbb{P}}_T$ for any $T > 0$ by:

$$\tilde{\mathbb{P}}_T(A) = \mathbb{E}\left[Z_T(X) \mathbf{1}_A\right]$$

, for all $A \in \mathcal{F}_T$, or in the language of R-N derivative:

$$\frac{d\tilde{\mathbb{P}}_T}{d\mathbb{P}} = Z_T(X)$$

Remark 5.2.2. If $A \in \mathcal{F}_t \subseteq \mathcal{F}_T$, for $0 \leq t \leq T$, notice that:

$$\begin{aligned} \tilde{\mathbb{P}}_T(A) &= \mathbb{E}\left[Z_T(X) \mathbf{1}_A\right] = \mathbb{E}\left[\mathbb{E}(\mathbf{1}_A Z_T(X) \mid \mathcal{F}_t)\right] \\ &= \mathbb{E}\left[\mathbf{1}_A \mathbb{E}(Z_T(X) \mid \mathcal{F}_t)\right] = \mathbb{E}\left[\mathbf{1}_A Z_t(X)\right] = \tilde{\mathbb{P}}_t(A) \end{aligned}$$

Hence, it is to say:

$$\tilde{\mathbb{P}}_T|_{\mathcal{F}_t} = \tilde{\mathbb{P}}_t$$

Theorem 5.2.1. (*Girsanov Theorem*)

Assume $Z(X)$ is a martingale. Define $\tilde{W} = \{\tilde{W}_t, t \geq 0\}$ by:

$$\tilde{W}_t^{(i)} = W_t^{(i)} - \int_0^t X_s^{(i)} ds$$

, for $1 \leq i \leq d$, and $t \geq 0$, then for each fixed $T > 0$,

$$\{\tilde{W}_t, 0 \leq t \leq T\}$$

is d -dimensional B.M., on $(\Omega, \mathcal{F}_T, \tilde{\mathbb{P}}_T)$, w.r.t. $\{\mathcal{F}_t, 0 \leq t \leq T\}$.

We postpone the proof of the theorem, and first see a lemma:

Lemma 5.2.2. Fix $T > 0$, if $0 \leq s \leq t \leq T$ and $Y \in \mathcal{F}_t$, with $\tilde{\mathbb{E}}_T|Y| < \infty$, then:

$$\tilde{\mathbb{E}}_T(Y \mid \mathcal{F}_s) = \frac{1}{Z_s} \mathbb{E}(Y Z_t \mid \mathcal{F}_s)$$

Proof. It suffices to prove, $\forall A \in \mathcal{F}_s$:

$$\tilde{\mathbb{E}}_T\left[\mathbf{1}_A \cdot Y\right] = \tilde{\mathbb{E}}_T\left[\mathbf{1}_A \cdot \frac{1}{Z_s} \cdot \mathbb{E}(Y Z_t \mid \mathcal{F}_s)\right]$$

To see this, recall the remark 5.2.2:

$$\begin{aligned} \tilde{\mathbb{E}}_T\left[\mathbf{1}_A \cdot \frac{1}{Z_s} \cdot \mathbb{E}(Y Z_t \mid \mathcal{F}_s)\right] &= \tilde{\mathbb{E}}_s\left[\mathbf{1}_A \cdot \frac{1}{Z_s} \cdot \mathbb{E}(Y Z_t \mid \mathcal{F}_s)\right] = \mathbb{E}\left[\mathbf{1}_A \mathbb{E}(Y Z_t \mid \mathcal{F}_s)\right] \\ &= \mathbb{E}\left[\mathbf{1}_A Y Z_t\right] = \tilde{\mathbb{E}}_t\left[\mathbf{1}_A Y\right] = \tilde{\mathbb{E}}_T\left[\mathbf{1}_A Y\right] \end{aligned}$$

, this completes the proof. □

Besides that, we provide another proposition:

Proposition 5.2.3. Define $c\mathcal{M}_T^{loc} = \{M \in c\mathcal{M}^{loc} : M \text{ is defined on } (\Omega, \mathcal{F}_T, \mathbf{P})\}$, and $c\widetilde{\mathcal{M}}_T^{loc} = \{M \in c\mathcal{M}^{loc} : M \text{ is defined on } (\Omega, \mathcal{F}_T, \widetilde{\mathbf{P}}_T)\}$, then:

(a) If $M \in c\mathcal{M}_T^{loc}$, then:

$$\widetilde{M}_t := M_t - \sum_{i=1}^d \int_0^t X_s^{(i)} d[M, W^{(i)}]_s$$

, for $0 \leq t \leq T$ is in $c\widetilde{\mathcal{M}}_T^{loc}$.

(b) If $N \in c\mathcal{M}_T^{loc}$ and define, $\widetilde{N}_t$ as above, then:

$$[\widetilde{M}, \widetilde{N}]_t = [M, N]_t$$

, for all $0 \leq t \leq T$, a.s. under $\mathbf{P}$ and $\widetilde{\mathbf{P}}_T$.

Proof. Fill in a proof. □

Powered by above proposition, we are ready to proof Girsanov Theorem:

Proof. of Theorem 5.2.1. Taking $M = W^{(j)}$, with $1 \leq j \leq d$, then:

$$\widetilde{W}_t^{(j)} = \widetilde{M}_t = W_t^{(j)} - \int_0^t X_s^{(j)} ds \in c\widetilde{\mathcal{M}}_T^{loc}$$

, similar to set $N = W^{(k)}$, we also have, $\widetilde{W}_t^{(k)} = \widetilde{N}_t \in c\widetilde{\mathcal{M}}_T^{loc}$, then:

$$[\widetilde{W}^{(j)}, \widetilde{W}^{(k)}]_t = [W^{(j)}, W^{(k)}]_t = \delta_{jk} \cdot t$$

, hence by Lévy criterion, we completed the proof. □

Now, we are ready to consider a simple case's SDE's existence of weak solution:

Theorem 5.2.4. Consider the following SDE:

$$dX_t = dW_t + b(t, X_t) dt$$

, for all $0 \leq t \leq T$, W is B.M., and $b(t, x)$ is borel-measurable with:

$$\|b(t, x)\| \leq K(1 + \|x\|)$$

, where $K > 0$, for all $0 \leq t \leq T$, $x \in \mathbb{R}^d$, then $\forall$ probability measure μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, the SDE has a weak solution, (X, W) , $(\Omega, \mathcal{F}, \mathbf{P})$, $\{\mathcal{F}_t, t \geq 0\}$, s.t.:

$$\mathbf{P}(X_0 \in \cdot) = \mu$$

Proof. Let X be a B.M., on $(\Omega, \mathcal{F})$, w.r.t. $\{\mathcal{F}_t, t \geq 0\}$, with law $(\mathbf{P}^x)_{x \in \mathbb{R}^d}$, s.t.:

$$\mathbf{P}^x(X_0 = x) = 1$$

You may verify that:

$$Z_t := \exp \left(\sum_{i=1}^d \int_0^t b_i(s, X_s) dX_s^{(i)} - \frac{1}{2} \int_0^t \|b(s, X_s)\|^2 ds \right)$$

is a martingale, then by Girsanov theorem, if we define:

$$Q^x(A) = \mathbf{P}^x(A \cdot Z_T)$$

, for all $A \in \mathcal{F}_T$, then:

$$W_t := X_t - X_0 - \int_0^t b(s, X_s) ds$$

, is a B.M. with $Q^x(W_0 = 0) = 1$, then clearly, it follows:

$$X_t = X_0 + W_t + \int_0^t b(s, X_s) ds$$

, for all $0 \leq t \leq T$. Notice that:

$$\mathbf{P}^x(X_0 = x) = 1 \implies Q^x(X_0 = x) = 1$$

, also we may define:

$$Q^\mu(A) = \int_{\mathbb{R}^d} Q^x(A) \mu(dx) \implies Q^\mu(W_0 = 0) = 1$$

, and more generally:

$$Q^\mu(X_0 \in A) = \int_{\mathbb{R}^d} Q^x(X_0 \in A) \mu(dx) = \int_{\mathbb{R}^d} \mathbf{1}_{\{x \in A\}} \mu(dx) = \mu(A)$$

, so under Q^μ we have $X_0 \sim \mu$ and $(X, W), (\Omega, \mathcal{F}, Q^\mu), \{\mathcal{F}_t, t \geq 0\}$ is weak solution. $\square$

In the end, we consider two relations:

Lemma 5.2.5. *Weak Existence + Pathwise Uniqueness $\implies$ Weak Uniqueness*

Proof. Assume we have two weak solution:

$$\left\{ \begin{array}{l} (X^{(1)}, W^{(1)}) \\ (\Omega_1, \mathcal{F}_1, \nu_1) \\ \{\mathcal{F}_t^{(1)}, t \geq 0\} \end{array} \right\}, \left\{ \begin{array}{l} (X^{(2)}, W^{(2)}) \\ (\Omega_2, \mathcal{F}_2, \nu_2) \\ \{\mathcal{F}_t^{(2)}, t \geq 0\} \end{array} \right\}$$

If the initial distribution is the same, we will show (using pathwise uniqueness), i.e.

$$\nu_1(X_0^{(1)} \in \cdot) = \nu_2(X_0^{(2)} \in \cdot) \implies \nu_1(X^{(1)} \in \cdot) = \nu_2(X^{(2)} \in \cdot)$$

In fact, we will prove that, denote $Y^{(j)} = X^{(j)} - X_0^{(j)}$, then:

$$\nu_1 \left(\left(X_0^{(1)}, W^{(1)}, Y^{(1)} \right) \in A \right) = \nu_2 \left(\left(X_0^{(2)}, W^{(2)}, Y^{(2)} \right) \in A \right)$$

, for all $A \in \mathcal{B}(\Theta)$, with:

$$\Theta = \mathbb{R}^d \times C[0, \infty)^r \times C[0, \infty)^d$$

Let $\mu(\cdot) = \nu_1(X_0^{(1)} \in \cdot) = \nu_2(X_0^{(2)} \in \cdot)$, and P_* denote wiener measure on $C[0, \infty)^r$, denote a new (bigger) probability space, $(\Omega, \mathcal{F})$ by:

$$\Omega = \Theta \times C[0, \infty)^r, \mathcal{F} = \mathcal{B}(\Theta) \otimes \mathcal{B}(C[0, \infty)^r)$$

For each $\omega \in \Omega$, we write, $\omega = (x, w, y_1, y_2)$, and define:

$$P(d\omega) = \mu(dx) P_*(dw) Q_1(x, w, dy_1) Q_2(x, w, dy_2)$$

, where $Q_j(x, w, dy_j)$ is the regular conditional probability, s.t.:

(1) $\forall x \in \mathbb{R}^d, w \in C[0, \infty)^r, Q_j(x, w, \cdot)$ is a prob. measure on $(C[0, \infty)^d, \mathcal{B}(C[0, \infty)^d))$.

(2) $\forall F \in \mathcal{B}(C[0, \infty)^d)$, the map:

$$(x, w) \mapsto Q_j(x, w, F)$$

, is measurable w.r.t. $\mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(C[0, \infty)^d)$.

(3) $\forall G \in \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(C[0, \infty)^d)$, and $\forall F \in \mathcal{B}(C[0, \infty)^d)$:

$$\nu_j \left(\left(X_0^{(j)}, W^{(j)} \right) \in G, Y^{(j)} \in F \right) = \int_G \mu(dx) P_*(d\omega) Q_j(x, w, F)$$

, i.e. $Q_j(x, w, F) = \nu_j(Y^{(j)} \in F \mid X_0^{(j)} = x, W^{(j)} = w)$

Then let:

$$\mathcal{G}_t := \sigma \{ (x, w_s, y_1^s, y_2^s) : x \in \mathbb{R}^d, 0 \leq s \leq t \}$$

And $\mathcal{F}_t$ be the augmentation of $\mathcal{G}_t$. Now we conclude that:

$$\begin{aligned} P(\{\omega \in \Omega : (x, w, y_1) \in A\}) &= \int P(d\omega) \mathbf{1}_{\{(x, w, y_1) \in A\}} \\ &= \int_A \mu(dx) P_*(d\omega) Q_1(x, w, dy_1) = \nu_1 \left(\left(X_0^{(1)}, W^{(1)}, Y^{(1)} \right) \in A \right) \end{aligned}$$

Therefore, $(x + y_1, w) \stackrel{d.}{=} (X^{(1)}, W^{(1)})$ under P & ν_1 , similarly we have, $(x + y_2, w) \stackrel{d.}{=} (X^{(2)}, W^{(2)})$ under P & ν_2 , and by pathwise uniqueness:

$$P(x + y_1 = x + y_2, \forall t \geq 0) = 1$$

, this information helps us to transfer between the two probability measure, i.e.:

$$\begin{aligned} \nu_1 \left(\left(X_0^{(1)}, W^{(1)}, Y^{(1)} \right) \in A \right) &= P(\{\omega \in \Omega : (x, w, y_1) \in A\}) \\ &= P(\{\omega \in \Omega : (x, w, y_2) \in A\}) = \nu_2 \left(\left(X_0^{(2)}, W^{(2)}, Y^{(2)} \right) \in A \right) \end{aligned}$$

The middle step is from pathwise uniqueness, hence, we completed the proof. $\square$

Chapter 6

Brownian Motion & PDE

6.1 Harmonic Function

Definition 6.1.1. $\forall$ smooth function, $f : \mathbb{R}^d \mapsto \mathbb{R}$, notice that:

$$\frac{\Delta}{2}f(x) = \lim_{t \downarrow 0} \frac{\mathbb{E}^x[f(W_t) - f(x)]}{t}$$

, and we call $\frac{\Delta}{2}$ that generator of B.M.

Remark 6.1.1. Recall the Laplacian:

$$\Delta u := \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2}$$

And the subscript x in (conditional) expectation means starting from x , i.e.:

$$\mathbb{E}^x[\cdot] = \mathbb{E}[\cdot \mid W_0 = x]$$

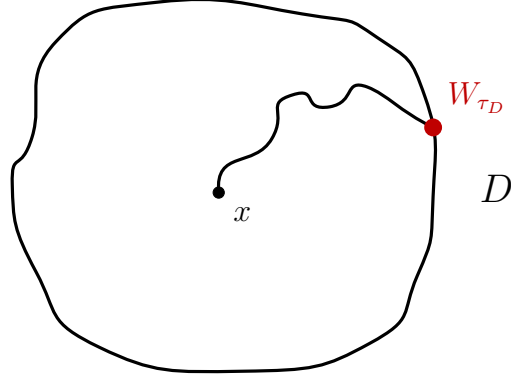
Then above is easy to show by applying Itô's formula:

$$f(W_t) = f(W_0) + \int_0^t \sum_{i=1}^d \frac{\partial}{\partial x_i} f(W_s) dW_s^{(i)} + \frac{1}{2} \int_0^t \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} f(W_s) ds$$

Simply rearranging the term and divide by t on both side:

$$\frac{1}{t} \mathbb{E}^x[f(W_t) - f(W_0)] = \frac{1}{2} \cdot \frac{1}{t} \int_0^t \mathbb{E}^x[\Delta f(W_s)] ds \xrightarrow{t \rightarrow 0} \frac{1}{2} \Delta f(x)$$

Definition 6.1.2. A function, u , defined on some open subset $D \subseteq \mathbb{R}^d$, is called **harmonic** if, (a) $u \in C^2(D)$, (b) $\Delta u = 0$.



Let $\{W_t, t \geq 0\}$ and its filtration, $\{\mathcal{F}_t, t \geq 0\}$, on probability space, $(\Omega, \mathcal{F})$, with its measure, $\{\mathbf{P}^x\}_{x \in \mathbb{R}^d}$, s.t.: $\mathbf{P}^x(W_0 = x) = 1$. For any open set, $D \in \mathbb{R}^d$, we set:

$$\tau_D = \inf\{t \geq 0 : W_t \notin D\}$$

If D is bounded, then we have, $\forall x \in D$:

$$\mathbf{P}^x(\tau_D < \infty) = 1$$

, since it will definitely move out from this region sometimes, and $W_{\tau_D} \in \partial D$, with:

$$\partial D = \text{topological boundary}$$

Also, $\overline{D} = D \cup \partial D$ is the closure of D . Considering the center ball:

$$B_r = \{x \in \mathbb{R}^d : \|x\| < r\}$$

And its neighboring ball, i.e., $\forall x \in \mathbb{R}^d$:

$$B_r(x) = \{y \in \mathbb{R}^d : \|y - x\| < r\}$$

, be the open ball centered at $0/x$ with radius $r > 0$, by a little detail, its volume:

$$V_r = \frac{2r^d \pi^{d/2}}{d \Gamma(d/2)}$$

And the surface area:

$$S_r = \frac{2r^{d-1} \pi^{d/2}}{\Gamma(d/2)}$$

With all this setup, we may define the measure, μ_r on ∂B_r , to be

$$\mu_r(dx) = \mathbf{P}^0(W_{\tau_{B_r}} \in dx)$$

And by **rotational invariance**, μ_r is a **uniform measure** on ∂B_r . In particular:

$$\mu_r = \frac{\sigma_r}{S_r}$$

, where $\sigma_r(dx)$ is the surface measure on ∂B_r . Recall for any function, f , in polar:

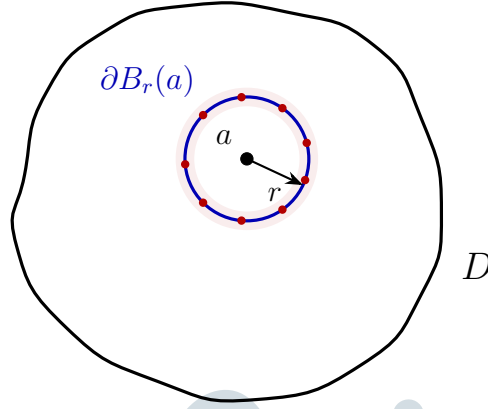
$$\int_{B_r} f(x) dx = \int_0^r d\rho \int_{\partial B_\rho} f(x) \sigma_\rho(dx) = \int_0^r d\rho \int_{\partial B_\rho} f(x) S_\rho \mu_\rho(dx)$$

Definition 6.1.3. (Mean Value Property)

A function, $u : D \mapsto \mathbb{R}^d$ has M.V.P. if $\forall a \in D, \forall r > 0$, with $\overline{B_r(a)} \subseteq D$, we have:

$$u(a) = \int_{\partial B_r} u(a+x) \mu_r(dx) = \int_{\partial B_r(a)} u(x) \mu(dx)$$

With this notation, the geometric meaning of the mean value property becomes transparent. Choose a point $a \in D$ and a radius $r > 0$ such that $\overline{B_r(a)} \subset D$. The value of a harmonic function at the centre is not determined by any preferred direction around a ; rather, it is recovered from the uniform average of its values on the surrounding sphere.



Assume M.V.P, we notice:

$$\int_{B_r(a)} u(x) dx = \int_0^t d\rho S_\rho \int_{\partial B_\rho(a)} u(x) \mu_\rho(dx) = V_r \cdot u(a)$$

Then it is equivalent to say:

$$u(a) = \frac{1}{V_r} \int_{B_r(a)} u(x) dx$$

And the last natural question is how this relates to harmonic function:

Proposition 6.1.1. Assume a function, $u : D \mapsto \mathbb{R}^d$, then:

- (a) If u is harmonic on D , then u has M.V.P.
- (b) If u has M.V.P. on D , then it is harmonic and $u \in C^\infty$.

Proof. Considering the first one, it is obvious we need to use Itô's formula:

- (a) Assume $\Delta u(x) = 0, \forall x \in D$, then $\forall r > 0$ with $\overline{B_r(a)} \subseteq D$, and $\forall a \in D$:

$$u(W_{t \wedge \tau_{B_r(a)}}) = u(W_0) + \int_0^{t \wedge \tau_{B_r(a)}} \nabla u(W_s) dW_s + \frac{1}{2} \int_0^{t \wedge \tau_{B_r(a)}} \Delta u(W_s) ds$$

Since $\Delta u = 0$ eliminates third term, and ∇u is bounded on $\overline{B_r(a)}$, then:

$$\mathbb{E}^a u(W_{t \wedge \tau_{B_r(a)}}) = u(a)$$

Now, we may use the compactness to do dominated convergence:

$$u(a) = \lim_{t \rightarrow \infty} \mathbb{E}^a u \left(W_{t \wedge \tau_{B_r(a)}} \right) = \mathbb{E}^a u \left(W_{\tau_{B_r(a)}} \right) = \int_{\partial B_r} u(x+a) \mu_r(dx)$$

(b) Considering use **mollifier**, $\forall \varepsilon > 0$, define, $g_\varepsilon : \mathbb{R}^d \mapsto \mathbb{R}$ by:

$$g_\varepsilon(x) = \begin{cases} c(\varepsilon) \exp \left(-\frac{1}{\varepsilon^2 - \|x\|^2} \right), & \|x\| < \varepsilon \\ 0, & \|x\| \geq \varepsilon \end{cases}$$

, where $c(\varepsilon) > 0$ is a constant (function), s.t.:

$$\int_{\mathbb{R}^d} g_\varepsilon(x) dx = \int_{B_\varepsilon} g_\varepsilon(x) dx = 1$$

Now, $\forall a \in D$ and $\varepsilon > 0$ with $\overline{B_\varepsilon(a)} \subseteq D$, define:

$$u_\varepsilon(a) := \int_{B_\varepsilon} u(x+a) g_\varepsilon(x) dx = \int_{\mathbb{R}^d} u(y) g_\varepsilon(y-a) dy$$

Then since mollifier has really good property, i.e. notice if we take derivative:

$$\begin{aligned} u'_\varepsilon(a) &= \lim_{\delta \downarrow 0} \frac{u(a+\delta) - u(a)}{\delta} \\ &= \lim_{\delta \downarrow 0} \int_{\mathbb{R}^d} u(y) \cdot \frac{g_\varepsilon(y-a-\delta) - g_\varepsilon(y-a)}{\delta} dy = - \int_{\mathbb{R}^d} u(y) g'_\varepsilon(y-a) dy \end{aligned}$$

Since mollifier is in C^∞ , then we have $u_\varepsilon \in C^\infty$, more interestingly, it coincides u :

$$\begin{aligned} u_\varepsilon(a) &= c(\varepsilon) \int_{B_\varepsilon} u(a+x) \exp \left(-\frac{1}{\varepsilon^2 - \|x\|^2} \right) dx \\ &= c(\varepsilon) \int_0^\varepsilon dS_\rho \exp \left(-\frac{1}{\varepsilon^2 - \rho^2} \right) \rho \int_{\partial B_\rho} u(a+x) \mu_\rho(dx) \\ &= u(a) \int_0^\varepsilon S_\rho c(\varepsilon) \exp \left(-\frac{1}{\varepsilon^2 - \rho^2} \right) d\rho \\ &= u(a) \int_{B_\varepsilon} c(\varepsilon) \exp \left(-\frac{1}{\varepsilon^2 - \|x\|^2} \right) dx = u(a) \end{aligned}$$

, $\forall a \in D$ with $\overline{B_\varepsilon(a)} \subseteq D$, by letting $\varepsilon \rightarrow 0$, then $u \in C^\infty$. Next we will prove, $\Delta u = 0$, consider $\forall a \in D$, $\exists \varepsilon > 0$, s.t. $\overline{B_\varepsilon(a)} \subseteq D$, by talor expansion at a :

$$u(a+y) = u(a) + \sum_{i=1}^d y_i \frac{\partial}{\partial x_i} u(a) + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d y_i y_j \frac{\partial^2}{\partial x_i \partial x_j} u(a) + o(\|y\|^2)$$

By (rotational) symmetry, we have, $\forall i \neq j$:

$$\int_{\partial B_\varepsilon} y_i \mu_\varepsilon(dy) = 0, \quad \int_{\partial B_\varepsilon} y_i y_j \mu_\varepsilon(dy) = 0$$

Besides, notice that if we integrate on the surface, i.e. ∂B_ε , of second order:

$$\int_{\partial B_\varepsilon} y_i^2 \mu_\varepsilon(dy) = \frac{1}{d} \sum_{i=1}^d \int_{\partial B_\varepsilon} y_i^2 \mu_\varepsilon(dy) = \frac{1}{d} \int_{\partial B_\varepsilon} \|y\|^2 \mu_\varepsilon(dy) = \frac{1}{d} \cdot \varepsilon^2$$

Hence if we integrate the taylor expansion around the surface, we conclude:

$$0 = \frac{\Delta}{2} u(a) \frac{1}{d} \varepsilon^2 + o(\varepsilon^2)$$

By letting $\varepsilon \rightarrow 0$, we land in $\Delta u(a) = 0, \forall a \in D$, then completes the proof. $\square$

6.2 Dirichlet Problem

Given an open set, $D \subseteq \mathbb{R}^d$, and a continuous function, f , on ∂D , the **Dirichlet Problem** (D, f) is to find $u \in C(\overline{D})$, s.t.:

$$\begin{cases} \Delta u = 0, & \text{on } D \\ u = f, & \text{in } \partial D \end{cases}$$

Remark 6.2.1. If $u(t, x)$ denotes temperature at $x \in D$ at time, t , then heat equation:

$$\frac{\partial}{\partial t} u(t, x) = \frac{\Delta}{2} u(t, x)$$

- **Uniqueness:** It suffices to show the claim:

Claim 6.2.1. *If f is bounded, and $\forall a \in D, P^a(\tau_D < \infty) = 1$, then any bounded solution, u to Dirichlet Problem (D, f) is given by:*

$$u(a) = E^a f(W_{\tau_D})$$

Proof. Let $u \in C(\overline{D})$ be bounded and:

$$\begin{cases} \Delta u = 0, & \text{on } D \\ u = f, & \text{in } \partial D \end{cases}$$

For each, $n \geq 1$, we let:

$$D_n = \left\{ x \in D : \inf_{y \in \partial D} \|x - y\| > \frac{1}{n} \right\} = \left\{ x \in D : d(x, \partial D) > \frac{1}{n} \right\}$$

Similar to previous discussion, applying Itô's formula:

$$u(W_{t \wedge \tau_{D_n} \wedge \tau_{B_n}}) = u(W_0) + \sum_{i=1}^d \int_0^{t \wedge \tau_{D_n} \wedge \tau_{B_n}} \frac{\partial}{\partial x_i} u(W_s) dW_s^{(i)}$$

Since the first derivative is bounded on, $\overline{B_n \cap D_n}$, and:

$$P^a(\tau_D < \infty) = 1 \implies u(W_{t \wedge \tau_{D_n} \wedge \tau_{B_n}}) \xrightarrow{a.s.} u(W_{\tau_D}) = f(W_{\tau_D})$$

, then by B.C.T:

$$u(a) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow \infty} \mathbb{E}^a u(W_{t \wedge \tau_{D_n} \wedge \tau_{B_n}}) = \mathbb{E}^a f(W_{\tau_D})$$

Hence, we completed the proof by solving out its form. $\square$

- **Existence:** Let $u(x) = \mathbb{E}^x f(W_{\tau_D})$, for all $x \in \overline{D}$, provided that:

$$\mathbb{E}^x |f(W_{\tau_D})| < \infty, \forall x \in D$$

Then, for special case, $x \in \partial D$, we get:

$$\mathbb{P}^x(\tau_D = 0) = 1$$

, where just recall the hitting time is defined as:

$$\tau_D := \inf \{t \geq 0 : W_t \notin D\} \implies \mathbb{P}^x(W_{\tau_D} = x) = 1 \implies \mathbb{E}^x f(W_{\tau_D}) = f(x)$$

Hence, we recover the boundary condition, next we want to show, $\Delta u = 1$, it suffice to show M.V.P., $\forall a \in D$, and $r > 0$, s.t., $\overline{B_r(a)} \subseteq D$, by spatial markov property:

$$\begin{aligned} u(a) &= \mathbb{E}^a f(W_{\tau_D}) = \mathbb{E}^a \left[\mathbb{E}^a \left(f(W_{\tau_D}) \mid \mathcal{F}_{\tau_{B_r(a)}} \right) \right] \\ &= \mathbb{E}^a \left[\mathbb{E}^{W_{\tau_{B_r(a)}}} [f(W_{\tau_D})] \right] = \mathbb{E}^a u(W_{\tau_{B_r(a)}}) = \int_{\partial B_r} u(a+y) \mu_r(dy) \end{aligned}$$

Therefore, u has M.V.P., it remains to prove, $u \in C(\overline{D})$, i.e.:

$$\lim_{D \ni x \rightarrow a} u(x) = u(a) = f(a), \forall a \in \partial D$$

Further to rewrite the equation, we land in, $\forall a \in \partial D$, we need:

$$\lim_{D \ni x \rightarrow a} \mathbb{E}^x f(W_{\tau_D}) = f(a)$$

Hence we curious how the point, starting from boundary, exit D , define:

$$\sigma_D = \inf \{t > 0 : W_t \notin D\}$$

It is obvious that, $\tau_D \leq \sigma_D$, then we define:

Definition 6.2.1. We say a point, $a \in \partial D$, is regular for D , if:

$$\mathbb{P}^a(\sigma_D = 0) = 1$$

Remark 6.2.2. To better understand this intuitively, we can see:

$$\begin{aligned} \sigma_D = 1/2 &\iff \forall 0 < t < 1/2, W_t \in D \\ \sigma_D = 0 &\iff \exists t_n \downarrow 0, \text{ s.t. } W_{t_n} \notin D \end{aligned}$$

Firstly, we consider one-dimensional case, $d = 1$, we choose the simplest open interval, $D = (a, b)$, then its boundary set, $\partial D = \{a, b\}$, now by L.I.L(Law of the Iterated Logarithm), a and b are both regular points, i.e.:

$$\limsup_{t \downarrow 0} \frac{W_t - W_0}{\sqrt{2t \log \log 1/t}} = 1$$

Then it will osculate around the points, hence has a sequences, i.e. $\exists t_n \downarrow 0$, s.t.:

$$W_t - \underbrace{W_0}_{a \text{ or } b} \geq (\sqrt{2} - \varepsilon) \sqrt{t_n \log \log 1/t_n} > 0$$

In other cases, if a point, $a \in \partial D$, is irregular, if:

$$\mathbf{P}^a(\sigma_D = 0) < 1 \iff \mathbf{P}^a(\sigma_D > 0) > 0$$

By Blumenthal 0-1 law, $\mathbf{P}^a(\sigma_D > 0) \in \{0, 1\}$, so if a is irregular, then:

$$\mathbf{P}^a(\sigma_D > 0) = 1$$

An easy example would be two dimensional B.M., taking:

$$D = \{x \in \mathbb{R}^d : 0 < \|x\| < 1\}$$

, then $0 \in \partial D$, but is irregular, since:

$$\begin{aligned} \sigma_D &= \inf \{t > 0 : \|W_t\| \geq 1 \text{ or } \|W_t\| = 0\} \\ &= \inf \{t > 0 : \|W_t\| \geq 1\} > 0 \end{aligned}$$

Hence, we will need a theorem to talk about the relations with regular:

Theorem 6.2.2. *For case, $d \geq 2$, and fix $a \in \partial D$, the followings are equivalent:*

(a) *$\forall$ bounded, measurable function, f , on ∂D s.t., f is continuous at a , it holds:*

$$\lim_{D \ni x \rightarrow a} \mathbf{E}^x f(W_{\tau_D}) = f(a)$$

(b) *a is regular.*

(c) *$\forall \varepsilon > 0$, we have:*

$$\lim_{D \ni x \rightarrow a} \mathbf{P}^x(\tau_D > \varepsilon) = 0$$

Proof. Firstly to consider the easiest direction:

(c) $\implies$ (a) By continuity of Brownian path, $\forall r > 0$, by letting $\varepsilon \rightarrow 0$:

$$\mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - x\| < r \right) = \mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - W_0\| < r \right) \longrightarrow 1$$

Then notice the following inequality:

$$\begin{aligned} \mathbf{P}^x \left(\|W_{\tau_D} - W_0\| < r \right) &\geq \mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - W_0\| < r, \tau_D \leq \varepsilon \right) \\ &\geq \mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - W_0\| < r \right) - \mathbf{P}^x(\tau_D > \varepsilon) \end{aligned}$$

By sending limit, $D \ni x \in a$ and $\varepsilon \rightarrow 0$, we conclude:

$$\lim_{D \ni x \in a} \mathbf{P}^x \left(\|W_{\tau_D} - W_0\| < r \right) = 1$$

Lastly, since the continuity of f at a , then, $\forall \varepsilon > 0, \exists r > 0$, s.t.:

$$\left| f(x) - f(a) \right| < \varepsilon, \forall |x - a| < r$$

Then on $\|W_{\tau_D} - x\| < r/2$, if $|x - a| < r/2$, then:

$$\|W_{\tau_D} - a\| \leq \|W_{\tau_D} - x\| + \|x - a\| < r \implies \left| f(W_{\tau_D}) - f(a) \right| < \varepsilon$$

Hence, we can separate the event into:

$$\begin{aligned} \mathbf{E}^x \left| f(W_{\tau_D}) - f(a) \right| &\leq \|f\|_\infty \mathbf{E}^x \mathbf{1}_{\{\|W_{\tau_D} - x\| > r/2\}} + \mathbf{E}^x \left| f(W_{\tau_D}) - f(a) \right| \mathbf{1}_{\{\|W_{\tau_D} - x\| < r/2\}} \\ &\leq \varepsilon + \|f\|_\infty \mathbf{P}^x \left(\|W_{\tau_D} - x\| > r/2 \right) \xrightarrow[\varepsilon \rightarrow 0]{D \ni x \rightarrow a} 0 \end{aligned}$$

(b) $\implies$ (c) W.l.o.g., we may assume, $a = 0$, then fix a $\varepsilon > 0$, define:

$$g(x) = \mathbf{P}^x (\sigma_D > \varepsilon) = \mathbf{P}^x (W_s \in D, \forall 0 < s \leq \varepsilon)$$

Then since, $\tau_D \leq \sigma_D$, provided the upper semicontinuous of g :

$$\limsup_{D \ni x \rightarrow 0} \mathbf{P}^x (\tau_D > \varepsilon) \leq \limsup_{D \ni x \rightarrow 0} \mathbf{P}^x (\sigma_D > \varepsilon) = \limsup_{D \ni x \rightarrow 0} g(x) \leq g(0) = 0$$

Then by the result from analysis, we may consider, $\forall 0 < \delta < \varepsilon$:

$$g_\delta(x) = \mathbf{P}^x (W_s \in D, \forall \delta \leq s \leq \varepsilon)$$

, it suffices to show:

(1) $\forall x \in \mathbb{R}, g_\delta \downarrow g(x)$. (clearly true)

(2) $x \mapsto g_\delta(x)$ is continuous, since, by markov property, fix $\delta \in (0, \varepsilon)$:

$$\begin{aligned} g_\delta(x) &= \mathbf{P}^x (W_s \in D, \forall \delta \leq s \leq \varepsilon) = \mathbf{E}^x \mathbf{P}^{W_\delta} (W_s \in D, \forall 0 \leq s \leq \varepsilon - \delta) \\ &= \mathbf{E}^x \mathbf{P}^{W_\delta} (\tau_D > \varepsilon - \delta) = \int_{\mathbb{R}^d} p_\delta(x, y) \mathbf{P}^y (\tau_D > \varepsilon - \delta) dy \end{aligned}$$

, where $p_\delta(\cdot, \cdot)$ is gaussian density, then g is continuous.

(a) $\implies$ (b) Prove by contradiction, we assume, 0 is irregular, i.e.:

$$\mathbf{P}^0 (\sigma_D = 0) = 0 \implies \mathbf{P}^0 (\sigma_D > 0) = 1$$

Since for dimension, $d \geq 2$, staring from 0, we know:

$$\mathbf{P}^0 (W_t \neq 0, \forall t > 0) = 1 \implies \mathbf{P}^0 (W_{\sigma_D} = 0) = 0 \implies \lim_{r \downarrow 0} \mathbf{P}^0 (W_{\sigma_D} \in B_r) = 0$$

Then we could always find a small fix $r > 0$, s.t.:

$$\mathbf{P}^0(W_{\sigma_D} \in B_r) < \frac{1}{4}$$

Let $\delta_n \downarrow 0$, and $0 < \delta_n < r$, we define:

$$\tau_n = \inf \{t \geq 0 : \|W_t\| \geq \delta_n\}$$

Then clearly, $\mathbf{P}^0(\tau_n \downarrow 0) = 1$, by irregularity of 0, we have:

$$1 = \mathbf{P}^0\left(\bigcup_{n=1}^{\infty} \{\tau_n < \sigma_D\}\right) = \lim_{n \rightarrow \infty} \mathbf{P}^0(\tau_n < \sigma_D)$$

Hence, $\exists N \geq 1$, s.t., $\forall n \geq N$,

$$\mathbf{P}^0(\tau_n < \sigma_D) \geq \frac{1}{2}$$

Then we may conclude our result, by noticing:

$$\begin{aligned} \frac{1}{4} &> \mathbf{P}^0(W_{\sigma_D} \in B_r) \geq \mathbf{P}^0(W_{\sigma_D} \in B_r, \tau_n < \sigma_D) \\ &= \mathbf{E}^0\left[\mathbf{1}_{\{\tau_n < \sigma_D\}} \mathbf{P}^0(W_{\sigma_D} \in B_r \mid \mathcal{F}_{\tau_n})\right] = \mathbf{E}^0\left[\mathbf{1}_{\{\tau_n < \sigma_D\}} \mathbf{P}^{W_{\tau_n}}(W_{\tau_D} \in B_r)\right] \\ &= \mathbf{E}^0\left[\mathbf{1}_{\{\tau_n < \sigma_D\}} \int_{\mathbb{R}_d} F_{W_{\tau_n}}(dx) \mathbf{P}^x(W_{\tau_D} \in B_r)\right] \\ &\geq \mathbf{P}^0(\tau_n < \sigma_D) \cdot \inf_{\substack{x \in D \\ \|x\| = \delta_n}} \mathbf{P}^x(W_{\tau_D} \in B_r) \geq \frac{1}{2} \inf_{\substack{x \in D \\ \|x\| = \delta_n}} \mathbf{P}^x(W_{\tau_D} \in B_r) \end{aligned}$$

Hence, we conclude that:

$$\inf_{\substack{x \in D \\ \|x\| = \delta_n}} \mathbf{P}^x(W_{\tau_D} \in B_r) < \frac{1}{2} \implies \exists x_n \in D, \|x_n\| = \delta_n, \mathbf{P}^{x_n}(W_{\tau_D} \in B_r) \leq \frac{1}{2}$$

Lastly, it is easy to find nice function, s.t.:

$$f(x) = \begin{cases} 0, & x \in B_r^c \\ \leq 1, & x \in B_r \setminus \{0\} \\ 1, & x = 0 \end{cases}$$

Then, it gives us the contradiction in a way:

$$\begin{aligned} \mathbf{E}^{x_n} f(W_{\tau_D}) &= \mathbf{E}^{x_n} f(W_{\tau_D}) \mathbf{1}_{\{W_{\tau_D} \in B_r\}} + \mathbf{E}^{x_n} f(W_{\tau_D}) \mathbf{1}_{\{W_{\tau_D} \notin B_r\}} \\ &\leq \mathbf{E}^{x_n} \mathbf{1}_{\{W_{\tau_D} \in B_r\}} \leq \frac{1}{2} < (\neq) f(0) = 1 \end{aligned}$$

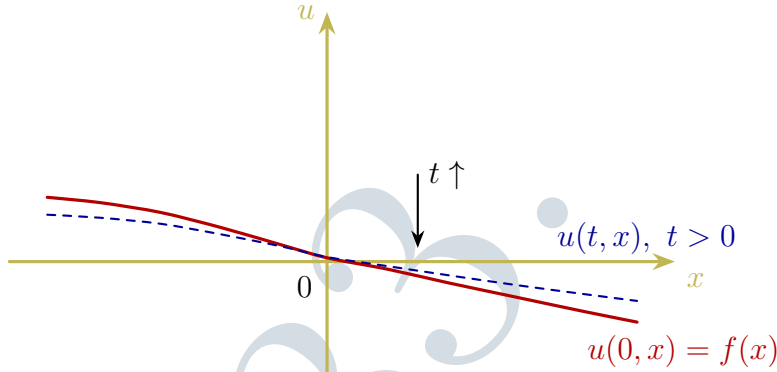
Then we completed the proof by showing all the implication. □

6.3 Heat Equation

The heat equation describes the diffusion of temperature in a medium. In one spatial dimension, if $u(t, x)$ denotes the temperature at location $x \in \mathbb{R}$ and time $t \geq 0$, then the basic physical principle is that heat flows from hotter regions to colder ones. As a result, the rate of change in time is governed by the spatial curvature of the temperature profile, which leads to the equation:

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \quad (6.1)$$

, with boundary condition, $u(0, x) = f(x)$. Here the factor $1/2$ is the probabilistic normalisation, chosen so that operator $\frac{1}{2}\Delta$ is precisely the generator of Brownian motion. Thus the heat equation may be understood in two complementary ways: physically, it describes the smoothing of an initial temperature profile $f(x)$; probabilistically, it describes the evolution obtained by averaging the initial data along Brownian paths.



Hence, a fundamental solution is, **Heat Kernel**:

$$p(t; x, y) = p_t(x, y) := \frac{1}{\sqrt{2\pi t}} \exp \left\{ -\frac{1}{2} (x - y)^2 \right\}$$

, where $t > 0, x, y \in \mathbb{R}$, it is easy to check it satisfies Eq. (6.1). Notice that, when $t \downarrow 0$:

$$p_t(x, y) = \begin{cases} 0, & \forall x \neq y \\ \infty, & \forall x = y \end{cases}$$

The reason why it is heat kernel is that other solution can be derived from involution:

$$u(t, x) = \mathbb{E}^x f(W_t) = \int_{-\infty}^{\infty} f(y) p_t(x, y) dy$$

If the tail delay not too slow, then solution defined above is well-defined, i.e.:

$$\int_{-\infty}^{\infty} e^{-ax^2} |f(x)| dx < \infty, \forall 0 < t < \frac{1}{2a}$$

Remark 6.3.1. If f is bounded and continuous, we may rewrite into:

$$u(t, x) = \mathbb{E}^x f(W_t) = \mathbb{E}^0 f(x + \widetilde{W}_t)$$

Now, by D.C.T., we may see the density degrade into "delta function", i.e.:

$$\lim_{\substack{t \downarrow 0 \\ y \rightarrow x}} u(t, y) = \mathbb{E}^0 f(x) = f(x)$$

Hence the existence we already showed, a natural following question is that, do this form holds uniqueness? or what conditions we need for uniqueness to holds?

Theorem 6.3.1. (Tychonoff Uniqueness Theorem)

Suppose $u \in C^{1,2}$ on $(0, T] \times \mathbb{R}$ and satisfies:

(a) The heat equation, Eq. (6.1), i.e.:

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$$

(b) , with the limiting boundary condition,

$$\lim_{\substack{t \downarrow 0 \\ y \rightarrow x}} u(t, y) = 0$$

(c) For some $K > 0$, and $a > 0$, it holds, $\forall x \in \mathbb{R}, 0 < t \leq T$ that:

$$\sup_{0 < t \leq T} |u(t, x)| \leq K e^{ax^2}$$

Then $u = 0$ on $(0, T] \times \mathbb{R}$.

Proof. Let $T_y = \inf\{t \geq 0 : W_t = y\}$ be the hitting time of $y \in \mathbb{R}$, now if we fix $x \in \mathbb{R}$, find a n s.t. $n > |x|$, then we restrict on $R_n = T_n \wedge T_{-n}$. Fix $t \in [0, T)$, define:

$$v(\theta, x) = u(T - t - \theta, x)$$

, for $0 \leq \theta < T - t$, and notice its derivative has following relations:

$$\begin{aligned} \frac{\partial}{\partial \theta} v(\theta, x) &= -\frac{\partial}{\partial t} u(T - t - \theta, x) \\ &= -\frac{1}{2} \frac{\partial^2}{\partial x^2} u(T - t - \theta, x) = -\frac{1}{2} \frac{\partial^2}{\partial x^2} v(\theta, x) \end{aligned}$$

Hence, inspired by this, by Itô's formula, on $0 \leq s < T - t$

$$v(s \wedge R_n, W_{s \wedge R_n}) = v(0, W_0) + \int_0^{s \wedge R_n} \frac{\partial}{\partial W} v(r, W_r) dW_r$$

By $|W_n| \leq n$ for all $r \leq R_n$, we have that:

$$\begin{aligned} u(T - t, x) &= v(0, x) = \lim_{s \uparrow T-t} \mathbb{E}^x v(s \wedge R_n, W_{s \wedge R_n}) \\ &= \lim_{s \uparrow T-t} \mathbb{E}^x v(s, W_s) \mathbf{1}_{\{s < R_n\}} + \mathbb{E}^x v(R_n, W_{R_n}) \mathbf{1}_{\{s \geq R_n\}} = \mathbb{E}^x v(R_n, W_{R_n}) \mathbf{1}_{\{R_n < T-t\}} \end{aligned}$$

The elimination comes from boundary condition, next we may check:

$$\begin{aligned} |u(T-t, x)| &\leq K e^{an^2} \mathbf{P}^x (R_n < T-t) \leq K e^{an^2} \left[\mathbf{P}^x (T_n < T-t) + \mathbf{P}^x (T_{-n} < T-t) \right] \\ &\leq K e^{an^2} \left[\mathbf{P}^0 (T_{n-x} \leq T) + \mathbf{P}^0 (T_{n+x} \leq T) \right] \leq K e^{an^2} C \left[e^{\frac{-(n+x)^2}{2T}} + e^{\frac{-(n-x)^2}{2T}} \right] \end{aligned}$$

By $1/2T > a$, we know it goes to 0. Hence we showed that, on $0 < t < T < 1/2a$:

$$u(T-t, x) = 0, \forall x \in R$$

Lastly, by iteration on interval, we may extend the result, hence completed the proof. $\square$

Remark 6.3.2. Notice that, if we write the form, $u(t, x) = \mathbf{E}^x f(W_t)$, then:

$$u(t, W_0) = \mathbf{E}^x u(0, W_t)$$

, the time and space is kind of reverse.

Theorem 6.3.2. (Widder 1994)

Let $v(t, x)$ be a non-negative function defined on $(0, T) \times \mathbb{R}$, where $0 < T < \infty$, then the followings are equivalent:

(a) $\exists$ nondecreasing function, $F : \mathbb{R} \mapsto \mathbb{R}$, s.t., $\forall x \in \mathbb{R}$:

$$v(t, x) = \int_{-\infty}^{\infty} p_{T-t}(x, y) dF(y)$$

(b) $\exists v \in C^{1,2}((0, T) \times \mathbb{R})$ satisfies the "backward" heat equation:

$$\frac{\partial v}{\partial t} + \frac{1}{2} \frac{\partial^2 v}{\partial t^2} = 0$$

(c) For a Brownian family, $\{W_s, 0 \leq s < \infty\}$ w.r.t. $\{\mathcal{F}_s, 0 \leq s < \infty\}$, and on $(\Omega, \mathcal{F})$, its measure, $\{\mathbf{P}^x\}_{x \in \mathbb{R}}$, also $\forall t \in (0, T)$, $\forall x \in \mathbb{R}$, the process:

$$\{v(t+s, W_s), \mathcal{F}_s, 0 \leq s < T-t\}$$

, is a martingale on $(\Omega, \mathcal{F}, \mathbf{P}^x)$.

(d) $\forall x \in \mathbb{R}, 0 < t \leq t+s < T$, it holds that:

$$v(t, x) = \mathbf{E}^x v(t+s, W_s)$$

Proof. We first consider the trivial few:

(a) $\implies$ (b) Just directly follows previous procedure, consider its derivative:

$$\frac{\partial}{\partial t} v(t, x) = \frac{\partial}{\partial t} \int_{-\infty}^{\infty} p_{T-t}(x, y) dF(y) = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} p_{T-t}(x, y) dF(y)$$

By Fubini, we do the same to others, and combine them to complete proof.

(c) $\implies$ (d) Follows directly from the martingale property.

(d) $\implies$ (c) Directly by markov property, [then \(c\) and \(d\) are equivalent](#).

(b) $\implies$ (d) With $x \in \mathbb{R}$ fixed, we let $a < x < b$, and using Itô's formula:

$$v(t + s \wedge T_a \wedge T_b, W_{s \wedge T_a \wedge T_b}) = v(t, x) + \int_0^{s \wedge T_a \wedge T_b} \frac{\partial}{\partial W} v(t + r, W_r) dW_r$$

, where $0 \leq s < T - t$, by $W_r \in [a, b]$ for all $r \leq T_a \wedge T_b$, then it is martingale, it remains to consider limiting cases, where firstly, we let $a \rightarrow -\infty$ and $b \rightarrow \infty$ to get:

$$\begin{aligned} \mathbb{E}^x v(t + s, W_s) &= \mathbb{E}^x \lim_{\substack{a \rightarrow -\infty \\ b \rightarrow \infty}} v(t + s \wedge T_a \wedge T_b, W_{s \wedge T_a \wedge T_b}) \\ &\leq \liminf_{\substack{a \rightarrow -\infty \\ b \rightarrow \infty}} \mathbb{E}^x v(t + s \wedge T_a \wedge T_b, W_{s \wedge T_a \wedge T_b}) = v(t, x) \end{aligned}$$

, where the inequality comes from Fatou's lemma, then for the other part:

$$\begin{aligned} v(t, x) &= \mathbb{E}^x v(t + s, W_s) \mathbf{1}_{\{s \leq T_a \wedge T_b\}} + \mathbb{E}^x v(t + T_a, W_{T_a}) \mathbf{1}_{\{T_s \leq s \wedge T_b\}} + \mathbb{E}^x v(t + T_b, W_{T_b}) \mathbf{1}_{\{T_b \leq s \wedge T_a\}} \\ &\leq \mathbb{E}^x v(t + s, W_s) + \mathbb{E}^x v(t + T_a, W_{T_a}) \mathbf{1}_{\{T_a \leq s\}} + \mathbb{E}^x v(t + T_b, W_{T_b}) \mathbf{1}_{\{T_b \leq s\}} \end{aligned}$$

Hence, by sending limit, it suffices to show:

$$\begin{cases} \lim_{a \rightarrow -\infty} \mathbb{E}^x v(t + T_a, W_{T_a}) \mathbf{1}_{\{T_a \leq s\}} = 0 \\ \lim_{b \rightarrow \infty} \mathbb{E}^x v(t + T_b, W_{T_b}) \mathbf{1}_{\{T_b \leq s\}} = 0 \end{cases}$$

We only consider the latter one, for which, we may consider even stronger:

Claim 6.3.3. *For some $B > 0$, the following integral converges:*

$$\int_B^\infty \mathbb{E}^x v(t + T_b, b) \mathbf{1}_{\{T_b \leq s\}} db < \infty$$

Recall the density function of T_b is given by, $f_{T_b}(t) = h(t, b - x)$, where $\forall t > 0$,

$$h(t, x) = \frac{|x|}{\sqrt{2\pi t^3}} e^{-\frac{x^2}{2t}}$$

Notice that, $\forall t < x^2/3$, it is increasing in t , since:

$$\frac{\partial}{\partial t} h(t, x) = C_0 e^{-\frac{x^2}{2t}} t^{-\frac{7}{2}} (x^2 - 3t) > 0$$

, for some constant, $C_0 > 0$. Then we may choose, $B > 0$ be large s.t. $b - x > 0$ is large and $h(\sigma, b - x)$ is increasing on $\sigma \in [0, s]$, thus, $h(\sigma, b - x) \leq h(s, b - x)$, hence we remain to estimate/bound the tail, now, let $s < r < t$, s.t.

$$\begin{aligned} h(s, b - x) &= \sqrt{\frac{r}{s^3}} \frac{1}{\sqrt{2\pi r}} e^{-\frac{(b-x)^2}{2s}} (b - x) \\ &\leq \sqrt{\frac{r}{s^3}} \frac{1}{\sqrt{2\pi r}} e^{-\frac{(b-x)^2}{2r}} \cdot \underbrace{e^{-(b-x)^2(\frac{1}{2s} - \frac{1}{2r})}}_{\leq C(>0)} (b - x) \\ &\leq C \sqrt{\frac{r}{s^2}} \cdot p_r(x, b) \end{aligned}$$

Then, we may bound the integral using this transition probability:

$$\begin{aligned}
& \int_B^\infty \mathbb{E}^x v(t + T_b, b) \mathbf{1}_{\{T_b \leq s\}} db \\
& \leq \int_B^\infty db \int_0^s v(t + \sigma, b) C \sqrt{\frac{r}{s^3}} p_r(x, b) d\sigma \\
& = \int_0^s d\sigma \sqrt{\frac{r}{s^3}} \int_B^\infty v(t + \sigma, b) p_r(x, b) db \\
& \leq \sqrt{\frac{r}{s^3}} \int_0^s d\sigma \int_{-\infty}^\infty v(t + \sigma, b) p_r(x, b) db \\
& = \sqrt{\frac{r}{s^3}} \int_0^s \mathbb{E}^x v(t + \sigma, W_r) d\sigma \leq \sqrt{\frac{r}{s^3}} \int_0^s v(t + \sigma - r, x) d\sigma < \infty
\end{aligned}$$

The boundedness due to the continuity of $v(\cdot)$, then $\forall 0 < t \leq t + s < T, s < t$:

$$v(t, x) = \mathbb{E}^x v(t + s, W_s)$$

To extend, we use the markov property, $\forall t \leq s < 2t$, pick, $s_1 < t$ and $s - s_1 < t$:

$$\mathbb{E}^x v(t + s, W_s) = \mathbb{E}^x \left[\mathbb{E}^x (v(t + s, W_s) \mid \mathcal{F}_{s_1}) \right] = \mathbb{E}^x v(t + s_1, W_{s_1}) = v(t, x)$$

(d) $\implies$ (a) Writing out expectation, we fix $T/2 < t < T$ and $\varepsilon \in (0, T/4)$:

$$\begin{aligned}
v(t - \varepsilon, x) &= \mathbb{E}^x v(T - \varepsilon, W_{T-t}) = \int_{-\infty}^\infty p_{T-t}(x, y) v(T - \varepsilon, y) dy \\
&= \int_{-\infty}^\infty \underbrace{\frac{p_{T-t}(x, y)}{p_{T/2}(0, y)} \cdot p_{T/2}(0, y) v(T - \varepsilon, y) dy}_{:= dF_\varepsilon(y)}
\end{aligned}$$

Denote $F_\varepsilon(y)$ as above, and notice that:

$$\begin{aligned}
F_\varepsilon(\infty) &= \int_{-\infty}^\infty p_{T/2}(0, z) v(T - \varepsilon, z) dz \\
&= \mathbb{E}^0 v(T - \varepsilon, W_{T/2}) = v\left(\frac{T}{2} - \varepsilon, 0\right) < \infty
\end{aligned}$$

Hence, $\sup_{0 < \varepsilon < T/4} F_\varepsilon(\infty) \leq \sup_{T/4 \leq s \leq T/2} v(s, 0) < \infty$, by Helly's Selection Principle:

$$\exists \varepsilon_n \downarrow 0, \text{ s.t. } \lim_{n \rightarrow \infty} F_{\varepsilon_n}(x) = F^*(x)$$

, for nondecreasing function, F^* at every continous point x . $\forall$ fixed $t \in (T/2, T)$:

$$\frac{p_{T-t}(x, y)}{p_{T/2}(0, y)} \sim \frac{e^{-\frac{(y-x)^2}{2(T-t)}}}{e^{-\frac{y^2}{2 \cdot T/2}}} = e^{-\frac{y^2}{2} \left(\underbrace{\frac{1}{T-t} - \frac{1}{T/2}}_{>0} \right)} \xrightarrow{y \rightarrow \infty} 0$$

Thus, by sending $\varepsilon \downarrow 0$, we showed:

$$v(t, x) = \int_{-\infty}^{\infty} p_{T-t}(x, y) \cdot \frac{1}{p_{T/2}(0, y)} dF^*(y) = \int_{-\infty}^{\infty} p_{T-t}(x, y) dF(y)$$

Lasly, $\forall 0 < t \leq T/2$, we choose, $t_1 \in (T/2, T)$ s.t.:

$$v(t_1, y) = \int_{-\infty}^{\infty} p_{T-t_1}(y, z) dF(z)$$

By Fubini, we could finish the last step:

$$\begin{aligned} v(t, x) &= \mathbf{E}^x v(t_1, W_{t_1-t}) = \int_{-\infty}^{\infty} v(t_1, y) p_{t_1-t}(x, y) dy \\ &= \int_{-\infty}^{\infty} p_{t_1-t}(x, y) dy \int_{-\infty}^{\infty} p_{T-t_1}(y, z) dF(z) \\ &= \int_{-\infty}^{\infty} p_{T-t}(x, z) dF(z) \end{aligned}$$

Therefore, we completed the proof by showing all the implication. □

Remark 6.3.3. From this theorem, we could also see the time reversion:

$$v(t, x) = \int_{-\infty}^{\infty} p_{T-t}(x, y) dF(y) \iff \frac{\partial v}{\partial t} + \frac{1}{2} \frac{\partial^2 v}{\partial x^2} = 0$$

Next, notice that the probablistic solution defined is:

$$u(t, x) = \int_{-\infty}^{\infty} p_t(x, y) dF(y) \iff \frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$$

Hence, we see that relation, that: $u(T - t, x) = v(t, x)$.

6.4 Feynman-Kac Formula

To extend the frontier, we could consider the parabolic equation:

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) + k(x)u(t, x) = \frac{1}{2} \Delta u(t, x) + g(t, x) \\ u(0, x) = f(x), \forall x \in \mathbb{R}^d \end{cases}$$

, for suitable function $k : \mathbb{R}^d \mapsto [0, \infty)$, $g : (0, \infty) \times \mathbb{R}^d \mapsto \mathbb{R}$ and $f : \mathbb{R}^d \mapsto \mathbb{R}$. Then Feynman-Kac formula is trying to provide a stochastic representation for solution of above parabolic equation. Firstly we check its "dual" problem:

Definition 6.4.1. Consider following continuous functions:

$$\begin{cases} f : \mathbb{R}^d \mapsto \mathbb{R} \\ k : \mathbb{R}^d \mapsto [0, \infty) \\ g : [0, T] \times \mathbb{R}^d \mapsto \mathbb{R} \end{cases}$$

Suppose that $v \in C^{1,2}([0, T] \times \mathbb{R}^d)$ and satisfies:

$$\begin{cases} -\frac{\partial v}{\partial t} + kv = \frac{1}{2}\Delta v + g \\ v(T, x) = f(x), \forall x \in \mathbb{R}^d \end{cases}$$

Then the function v is called to be a solution of the **Cauchy problem** for the backward heat equation, with corresponding terminal condition.

And Feynman-Kac established the result based on this Cauchy problem:

Theorem 6.4.1. (Feynman (1948), Kac (1949))

Let v defined as Def. 6.4.1, assume that, $\forall x \in \mathbb{R}^d$:

$$\max_{0 \leq t \leq T} |v(t, x)| + \max_{0 \leq t \leq T} |g(t, x)| \leq Ke^{a\|x\|^2}$$

, for some constant $K > 0$ and $0 < a < \frac{1}{2Td}$, then v admits, $\forall x \in \mathbb{R}^d$:

$$\begin{aligned} v(t, x) = & \mathbb{E}^x \left[f(W_{T-t}) \exp \left\{ - \int_0^{T-t} k(W_s) ds \right\} \right. \\ & \left. + \int_0^{T-t} g(t + \theta, W_\theta) \exp \left\{ - \int_0^\theta k(W_s) ds \right\} d\theta \right] \end{aligned}$$

, $\forall 0 \leq t \leq T$. In particular, such a solution is unique.

Proof. Fill in a proof. □

Corollary 6.4.2. Assume $u \in C^{1,2}((0, \infty) \times \mathbb{R}^d)$ and satisfies:

$$\begin{cases} \frac{\partial u}{\partial t} + ku = \frac{1}{2}\Delta u + g \\ u(0, x) = f(x), \forall x \in \mathbb{R}^d \end{cases}$$

And for each finite $T > 0$, $\exists$ a constant $K > 0$ and $0 < a < \frac{1}{2Td}$, s.t., $\forall x \in \mathbb{R}^d$:

$$\max_{0 \leq t \leq T} |u(t, x)| + \max_{0 \leq t \leq T} |g(t, x)| \leq Ke^{a\|x\|^2}$$

, then u admits the stochastic representation:

$$\begin{aligned} u(t, x) = & \mathbb{E}^x \left[f(W_t) \exp \left\{ - \int_0^t k(W_s) ds \right\} \right. \\ & \left. + \int_0^{T-t} g(t + \theta, W_\theta) \exp \left\{ - \int_0^\theta k(W_s) ds \right\} d\theta \right] \end{aligned}$$

Proof. Directly follow from Theorem 6.4.1, recall that:

$$\frac{\partial}{\partial t} v(t, x) = -\frac{\partial}{\partial t} u(T - t, x)$$

Similar to apply Itô's formula, we completed the proof. $\square$

Next, we lastly consider one-dimensional case, i.e.:

$$(\alpha + k(x)) z(x) = \frac{1}{2} z''(x) + f(x)$$

, for $x \in \mathbb{R}$. Here, $\alpha > 0$, k and f are piecewise continuous function on $\mathbb{R}$.

Definition 6.4.2. A Borel-measurable function, $f : \mathbb{R} \mapsto \mathbb{R}$ is called piecewise continuous if on every bounded interval, there are at most finitely many points of discontinuity and $\forall t \in \mathbb{R}$, $f(t+)$ & $f(t-)$ $\exists$.

Remark 6.4.1. A continuous function, f , is called piecewise, C^j , ($j \geq 1$) if its derivatives, $f^{(i)}$, $1 \leq i \leq j - 1$ are continuous, and $f^{(j)}$ is piecewise continuous.

A natural example is to be the, $f(x) = |x|$, which is piecewise C^1 .

Theorem 6.4.3. (Kac 1951) Let f and k be piecewise continuous, s.t., $\forall x \in \mathbb{R}$:

$$\mathbb{E}^x \int_0^\infty e^{-\alpha t} |f(W_t)| dt < \infty$$

Then,

$$z(x) = \mathbb{E}^x \int_0^\infty f(W_t) \exp \left\{ -\alpha t - \int_0^t k(W_s) ds \right\} dt$$

, is piecewise C^2 and satisfies the one-dimensional parabolic equation, i.e.:

$$(\alpha + k(x)) z(x) = \frac{1}{2} z''(x) + f(x)$$

, for all $x \in \mathbb{R} / (D_f \cup D_k)$, where $D_g := \{\text{discontinuity points of } g\}$.

Proof. For piecewise continuous function, g , we introduce resolvent operator, G_α :

$$\begin{aligned} (G_\alpha g)(x) &:= \mathbb{E}^x \int_0^\infty e^{-\alpha t} g(W_t) dt = \int_0^\infty e^{-\alpha t} dt \int_{-\infty}^\infty g(y) p_t(x, y) dy \\ &= \int_{-\infty}^\infty dy g(y) \int_0^\infty e^{-\alpha t} \frac{1}{\sqrt{2\pi t}} e^{-\frac{|y-x|^2}{2t}} dt \\ &= \int_{-\infty}^\infty g(y) \frac{1}{\sqrt{2\alpha}} e^{-|y-x|\sqrt{2\alpha}} dy \end{aligned}$$

Then by the definition of derivative, and D.C.T, we can interchange the limit to get:

$$\begin{aligned} (G_\alpha g)'(x) &= \frac{1}{\sqrt{2\alpha}} \int_{-\infty}^\infty g(y) \text{sgn}(y - x) e^{-|y-x|\sqrt{2\alpha}} dy \\ (G_\alpha g)''(x) &= 2\alpha (G_\alpha g)(x) - 2g(x) \end{aligned}$$

Next, notice that:

$$\frac{d}{ds} \left(e^{-\int_s^t k(W_r) dr} \right) = k(W_s) e^{-\int_s^t k(W_r) dr}$$

Hence, we have following observation:

$$1 - \exp \left\{ - \int_0^t k(W_r) dr \right\} = \int_0^t k(W_s) \exp \left\{ - \int_s^t k(W_r) dr \right\} ds$$

And by Fubini with markov property, we could show (you may verify):

$$G_\alpha(kz) = G_\alpha f - z, \quad G_\alpha(|kz|)(x) < \infty$$

Then we could see:

$$\begin{aligned} -z''(x) &= (G_\alpha kz - G_\alpha f)''(x) = 2\alpha (G_\alpha kz - G_\alpha f)(x) - 2kz(x) + 2f(x) \\ &= -2\alpha z(x) - 2kz(x) + 2f(x) \end{aligned}$$

Rearranging the terms, we completed the proof. □

Remark 6.4.2. Empowered by Laplace transform computation:

$$\int_0^\infty e^{-\alpha t} \frac{1}{\sqrt{2\pi t}} e^{-\xi^2/2t} dt = \frac{1}{\sqrt{2\alpha}} e^{-|\xi|\sqrt{2\alpha}}$$

, enables us to replace the condition by the equivalent condition, $\forall x \in \mathbb{R}$:

$$\int_{-\infty}^\infty |f(x+y)| e^{-|y|\sqrt{2\alpha}} dy$$

Proposition 6.4.4. (Arc-Sine Law for Occupation Time) *Let:*

$$\Gamma_+(t) := \int_0^t \mathbf{1}_{(0,\infty)}(W_s) ds$$

Then, $\forall 0 \leq \theta \leq t$:

$$\mathbb{P}^0(\Gamma_+(t) \leq \theta) = \int_0^{\theta/t} \frac{ds}{\pi \sqrt{s(1-s)}} = \frac{2}{\pi} \arcsin \sqrt{\frac{\theta}{t}}$$

Proof. An application of Theorem 6.4.3, we omit the proof. □